COMP2711 / DISCRETE MATHEMATICAL TOOLS

Logic

Past Paper Solutions

202	3	2013-25
verified exercises	topic groups	source papers

English solutions aligned with the supplied official answers.

Each answer has two parts: the original Official Answer and the existing Supplementary Explanation. Source PDF pages are retained for verification.

Generated 2 October 2026

All verified, in-scope questions linked to Logic in the knowledge base.

Each question is included once, including cross-topic applications.

READING GUIDE

Contents

Propositional Logic
50 exercises — L001-L050

Predicate Logic
100 exercises — L051-L150

Inference and Proofs
52 exercises — L151-L202

How to use this booklet

Questions are grouped by their first Logic subtopic, then ordered by difficulty and year (newest first). Each question appears once as L001-L202. Difficulty ranges from 1 (introductory) to 4 (challenging).

Official Answer reproduces the answer in the source paper, with its physical PDF page reference.

Supplementary Explanation retains the existing independently authored reasoning, including the stated result and derivation. The two parts remain separate throughout the booklet.

Source references use physical PDF pages. File links work with the local source collection at its existing location.

The answer provenance shown for each exercise retains its knowledge-base classification; attaching an official quotation does not reclassify a derived correction as official.

L001

Propositional Logic

2024 Spring Midterm | Question 2(a) | Source page 4 | Difficulty 1 | 1 mark
Q-2024S-MID-P02-a | Answer provenance: official

> Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.
>
> (a) $(p \vee q) \rightarrow (p \wedge q)$

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Midterm.pdf).

> Answer. (a) N

Supplementary Explanation

Answer. N.

Derivation/proof. With $p = T, q = F$, the antecedent is true and the consequent false, so the formula is false and cannot be a tautology. With $p = q = T$, both are true, so the implication is true and cannot be a contradiction. Both assignments are permitted, establishing the classification N.

L002

Propositional Logic

2024 Spring Midterm | Question 2(b) | Source page 4 | Difficulty 1 | 1 mark
Q-2024S-MID-P02-b | Answer provenance: official

> Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.
>
> (b) $(p \vee \neg q) \wedge (\neg p \wedge q)$

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Midterm.pdf).

> Answer. (b) F

Supplementary Explanation

Answer. F.

Derivation/proof. If the whole conjunction were true, its right side would force $p = F, q = T$. Then $p \vee \neg q = F \vee F = F$, a contradiction. Thus there is no true assignment, so the formula is a contradiction, classified F. Equivalently, distributing gives $(p \wedge \neg p \wedge q) \vee (\neg q \wedge \neg p \wedge q) = F$.

L003

Propositional Logic

2024 Spring Midterm | Question 2(c) | Source page 4 | Difficulty 1 | 1 mark
Q-2024S-MID-P02-c | Answer provenance: official

> Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.
>
> (c) $((p \vee q) \wedge r) \leftrightarrow ((p \wedge r) \vee (q \wedge r))$

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Midterm.pdf).

> Answer. (c) T

Supplementary Explanation

Answer. T.

Derivation/proof. The distributive law gives $(p \vee q) \wedge r \equiv (p \wedge r) \vee (q \wedge r)$. The left side says that r is true and at least one of p, q is true; the right side says exactly the same. Both sides have the same truth value, so their biconditional is true under every assignment. The answer is T.

L004

Propositional Logic

2024 Spring Midterm | Question 2(d) | Source page 4 | Difficulty 1 | 1 mark
Q-2024S-MID-P02-d | Answer provenance: official

> Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.
>
> (d) $(p \leftrightarrow q) \wedge (q \leftrightarrow r) \wedge \neg(p \leftrightarrow r)$

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Midterm.pdf).

> Answer. (d) F

Supplementary Explanation

Answer. F.

Derivation/proof. If the first two biconditionals hold, p, q, r all have the same truth value. Then $p \leftrightarrow r$ is true, and the final negation is false. If either of the first two biconditionals is false, the whole conjunction is false already. These two cases exhaust all assignments, so the formula is always false and is classified F.

L005

Propositional Logic

2024 Spring Midterm | Question 2(e) | Source page 4 | Difficulty 1 | 1 mark
Q-2024S-MID-P02-e | Answer provenance: official

> Assume p, q and r are logical propositions. Which of the following statements are tautologies (T), which are contradictions (F), and which are neither of them (N)? No explanation is needed.
>
> (e) $((p \to (q \vee r)) \leftrightarrow ((p \to q) \vee (p \to r))$

Official Answer

Source: physical PDF page 4 (COMP2711_S2024_Midterm.pdf).

> Answer. (e) T

Supplementary Explanation

Answer. T.

Derivation/proof. The left operand satisfies $p \to (q \vee r) \equiv \neg p \vee q \vee r$. The right satisfies $(p \to q) \vee (p \to r) \equiv (\neg p \vee q) \vee (\neg p \vee r) \equiv \neg p \vee q \vee r$, using associativity, commutativity and the idempotent law. The operands are equivalent, making the biconditional a tautology, classified T. The extra opening parenthesis in the printed expression adds no operand or connective.

L006

Propositional Logic

2023 Spring Midterm | Question 1(a)(2) | Source page 3 | Difficulty 1
Q-2023S-MID-P01-a-2 | Answer provenance: official

> Let
>
> 1. r: It rains.
> 2. f: It is foggy.
> 3. s: The sailing race will be held.
> 4. l: The life saving demonstration will go on.
> 5. t: The trophy will be awarded.
>
> Hypotheses (the printed premises, retained for the translation and inference atoms):
>
> 1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.
> 2. If the sailing race is held, then the trophy will be awarded.
> 3. The trophy was not awarded.
>
> First translate the following hypotheses into propositional logic statements:
>
> 2. If the sailing race is held, then the trophy will be awarded.

Official Answer

Source: physical PDF page 4 (COMP2711_S2023_Midterm.pdf).

> (a) We are given premises $(\neg r \vee \neg f) \rightarrow (s \wedge l)$, $s \rightarrow t$, and $\neg t$. We want to conclude r.

Supplementary Explanation

Answer. $s \rightarrow t$.

Derivation/proof. By the definitions in the shared context, “the sailing race is held” is s and “the trophy will be awarded” is t. The English construction “If A, then B” translates to $A \rightarrow B$, so the formula is $s \rightarrow t$. Its only falsifying case is a held race together with no trophy award. A trophy without the race is not prohibited by this hypothesis.

The only falsifying case is a race with no trophy; a trophy without a race is allowed. Thus the statement cannot be strengthened to $s \leftrightarrow t$.

L007

Propositional Logic

2023 Spring Midterm | Question 1(a)(3) | Source page 3 | Difficulty 1
Q-2023S-MID-P01-a-3 | Answer provenance: official

> Let
>
> 1. r: It rains.
> 2. f: It is foggy.
> 3. s: The sailing race will be held.
> 4. l: The life saving demonstration will go on.
> 5. t: The trophy will be awarded.
>
> Hypotheses (the printed premises, retained for the translation and inference atoms):
>
> 1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.
> 2. If the sailing race is held, then the trophy will be awarded.
> 3. The trophy was not awarded.
>
> First translate the following hypotheses into propositional logic statements:
>
> 3. The trophy was not awarded.

Official Answer

Source: physical PDF page 4 (COMP2711_S2023_Midterm.pdf).

> (a) We are given premises $(\neg r \vee \neg f) \rightarrow (s \wedge l)$, $s \rightarrow t$, and $\neg t$. We want to conclude r.

Supplementary Explanation

Answer. $\neg t$.

Derivation/proof. The shared definition assigns t to the trophy being awarded. The third hypothesis explicitly says this event did not occur, so its translation is $\neg t$. This premise fixes the value of t to false and does not itself fix r, f, s, or l. Combined with $s \rightarrow t$ it can later support modus tollens, but that later inference is not part of this translation.

This makes the truth value of t false. Combining it with another premise is a separate inference step, not part of the translation.

L008

Propositional Logic

2022 Midterm | Question 1(a)(i) | Source page 1 | Difficulty 1
Q-2022-MID-P01-a-i | Answer provenance: official

> Given $\neg a \vee (b \rightarrow \neg c)$, express its equivalent statements using only:
>
> NOT ($\neg$) and Implication ($\rightarrow$).

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (a) (i) $a \rightarrow (b \rightarrow \neg c)$

Supplementary Explanation

Answer. $a \rightarrow (b \rightarrow \neg c)$.

Derivation/proof. The equivalence $\neg a \vee X \equiv a \rightarrow X$ applies with X=(b→¬c). The inner formula already uses only implication and negation, so no further transformation is needed. Expanding both arrows produces ¬a∨¬b∨¬c, which verifies equivalence to the original expression.

Expanding both implications yields $\neg a \vee \neg b \vee \neg c$.

L009

Propositional Logic

2022 Midterm | Question 1(a)(ii) | Source page 1 | Difficulty 1
Q-2022-MID-P01-a-ii | Answer provenance: official

> Given $\neg a \vee (b \rightarrow \neg c)$, express its equivalent statements using only:
>
> NOT ($\neg$) and OR ($\vee$).

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (ii) $\neg a \vee \neg b \vee \neg c$

Supplementary Explanation

Answer. $\neg a \vee \neg b \vee \neg c$.

Derivation/proof. Replace b→¬c by ¬b∨¬c using implication elimination. The original becomes ¬a∨(¬b∨¬c), and associativity removes the unnecessary parentheses. Only negation and OR remain. The expression is false precisely when all three variables are true, matching the original.

The elimination is $b \rightarrow \neg c \equiv \neg b \vee \neg c$. The formula is false when all three are true, its only false valuation.

L010

Propositional Logic

2019 Midterm | Question 1(a) | Source page 3 | Difficulty 1 | 2 marks
Q-2019-MID-P01-a | Answer provenance: official

> Let c, d, r, and w be the propositiions
>
> c: “It is cold”
>
> d: “It is dry”
>
> r: “It is rainy”
>
> w: “It is windy”
>
> Write the following propositions using c, d, r and w and the Boolean connectives.
>
> (a) It is neither cold nor dry.

Official Answer

Source: physical PDF page 3 (2019-midterm.pdf).

$\neg c \wedge \neg d$.

Supplementary Explanation

Answer. $\neg c \wedge \neg d$.

Derivation/proof. “Neither A nor B” requires both A and B to be false. Here that is not cold and not dry, so the two negative requirements are joined with $\wedge$. By De Morgan's law, $\neg(c \vee d)$ is equivalent, whereas $\neg(c \wedge d)$ would also allow exactly one to be true.

L011

Propositional Logic

2019 Midterm | Question 1(b) | Source page 3 | Difficulty 1 | 2 marks
Q-2019-MID-P01-b | Answer provenance: official

> Let c, d, r, and w be the propositiions
>
> c: “It is cold”
>
> d: “It is dry”
>
> r: “It is rainy”
>
> w: “It is windy”
>
> Write the following propositions using c, d, r and w and the Boolean connectives.
>
> (b) It is rainy if it is not cold.

Official Answer

Source: physical PDF page 3 (2019-midterm.pdf).

$\neg c \to r$.

Supplementary Explanation

Answer. $\neg c \to r$.

Derivation/proof. A sentence “A if B” means “if B, then A.” Taking A as rainy and B as not cold places $\neg c$ in the antecedent and r in the consequent. It makes no claim about rain when it is cold.

The only falsifying case is not cold and not rainy: $\neg c$ is then true while r is false. This confirms the direction of the implication $\neg c \to r$.

L012

Propositional Logic

2018 Midterm | Question 2(a) | Source page 4 | Difficulty 1
Q-2018-MID-P02-a | Answer provenance: official

> Some of the following propositions and predicates are tautologies. For each statement that is not a tautology, prove it by providing a counterexample. For each statement that is a tautology, just state it is a tautology; no proof is required.
>
> (a) $(p \wedge q) \vee (q \wedge \neg p) \vee \neg q \vee (r \wedge T)$

Official Answer

Source: physical PDF page 4 (2018-midterm.pdf).

Tautology. (No proof required.)

Supplementary Explanation

Answer. It is a tautology.

Derivation/proof. The first two terms simplify to $q \wedge (p \vee \neg p) = q$. The expression therefore becomes $q \vee \neg q \vee r = T \vee r = T$. This checks every valuation without a truth table; the exam itself requires only the classification.

L013

Propositional Logic

2018 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2018-MID-P04-a-contrapositive | Answer provenance: official

> Consider finite sets S and T, and a function $f : S \to T$.
>
> "If f is injective, then $|T| \geq |S|$."
>
> (a) Consider finite sets S and T, and a function $f : S \to T$. Write down the contrapositive of "If f is injective, then $|T| \geq |S|$."

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

The contrapositive is "If $|T| < |S|$, then f is not injective". We can prove it by the pigeonhole principle. f maps pigeons in S to holes in T. Since $|T| < |S|$, there must be two pigeons that share a same hole, say $a, b \in S$. Then $a \neq b$ but $f(a) = f(b)$, so f is not injective.

Supplementary Explanation

Answer. If $|T| < |S|$, then f is not injective.

Derivation/proof. Write the original statement as $A \to B$, where A is injectivity and B is $|T| \geq |S|$. Its contrapositive is $\neg B \to \neg A$. For finite cardinalities, $\neg(|T| \geq |S|)$ is exactly $|T| < |S|$, yielding the stated sentence.

L014

Propositional Logic

2017 Midterm | Question 1(b) | Source page 3 | Difficulty 1
Q-2017-MID-P01-b | Answer provenance: official

> Some of the following propositions and predicates are tautologies. Give a proof for each of the statements that are tautologies. You do not need to disprove the ones that are not tautologies.
>
> (b) $(p \to q) \vee r \vee q$

Official Answer

Source: physical PDF page 3 (2017-midterm.pdf).

Not a tautology. It is false when p is true, q is false and r is false. (No counter example required)

Supplementary Explanation

Answer. It is not a tautology.

Derivation/proof. Take $p = T, q = F, r = F$. The implication $p \to q$ is false, so the whole expression is $F \vee F \vee F = F$. This additional check verifies the classification even though a counterexample is not required in the exam.

L015

Propositional Logic

2013 Final | Question 5(a) | Source page 8 | Difficulty 1
Q-2013-FINAL-P05-a | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (a) $(p \wedge q) \Rightarrow r$ and $\neg p \vee \neg q \vee r$

Official Answer

Source: PDF physical page 8 (2013-final.pdf).

(a) Equivalent.

$$\begin{aligned}(p \wedge q) \Rightarrow r &\equiv \neg(p \wedge q) \vee r && (s \Rightarrow t \equiv \neg s \vee t)\\ &\equiv \neg p \vee \neg q \vee r && (\text{by DeMorgan's law})\end{aligned}$$

Supplementary Explanation

Answer. Yes.

Derivation/proof. $(p \wedge q) \Rightarrow r \equiv \neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$ by implication elimination and De Morgan's law.

L016

Propositional Logic

2013 Final | Question 5(b) | Source page 8 | Difficulty 1
Q-2013-FINAL-P05-b | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (b) $(p \wedge q) \Rightarrow r$ and $\neg r \Rightarrow (p \Rightarrow \neg q)$

Official Answer

Source: PDF physical page 8 (2013-final.pdf).

(b) Equivalent.

$$\begin{aligned}\neg r \Rightarrow (p \Rightarrow \neg q) &\equiv r \vee (\neg p \vee \neg q) && (s \Rightarrow t \equiv \neg s \vee t)\\ &\equiv (p \wedge q) \Rightarrow r && (\text{by part (a)})\end{aligned}$$

Supplementary Explanation

Answer. Yes.

Derivation/proof. $\neg r \Rightarrow (p \Rightarrow \neg q) \equiv r \vee (\neg p \vee \neg q) = \neg p \vee \neg q \vee r$, which is the expression obtained from $(p \wedge q) \Rightarrow r$ in part (a).

The first expression also becomes $\neg(p \wedge q) \vee r \equiv \neg p \vee \neg q \vee r$.

L017

Propositional Logic

2025 Fall Final | Question 1(a) | Source page 3 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P01-a | Answer provenance: official

> (a) [3 points] Without using truth table prove that:
>
> $$[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \equiv p \wedge q$$

Official Answer

Source: physical PDF page 3 (COMP2711_F2025_Final.pdf), physical PDF page 4 (COMP2711_F2025_Final.pdf).

(a)

$$\begin{aligned}
(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r) &\equiv \neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r) && \text{(implication equivalence)} \\
&\equiv (p \wedge q) \vee (p \wedge q \wedge r) && \text{(De Morgan's law)} \\
&\equiv (p \wedge q) \vee \big((p \wedge q) \wedge r\big) && \text{(associativity)} \\
&\equiv (p \wedge q) \wedge (1 \vee r) && \text{(distributivity: } A \vee (A \wedge B) \equiv A \wedge (1 \vee B)\text{)} \\
&\equiv (p \wedge q) \wedge 1 && \text{(since } 1 \vee r \equiv 1\text{)} \\
&\equiv p \wedge q && \text{(identity law for } \wedge\text{)}
\end{aligned}$$

Thus,

$$[(\neg p \vee \neg q) \rightarrow (p \wedge q \wedge r)] \equiv p \wedge q.$$

(Note: the proposition r cancels out completely and does not affect the equivalence.)

Supplementary Explanation

Answer. The equivalence holds.

Derivation/proof. The implication law rewrites the left side as $\neg(\neg p \vee \neg q) \vee (p \wedge q \wedge r)$. De Morgan's law and double negation give $(p \wedge q) \vee ((p \wedge q) \wedge r)$. Put $A = p \wedge q$; absorption states $A \vee (A \wedge r) \equiv A$, so the expression equals $p \wedge q$. Every transformation preserves logical equivalence. No truth table or assumption about the truth value of r is used.

L018

Propositional Logic

2025 Fall Midterm | Question 1(a) | Source page 3 | Difficulty 2 | 3 marks
Q-2025F-MID-P01-a | Answer provenance: official

> We define NAND operator ($\overline{\wedge}$) as $p \mathbin{\overline{\wedge}} q = \neg(p \wedge q)$.
>
> (a) [3 points] Use propositional laws to prove that
>
> $$p \mathbin{\overline{\wedge}} (q \to r) \equiv (p \to q) \wedge (p \to \neg r)$$
>
> (you are NOT allowed to use truth table)

Official Answer

Source: physical PDF page 4 (COMP2711_F2025_Midterm.pdf).

> (a) Start with the left-hand side:
>
> $$p \mathbin{\overline{\wedge}} (q \to r) = \neg(p \wedge (q \to r)).$$
>
> Since $q \to r \equiv \neg q \vee r$:
>
> $$p \wedge (q \to r) = p \wedge (\neg q \vee r).$$
>
> Apply De Morgan's law to the NAND:
>
> $$\neg(p \wedge (\neg q \vee r)) = \neg p \vee \neg(\neg q \vee r) = \neg p \vee (q \wedge \neg r).$$
>
> Now consider the right-hand side:
>
> $$(p \to q) \wedge (p \to \neg r) = (\neg p \vee q) \wedge (\neg p \vee \neg r).$$
>
> Apply the distributive law:
>
> $$(\neg p \vee q) \wedge (\neg p \vee \neg r) = \neg p \vee (q \wedge \neg r).$$
>
> Both sides are equal:
>
> $$\neg p \vee (q \wedge \neg r) \equiv \neg p \vee (q \wedge \neg r).$$
>
> Thus, $p \mathbin{\overline{\wedge}} (q \to r) \equiv (p \to q) \wedge (p \to \neg r)$.

Supplementary Explanation

Answer. The equivalence holds.

Derivation/proof. By definition, the left side is $\neg(p \wedge (q \to r))$. Eliminate implication to obtain $\neg(p \wedge (\neg q \vee r))$, and use De Morgan's laws to obtain $\neg p \vee (q \wedge \neg r)$. On the right, eliminating both implications gives $(\neg p \vee q) \wedge (\neg p \vee \neg r)$, which equals the same expression by distribution. Every step uses equivalence laws, without a truth table.

L019

Propositional Logic

2025 Fall Midterm | Question 1(b) | Source page 3 | Difficulty 2 | 2 marks
Q-2025F-MID-P01-b | Answer provenance: official

> We define NAND operator ($\overline{\wedge}$) as $p \mathbin{\overline{\wedge}} q = \neg(p \wedge q)$.
>
> (b) [2 points] Argue whether the following two expressions are equivalent or not. If they are, prove it without using a truth table. If they are not, provide a counterexample.
>
> Expression A: $p \mathbin{\overline{\wedge}} (p \mathbin{\overline{\wedge}} q)$
>
> Expression B: $(p \mathbin{\overline{\wedge}} p) \mathbin{\overline{\wedge}} q$

Official Answer

Source: physical PDF page 4 (COMP2711_F2025_Midterm.pdf).

> (b) Consider a counterexample: let $p = \text{true}, q = \text{false}$. - Expression A: False. - Expression B: True. Since the expressions differ, they are not equivalent.

Supplementary Explanation

Answer. Not equivalent: at $p = T, q = F$, $A = F$ and $B = T$.

Derivation/proof. For A, the inner NAND is $\neg(T \wedge F) = T$, so the outer NAND is $\neg(T \wedge T) = F$. For B, the inner NAND is $\neg(T \wedge T) = F$, so the outer NAND is $\neg(F \wedge F) = T$. One assignment giving opposite values disproves equivalence. Changing the parentheses changes which conjunction is negated first; ordinary AND's associativity cannot be transferred to NAND.

L020

Propositional Logic

2025 Spring Final | Question 1(b) | Source page 2 | Difficulty 2 | 2 marks
Q-2025S-FINAL-P01-b | Answer provenance: official

> (b) [2 pts] Prove or disprove the following statement. In order to prove, you must use propositional laws. In order to disprove, you need to come up with values that this statement is false.
>
> $$(x \vee y \vee \neg z) \wedge (\neg x \vee y \vee z) \equiv \neg y \rightarrow (x \leftrightarrow z)$$

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(b) $(x \vee y \vee \neg z) \wedge (\neg x \vee y \vee z) \equiv y \vee ((x \vee \neg z) \wedge (\neg x \vee z)) \equiv y \vee ((z \rightarrow x) \wedge (x \rightarrow z)) \equiv y \vee (x \leftrightarrow z) \equiv \neg y \rightarrow (x \leftrightarrow z)$

Supplementary Explanation

Answer. The equivalence holds.

Derivation/proof. Commutativity and distributivity transform the left side into $y \vee ((x \vee \neg z) \wedge (\neg x \vee z))$. Here $x \vee \neg z \equiv z \rightarrow x$ and $\neg x \vee z \equiv x \rightarrow z$; their conjunction is the biconditional, giving $y \vee (x \leftrightarrow z)$. By the implication law the right side is $\neg(\neg y) \vee (x \leftrightarrow z) = y \vee (x \leftrightarrow z)$. Both sides therefore have the same expression by propositional laws.

L021

Propositional Logic

2025 Spring Midterm | Question 1(a) | Source page 2 | Difficulty 2
Q-2025S-MID-P01-a | Answer provenance: official

> (a) Prove that the following expression is a tautology without using the truth table
>
> $$\neg(\neg p \to (q \wedge \neg r)) \vee \neg(\neg p \wedge r)$$

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Midterm.pdf).

(a)

$$\begin{aligned}&\neg(\neg P \to (Q \wedge \neg R)) \vee \neg(\neg P \wedge R)\\ &\equiv \neg(P \vee (Q \wedge \neg R)) \vee \neg(\neg P \wedge R)\\ &\equiv \neg((P \vee Q) \wedge (P \vee \neg R)) \vee \neg(\neg P \wedge R)\\ &\equiv \neg(P \vee Q) \vee \neg(P \vee \neg R) \vee (P \vee \neg R) \equiv \neg(P \vee Q) \vee T \equiv T.\end{aligned}$$

Supplementary Explanation

Answer. T (a tautology).

Derivation/proof. Replacing the inner implication by a disjunction gives $\neg(p \vee (q \wedge \neg r)) \vee (p \vee \neg r)$. De Morgan's laws give $(\neg p \wedge (\neg q \vee r)) \vee p \vee \neg r$. Applying $p \vee (\neg p \wedge X) \equiv p \vee X$ yields $p \vee \neg q \vee r \vee \neg r$. Since $r \vee \neg r = T$, the whole disjunction is true. Every step is a propositional equivalence; no truth table is used.

L022

Propositional Logic

2025 Spring Midterm | Question 1(b) | Source page 2 | Difficulty 2
Q-2025S-MID-P01-b | Answer provenance: official

> (b) Argue whether the following two expressions are equivalent or not. If they are, prove it without using truth table. If they are not, provide a counterexample.
>
> Expression A: $(p \wedge \neg r) \vee (\neg q \wedge r)$
>
> Expression B: $(p \vee \neg q) \wedge (r \vee \neg r)$

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Midterm.pdf).

(b) They are not equivalent. $P = T, Q = T, R = T$ is a counter-example. It makes A false and makes B true.

Supplementary Explanation

Answer. Not equivalent; $p = q = r = T$ is a counterexample.

Derivation/proof. Use the same assignment on both sides. In A, $T \wedge F = F$ and $F \wedge T = F$, so $A = F \vee F = F$. In B, $(T \vee F) \wedge (T \vee F) = T$. One assignment with different truth values disproves equivalence. The tautological factor in B removes the restriction involving r, but does not make B itself a tautology.

L023

Propositional Logic

2024 Fall Final | Question 1(a) | Source page 3 | Difficulty 2 | 3 marks
Q-2024F-FINAL-P01-a | Answer provenance: official

> (a) Let p, q, r denote primitive statements. Use the laws of logic to show that $p \to (q \vee r)$ is equivalent with $(p \wedge \neg q) \to r$.

Official Answer

Source: physical PDF page 4 (COMP2711_F2024_Final.pdf).

> Answer: (a) Let p, q, r denote primitive statements. We aim to show that: $p \to (q \vee r) \equiv (p \wedge \neg q) \to r$.
>
> 1. $p \to (q \vee r) \equiv \neg p \vee (q \vee r)$ (Definition of implication)
> 2. $\equiv (\neg p \vee q) \vee (\neg p \vee r)$ (Distributive law)
> 3. Now consider the other side: $(p \wedge \neg q) \to r \equiv \neg(p \wedge \neg q) \vee r$ (Definition of implication)
> 4. $\equiv (\neg p \vee q) \vee r$ (De Morgan's Law)
> 5. $\equiv (\neg p \vee q) \vee (\neg p \vee r)$ (Since $(\neg p \vee q) \vee r$ is equivalent to $(\neg p \vee q) \vee (\neg p \vee r)$) Thus, $p \to (q \vee r)$ is logically equivalent to $(p \wedge \neg q) \to r$.

Supplementary Explanation

Answer. $p \to (q \vee r) \equiv (p \wedge \neg q) \to r$.

Derivation/proof. The implication law changes the left expression to $\neg p \vee (q \vee r) \equiv \neg p \vee q \vee r$. On the right, $(p \wedge \neg q) \to r \equiv \neg(p \wedge \neg q) \vee r$. De Morgan gives $(\neg p \vee \neg\neg q) \vee r$; double negation gives $\neg p \vee q \vee r$. Both reduce to the same expression, so their truth values agree for every assignment. Repeating $\neg p$ in the printed derivation also preserves equivalence, but uses idempotence and associativity rather than its stated distributive law.

L024

Propositional Logic

2024 Fall Midterm | Question 1(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2024F-MID-P01-a | Answer provenance: official

> (a) Briefly argue whether
>
> $$(p \to q) \vee (q \to r)$$
>
> is a tautology or not. If Yes, prove it. If No, provide a counter example. (You are not allowed to use truth table)

Official Answer

Source: physical PDF page 3 (COMP2711_F2024_Midterm.pdf).

> Answer: (a)
>
> $(p \to q) \vee (q \to r) \equiv (\neg p \vee q) \vee (\neg q \vee r) \equiv \neg p \vee q \vee \neg q \vee r \equiv \neg p \vee (q \vee \neg q) \vee r \equiv \neg p \vee T \vee r \equiv \text{True}.$

Supplementary Explanation

Answer. It is a tautology.

Derivation/proof. Logical equivalences give $(p \to q) \vee (q \to r) \equiv (\neg p \vee q) \vee (\neg q \vee r) \equiv \neg p \vee (q \vee \neg q) \vee r$. The excluded-middle law gives $q \vee \neg q = T$, so this is $\neg p \vee T \vee r = T$. These equivalences hold for every assignment of p, q, r, proving the tautology without the forbidden truth table.

L025

Propositional Logic

2024 Fall Midterm | Question 1(b) | Source page 3 | Difficulty 2 | 4 marks
Q-2024F-MID-P01-b | Answer provenance: official

> (b) Consider these two statements:
>
> $$p \to (q \to r)$$
>
> $$(p \to q) \to r$$
>
> If they are equivalent prove it. Otherwise, provide a counter example. (You are not allowed to use truth table)

Official Answer

Source: physical PDF page 4 (COMP2711_F2024_Midterm.pdf).

> (b) These two are not equivalent. Consider the following counter example: $p = F, q = T, r = F$. The top equation is True however the bottom equation is False.

Supplementary Explanation

Answer. Not equivalent; take $p = F, q = T, r = F$.

Derivation/proof. The first expression has inner value $q \to r = T \to F = F$ and outer value $p \to (q \to r) = F \to F = T$. The second has $p \to q = F \to T = T$ and then $(p \to q) \to r = T \to F = F$. The same valid assignment makes the first true and the second false, disproving equivalence. One counterexample suffices without a truth table.

L026

Propositional Logic

2024 Fall Midterm | Question 3(c) | Source page 7 | Difficulty 2 | 4 marks
Q-2024F-MID-P03-c | Answer provenance: official

> (c) Prove that $A - (B \cap C) = (A - B) \cup (A - C)$. You are Not allowed to use Venn diagram to prove it.

Official Answer

Source: physical PDF page 7 (COMP2711_F2024_Midterm.pdf).

> (c) Assume $x \in A - (B \cap C) \leftrightarrow x \in A \wedge x \notin (B \cap C) \leftrightarrow x \in A \wedge (x \notin B \vee x \notin C) \leftrightarrow (x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C) \leftrightarrow (x \in (A - B)) \vee (x \in (A - C)) \leftrightarrow x \in ((A - B) \cup (A - C))$.

Supplementary Explanation

Answer. $A - (B \cap C) = (A - B) \cup (A - C)$.

Derivation/proof. For arbitrary x, membership $x \in A \setminus (B \cap C)$ is equivalent to $x \in A \wedge \neg(x \in B \wedge x \in C)$. De Morgan gives $x \in A \wedge (x \notin B \vee x \notin C)$. The distributive law turns this into $(x \in A \wedge x \notin B) \vee (x \in A \wedge x \notin C)$, exactly $x \in (A \setminus B) \cup (A \setminus C)$. Every step is a two-way equivalence and x was arbitrary, so the sets have exactly the same members without a Venn diagram.

L027

Propositional Logic

2023 Fall Midterm | Question 1(a) | Source page 3 | Difficulty 2 | 4 marks
Q-2023F-MID-P01-a | Answer provenance: official

> Prove the following propositional equivalence without using a truth table:
>
> $$(p \to r) \vee (q \to r) \equiv (p \wedge q) \to r.$$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Midterm.pdf).

> (a) $(p \to r) \vee (q \to r) \equiv (\neg p \vee r) \vee (\neg q \vee r) \equiv \neg p \vee \neg q \vee r \equiv (\neg p \vee \neg q) \vee r \equiv \neg(p \wedge q) \vee r \equiv (p \wedge q) \to r$

Supplementary Explanation

Answer. The equivalence holds.

Derivation/proof. Using implication elimination, associativity, idempotence, and De Morgan in succession,

$$(p \to r) \vee (q \to r) \equiv (\neg p \vee r) \vee (\neg q \vee r) \equiv \neg p \vee \neg q \vee r \equiv \neg(p \wedge q) \vee r \equiv (p \wedge q) \to r.$$

Each step preserves truth under every valuation, and the final expression is exactly the requested right-hand side. No truth table is used.

L028

Propositional Logic

2023 Fall Midterm | Question 1(c) | Source page 3 | Difficulty 2 | 3 marks
Q-2023F-MID-P01-c | Answer provenance: official

> Determine if the two compound propositions $(p \to q) \to (r \to s)$ and $(p \to r) \to (q \to s)$ are logically equivalent. Show a proof if they are, or give a counter-example if they are not.

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Midterm.pdf).

> (c) Not equivalent. Counter-example: $p = F, q = T, r = F, s = F$

Supplementary Explanation

Answer. Not equivalent; use $p = F, q = T, r = F, s = F$.

Derivation/proof. For this valuation, $p \to q = T$ and $r \to s = T$, so the first expression is $T \to T = T$. Meanwhile $p \to r = T$ but $q \to s = F$, so the second expression is $T \to F = F$. Because logical equivalence requires equality under every common valuation, this one disagreement suffices to disprove it. The displayed values also independently verify the published counterexample.

L029

Propositional Logic

2023 Spring Midterm | Question 1(a)(1) | Source page 3 | Difficulty 2
Q-2023S-MID-P01-a-1 | Answer provenance: official

> Let
>
> 1. r: It rains.
> 2. f: It is foggy.
> 3. s: The sailing race will be held.
> 4. l: The life saving demonstration will go on.
> 5. t: The trophy will be awarded.
>
> Hypotheses (the printed premises, retained for the translation and inference atoms):
>
> 1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.
> 2. If the sailing race is held, then the trophy will be awarded.
> 3. The trophy was not awarded.
>
> First translate the following hypotheses into propositional logic statements:
>
> 1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.

Official Answer

Source: physical PDF page 4 (COMP2711_S2023_Midterm.pdf).

> (a) We are given premises $(\neg r \vee \neg f) \to (s \wedge l)$, $s \to t$, and $\neg t$. We want to conclude r.

Supplementary Explanation

Answer. $(\neg r \vee \neg f) \to (s \wedge l)$.

Derivation/proof. The antecedent is “it does not rain or it is not foggy”, so it is $\neg r \vee \neg f$. The consequent requires both the race and the demonstration, hence $s \wedge l$. Joining these with implication gives the stated formula. In particular, the hypothesis permits the implication to be vacuously true when both weather conditions are bad; it does not assert that either event actually occurs.

When both bad weather conditions hold, the antecedent is false, so the conditional does not assert that either activity actually occurs.

L030

Propositional Logic

2022 Midterm | Question 1(a)(iii) | Source page 1 | Difficulty 2
Q-2022-MID-P01-a-iii | Answer provenance: official

> Given $\neg a \vee (b \rightarrow \neg c)$, express its equivalent statements using only:
>
> NOT ($\neg$) and AND ($\wedge$).

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (iii) $\neg(a \wedge b \wedge c)$

Supplementary Explanation

Answer. $\neg(a \wedge b \wedge c)$.

Derivation/proof. First remove the implication to obtain ¬a∨¬b∨¬c. De Morgan's law in reverse gives ¬(a∧b∧c). This uses only the permitted NOT and AND operators, and its sole false valuation is a=b=c=true. Negating the entire conjunction is essential; moving the negation inside changes the formula.

The formula is false only when all three variables are true.

L031

Propositional Logic

2019 Midterm | Question 1(c) | Source page 3 | Difficulty 2 | 2 marks
Q-2019-MID-P01-c | Answer provenance: official

> Let c, d, r, and w be the propositiions
>
> c: “It is cold”
>
> d: “It is dry”
>
> r: “It is rainy”
>
> w: “It is windy”
>
> Write the following propositions using c, d, r and w and the Boolean connectives.
>
> (c) To be windy it is necessary that it be cold.

Official Answer

Source: physical PDF page 3 (2019-midterm.pdf).

$w \rightarrow c$.

Supplementary Explanation

Answer. $w \rightarrow c$.

Derivation/proof. Cold is necessary for wind, meaning wind cannot occur without cold. Thus every valuation with $w = T$ must have $c = T$, exactly the implication $w \rightarrow c$. The equivalent contrapositive is $\neg c \rightarrow \neg w$. Cold alone does not guarantee wind.

The implication permits cold without wind: w false and c true still satisfies $w \rightarrow c$. It forbids wind without cold, so reversing the implication would change the statement.

L032

Propositional Logic

2019 Midterm | Question 1(d) | Source page 3 | Difficulty 2 | 2 marks
Q-2019-MID-P01-d | Answer provenance: official

> Let c, d, r, and w be the propositiions
>
> c: "It is cold"
>
> d: "It is dry"
>
> r: "It is rainy"
>
> w: "It is windy"
>
> Write the following propositions using c, d, r and w and the Boolean connectives.
>
> (d) It is rainy only if it is windy and cold.

Official Answer

Source: physical PDF page 3 (2019-midterm.pdf).

$r \to (w \wedge c)$.

Supplementary Explanation

Answer. $r \to (w \wedge c)$.

Derivation/proof. "A only if B" makes B necessary for A. Here B itself consists of two simultaneous requirements, windy and cold. Therefore rain implies both w and c, expressed as a conjunction inside the consequent. The claim says nothing requiring rain whenever those two conditions hold.

Equivalently, $(r \to w) \wedge (r \to c)$. When it is not rainy, the sentence imposes no condition on wind or cold.

L033

Propositional Logic

2019 Midterm | Question 3(a) | Source page 5 | Difficulty 2 | 3 marks
Q-2019-MID-P03-a | Answer provenance: official

> For each of the following propositions, determine whether the proposition is a tautology. If yes, prove it using propositional equivalence and the laws of logic. If not, give a counterexample which consists of the truth values of the propositions under the setting.
>
> (a) $((p \to q) \wedge \neg p) \to \neg q$

Official Answer

Source: physical PDF page 5 (2019-midterm.pdf).

No. Set $p = F$ and $q = T$. We have $((F \to T) \wedge T) \to F \equiv T \to F \equiv F$.

Supplementary Explanation

Answer. No; take $p = F, q = T$.

Derivation/proof. With $p = F, q = T$, the implication $p \to q$ is true and $\neg p$ is true, so the outer antecedent is true. But $\neg q = F$, making the full proposition $T \to F = F$. This is the invalid pattern of denying the antecedent.

L034

Propositional Logic

2019 Midterm | Question 3(b) | Source page 5 | Difficulty 2 | 3 marks
Q-2019-MID-P03-b | Answer provenance: official

> For each of the following propositions, determine whether the proposition is a tautology. If yes, prove it using propositional equivalence and the laws of logic. If not, give a counterexample which consists of the truth values of the propositions under the setting.
>
> (b) $((p \to \neg q) \wedge q) \to \neg p$

Official Answer

Source: physical PDF page 5 (2019-midterm.pdf).

Yes.

$$\begin{aligned} ((p \to \neg q) \wedge q) \to \neg p &\equiv ((\neg p \vee \neg q) \wedge q) \to \neg p \\ &\equiv (\neg p \wedge q) \to \neg p \\ &\equiv \neg(\neg p \wedge q) \vee \neg p \\ &\equiv (p \vee \neg q) \vee \neg p \\ &\equiv T. \end{aligned}$$

Supplementary Explanation

Answer. It is a tautology.

Derivation/proof. Replace $p \to \neg q$ by $\neg p \vee \neg q$. Its conjunction with q becomes $(\neg p \wedge q) \vee (\neg q \wedge q) = \neg p \wedge q$. Therefore the whole expression is $(\neg p \wedge q) \to \neg p \equiv \neg(\neg p \wedge q) \vee \neg p \equiv p \vee \neg q \vee \neg p \equiv T$.

Using commutativity and associativity exposes $(p \vee \neg p) \vee \neg q \equiv T$, so this part of the formula is a tautology by the excluded-middle law.

L035

Propositional Logic

2018 Midterm | Question 1(b) | Source page 3 | Difficulty 2
Q-2018-MID-P01-b | Answer provenance: official

> There are only two kinds of inhabitants on an island: knights, who always tell the truth; and knaves, who always lie. You encounter two of the island's inhabitants, Alice and Bob. Alice tells you, "Bob is a knave." Bob clarifies, "Actually, Alice and I are both knaves."
>
> (b) Solve your proposition in (a) to show whether Alice is a knight or a knave, and Bob is a knight or a knave. (There may be multiple or no correct answers; list all of them.)

Official Answer

Source: physical PDF page 3 (2018-midterm.pdf).

Simplifying Bob's two cases:

$$\begin{aligned}((q \wedge (\neg q \wedge \neg p)) \vee (\neg q \wedge (p \vee q))) &\equiv F \vee (\neg q \wedge (p \vee q)) \\ &\equiv \neg q \wedge (p \vee q) \\ &\equiv \neg q \wedge p.\end{aligned}$$

This is in fact the first of Alice's cases, so Alice must be a knight and Bob must be a knave.

Supplementary Explanation

Answer. Alice is a knight and Bob is a knave; this is the only solution.

Derivation/proof. If $q = T$, Bob's statement $\neg p \wedge \neg q$ is false, contradicting his being a knight. Hence $q = F$. His statement must then be false, so $\neg p \wedge T$ is false and $p = T$. Alice's assertion $\neg q$ is true, consistent with her being a knight. Both statements are now checked and all alternatives excluded.

Thus the only solution is $p = T, q = F$. Bob's entire conjunction is false; its individual conjuncts need not both be false.

L036

Propositional Logic

2018 Midterm | Question 2(b) | Source page 4 | Difficulty 2
Q-2018-MID-P02-b | Answer provenance: official

> Some of the following propositions and predicates are tautologies. For each statement that is not a tautology, prove it by providing a counterexample. For each statement that is a tautology, just state it is a tautology; no proof is required.
>
> (b) $((p \to q) \to r) \to p$

Official Answer

Source: physical PDF page 4 (2018-midterm.pdf).

Not a tautology. It is false when p is false, q is true and r is True:

$$\begin{aligned}((p \to q) \to r) \to p &\equiv ((F \to T) \to T) \to F \\ &\equiv (T \to T) \to F \\ &\equiv T \to F \\ &\equiv F.\end{aligned}$$

Supplementary Explanation

Answer. Not a tautology: $p = F, q = T, r = T$.

Derivation/proof. At this valuation, $p \to q = F \to T = T$, then $(p \to q) \to r = T \to T = T$. The outer implication is consequently $T \to F = F$. Both levels of parentheses are essential to this evaluation.

L037

Propositional Logic

2017 Midterm | Question 1(a) | Source page 3 | Difficulty 2
Q-2017-MID-P01-a | Answer provenance: official

> Some of the following propositions and predicates are tautologies. Give a proof for each of the statements that are tautologies. You do not need to disprove the ones that are not tautologies.
>
> (a) $(p \wedge q) \vee (q \wedge \neg p) \vee \neg q \vee (p \wedge T)$

Official Answer

Source: physical PDF page 3 (2017-midterm.pdf).

Tautology.

$$\begin{aligned}(p \wedge q) \vee (q \wedge \neg p) \vee \neg q \vee (p \wedge T) &\equiv (q \wedge (p \vee \neg p)) \vee \neg q \vee p \\ &\equiv (q \wedge T) \vee \neg q \vee p \\ &\equiv (q \vee \neg q) \vee p \\ &\equiv T \vee p \\ &\equiv T.\end{aligned}$$

Supplementary Explanation

Answer. It is a tautology.

Derivation/proof. By distributivity, $(p \wedge q) \vee (q \wedge \neg p) = q \wedge (p \vee \neg p) = q$. Also $p \wedge T = p$. Hence the whole expression is $q \vee \neg q \vee p = T \vee p = T$, independently of the values of p, q.

L038

Propositional Logic

2016 Midterm | Question 4(a) | Source page 7 | Difficulty 2 | 4 marks
Q-2016-MID-P04-a | Answer provenance: official

> (a) Construct a truth table for the statement $(p \oplus q) \Leftrightarrow r$.

Official Answer

Source: PDF physical page 7 (2016-midterm.pdf).

(a) Truth Table:

p	q	r	$p \oplus q$	$(p \oplus q) \Leftrightarrow r$
T	T	T	F	F
T	T	F	F	T
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	F
F	F	F	F	T

Supplementary Explanation

Answer. The final column in TTT, TTF, TFT, TFF, FTT, FTF, FFT, FFF order is F, T, T, F, T, F, F, T.

Derivation/proof. Evaluate XOR first, then compare that result with r in each row.

p	q	r	$p \oplus q$	$(p \oplus q) \leftrightarrow r$
T	T	T	F	F
T	T	F	F	T
T	F	T	T	T
T	F	F	T	F
F	T	T	T	T
F	T	F	T	F
F	F	T	F	F
F	F	F	F	T

The XOR column tests whether $p \neq q$; the last column tests whether that Boolean result equals r.

The XOR column is F,F,T,T,T,T,F,F, and the final column is F,T,T,F,T,F,F,T. For example, TTF has XOR F equal to $r = F$, so its final entry is T.

L039

Propositional Logic

2015 Midterm | Question 1(a) | Source page 3 | Difficulty 2 | 3 marks
Q-2015-MID-P01-a | Answer provenance: official

> For each of the following parts, circle the correct answer or answers. There is no need to explain how your answers are derived.
>
> (a) Let p and q be two propositions. Which of the following propositions is/are equivalent to $p \leftrightarrow q$?
>
> 1. $(p \to q) \wedge (q \to p)$
> 2. $(\neg p \to \neg q) \wedge (\neg q \to \neg p)$
> 3. $(p \wedge q) \wedge (\neg p \wedge \neg q)$
> 4. $(p \wedge q) \vee (\neg p \wedge \neg q)$
> 5. None of above.

Official Answer

Source: PDF physical page 4 (2015-midterm.pdf).

(a) 1, 2, 4

Supplementary Explanation

Answer. 1, 2, 4.

Derivation/proof. Option 1 is the definition of the biconditional. Contraposition changes its two implications to the two factors in option 2. The true rows of $p \leftrightarrow q$ are (T, T) and (F, F), exactly option 4. Option 3 is always false, so it fails on both those rows; consequently option 5 is false.

Option 2 uses $\neg p \to \neg q \equiv q \to p$ and $\neg q \to \neg p \equiv p \to q$. Option 3 requires both p and $\neg p$, so it is false even on the target's true rows.

The two true rows of the biconditional are (T, T), (F, F).

L040

Propositional Logic

2015 Midterm | Question 1(b) | Source page 3 | Difficulty 2 | 3 marks
Q-2015-MID-P01-b | Answer provenance: official

> For each of the following parts, circle the correct answer or answers. There is no need to explain how your answers are derived.
>
> (b) Let p and q be two propositions. Which of the following propositions is/are tautology/tautologies?
>
> 1. $p \to (p \to q)$
> 2. $\neg p \to (p \to q)$
> 3. $q \to (p \to q)$
> 4. $\neg q \to (p \to q)$
> 5. None of above.

Official Answer

Source: PDF physical page 4 (2015-midterm.pdf).

(b) 2, 3

Supplementary Explanation

Answer. 2, 3.

Derivation/proof. The four formulas simplify respectively to $\neg p \vee q$, $p \vee \neg p \vee q$, $\neg q \vee \neg p \vee q$, and $q \vee \neg p \vee q$. The middle two contain a complementary pair and are true for all assignments. The first and fourth both reduce to $\neg p \vee q$, false at $p = T, q = F$.

L041

Propositional Logic

2014 Final | Question 5(a) | Source page 9 | Difficulty 2
Q-2014-FINAL-P05-a | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (a) $(p \wedge \neg q) \Rightarrow (r \wedge \neg r)$ and $p \Rightarrow q$

Official Answer

Source: PDF physical page 9 (2014-final.pdf).

(a) Equivalent.

$$\begin{aligned}(p \wedge \neg q) \Rightarrow (r \wedge \neg r) &\equiv \neg(p \wedge \neg q) \vee (r \wedge \neg r) \\ &\equiv \neg p \vee q \vee F \\ &\equiv \neg p \vee q \\ &\equiv p \Rightarrow q\end{aligned}$$

Supplementary Explanation

Answer. Yes.

Derivation/proof. Because $r \wedge \neg r = F$, the first formula is $\neg(p \wedge \neg q) \vee F = \neg p \vee q$, which is exactly $p \Rightarrow q$.

After De Morgan and the contradiction law, the intermediate disjunction is $\neg p \vee q \vee F$.

L042

Propositional Logic

2013 Final | Question 5(c) | Source page 8 | Difficulty 2
Q-2013-FINAL-P05-c | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (c) $(p \Rightarrow r) \wedge (q \Rightarrow r)$ and $(p \wedge q) \Rightarrow r$

Official Answer

Source: PDF physical page 8 (2013-final.pdf).

(c) Not equivalent. Counter example: $p = T, q = F, r = F$. The first statement is false, while the second statement is true.

Supplementary Explanation

Answer. No; take $p = T, q = F, r = F$.

Derivation/proof. At $p = T, q = F, r = F$, the first conjunction is $(T \Rightarrow F) \wedge (F \Rightarrow F) = F$, whereas $(p \wedge q) \Rightarrow r = F \Rightarrow F = T$. The differing truth values disprove equivalence.

Explicitly, the first expression is $F \wedge T = F$, whereas the second is $F \Rightarrow F = T$.

L043

Propositional Logic

2024 Fall Midterm | Question 1(c) | Source page 3 | Difficulty 3 | 6 marks
Q-2024F-MID-P01-c | Answer provenance: official

> (c) Let p, q, r, s, t denote five propositions. You are given the following set of premises:
>
> 1. $s \vee \neg t$
> 2. $p \rightarrow (q \wedge \neg r)$
> 3. $s \rightarrow p$
> 4. $q \rightarrow r$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $(\neg p \wedge \neg t)$ is true. Hint: Start by combining premises 4 and 2.

Official Answer

Source: physical PDF page 4 (COMP2711_F2024_Midterm.pdf).

> (c)
>
> 1. $s \vee \neg t$
> 2. $p \rightarrow (q \wedge \neg r)$
> 3. $s \rightarrow p$
> 4. $q \rightarrow r$
> 5. $\neg q \vee r$ from 4
> 6. $\neg(q \wedge \neg r)$ from 5
> 7. $\neg(q \wedge \neg r) \rightarrow \neg p$ from 2
> 8. $\neg p$ from 7,2
> 9. $\neg p \rightarrow \neg s$ from 3
> 10. $\neg s$ from 9,8
> 11. $\neg t$ from 10,1
> 12. $\neg p \wedge \neg t$ from 11,8

Supplementary Explanation

Answer. $\neg p \wedge \neg t$.

Derivation/proof. 5. Eliminate implication in 4 to obtain $\neg q \vee r$. 6. De Morgan gives $\neg(q \wedge \neg r)$. 7. Contrapose 2 to obtain $\neg(q \wedge \neg r) \rightarrow \neg p$. 8. From 6 and 7, modus ponens yields $\neg p$. 9. Contrapose 3 to obtain $\neg p \rightarrow \neg s$. 10. From 8 and 9 obtain $\neg s$. 11. From 1 and 10, disjunctive syllogism yields $\neg t$. 12. Conjoin 8 and 11 to obtain $\neg p \wedge \neg t$. Every corrected reference supplies a previously established antecedent.

L044

Propositional Logic

2024 Spring Midterm | Question 5 | Source page 7 | Difficulty 3 | 5 marks
Q-2024S-MID-P05 | Answer provenance: official

> Let p, q, r, s, t, u denote five propositions. You are given the following set of premises:
>
> 1. $(\neg p \vee q) \rightarrow r$
> 2. $r \rightarrow (s \vee t)$
> 3. $\neg s \wedge \neg u$
> 4. $\neg u \rightarrow \neg t$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion p is true. You do not need to write the names of the rules you are using; instead, you can just number each statement and write down from which statement(s) a new statement is derived.

Official Answer

Source: physical PDF page 7 (COMP2711_S2024_Midterm.pdf).

> Answer. 5. from 1 $\neg(\neg p \vee q) \vee r$ 6. from 5 $(p \wedge \neg q) \vee r$ 7. from 2 $\neg r \vee s \vee t$ 8. from 3 $\neg s$ 9. from 3 $\neg u$ 10. from 7 and 8 $\neg r \vee t$ 11. from 4 and 9 $\neg t$ 12. from 10 and 11 $\neg r$ 13. from 6 and 12 $p \wedge \neg q$ 14. from 13 p

Supplementary Explanation

Answer. The premises imply p.

Derivation/proof. 5. From 3, conjunction elimination gives $\neg s$. 6. From 3 likewise obtain $\neg u$. 7. From 4 and 6, modus ponens gives $\neg t$. 8. From 5 and 7 obtain $\neg s \wedge \neg t$, equivalently $\neg(s \vee t)$. 9. From 2 and 8, modus tollens gives $\neg r$. 10. From 1 and 9, modus tollens gives $\neg(\neg p \vee q)$. 11. De Morgan's law and double negation give $p \wedge \neg q$. 12. Conjunction elimination gives p. Every step follows from a premise or an earlier step.

L045

Propositional Logic

2023 Fall Midterm | Question 1(b) | Source page 3 | Difficulty 3 | 4 marks
Q-2023F-MID-P01-b | Answer provenance: official

> Let p, q, r, s, t denote five propositions. You are given the following set of premises:
>
> 1. $p \to (q \to r)$
> 2. $p \vee s$
> 3. $t \to q$
> 4. $\neg s$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $\neg r \to \neg t$ is true.

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Midterm.pdf).

> (b)
>
> 1. $\neg s$ — Premise
> 2. $p \vee s$ — Premise
> 3. p — (1,2)
> 4. $p \to (q \to r)$ — Premise
> 5. $q \to r$ — (3,4)
> 6. $t \to q$ — Premise
> 7. $t \to r$ — (5,6)
> 8. $\neg r \to \neg t$ — (7)

Supplementary Explanation

Answer. $\neg r \to \neg t$.

Derivation/proof. 1. $p \to (q \to r)$ (premise). 2. $p \vee s$ (premise). 3. $t \to q$ (premise). 4. $\neg s$ (premise). 5. p (2,4; disjunctive syllogism). 6. $q \to r$ (1,5; modus ponens). 7. $t \to r$ (3,6; hypothetical syllogism). 8. $\neg r \to \neg t$ (7; contraposition).

The inference uses only the four premises; no truth value for q, r, or t is assumed.

In the supplementary numbering, step 5 gives p, step 6 gives $q \to r$, and step 8 gives $\neg r \to \neg t$.

L046

Propositional Logic

2023 Spring Midterm | Question 1(a) | Source page 3 | Difficulty 3
Q-2023S-MID-P01-a-proof | Answer provenance: official

> Let
>
> 1. r: It rains.
> 2. f: It is foggy.
> 3. s: The sailing race will be held.
> 4. l: The life saving demonstration will go on.
> 5. t: The trophy will be awarded.
>
> Hypotheses (the printed premises, retained for the translation and inference atoms):
>
> 1. If it does not rain or if it is not foggy, then the sailing race will be held and the lifesaving demonstration will go on.
> 2. If the sailing race is held, then the trophy will be awarded.
> 3. The trophy was not awarded.
>
> Then use rules of inference to prove the conclusion “it rains”. For each step, you need to write from which earlier statement this step is derived from, but you don't have to write the name of the rule used.

Official Answer

Source: physical PDF page 4 (COMP2711_S2023_Midterm.pdf).

> (a) We are given premises $(\neg r \vee \neg f) \to (s \wedge l)$, $s \to t$, and $\neg t$. We want to conclude r.
>
> | Step | Reason |
> |---|---|
> | 1. $\neg t$ | Premise |
> | 2. $s \to t$ | Premise |
> | 3. $\neg s$ | Modus tollens using (1) and (2) |
> | 4. $(\neg r \vee \neg f) \to (s \wedge l)$ | Premise |
> | 5. $(\neg(s \wedge l)) \to \neg(\neg r \vee \neg f)$ | Contrapositive of (4) |
> | 6. $(\neg s \vee \neg l) \to (r \wedge f)$ | Simplification using (5) |
> | 7. $\neg s \vee \neg l$ | Simplification using (2) |
> | 8. $r \wedge f$ | Modus ponens using (6,7) |
> | 9. r | Simplification from (8) |

Supplementary Explanation

Answer. r.

Derivation/proof. 1. $(\neg r \vee \neg f) \to (s \wedge l)$ (premise). 2. $s \to t$ (premise). 3. $\neg t$ (premise). 4. $\neg s$ (2,3; modus tollens). 5. $\neg s \vee \neg l$ (4; addition). 6. $\neg(s \wedge l)$ (5; De Morgan). 7. $\neg(\neg r \vee \neg f)$ (1,6; modus tollens). 8. $r \wedge f$ (7; De Morgan and double negation). 9. r (8; simplification).

Every step cites existing premises or earlier conclusions; none assumes the desired conclusion.

Addition proves the disjunction without establishing $\neg l$ individually. Step 9 concludes r.

L047

Propositional Logic

2022 Midterm | Question 1(b) | Source page 1 | Difficulty 3 | 3 marks
Q-2022-MID-P01-b | Answer provenance: official

> Given the statement $(a \vee b) \wedge (\neg b \vee c)$, express its equivalent statement using only NOT ($\neg$) and Implication ($\rightarrow$).

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (b)
>
> $$\begin{aligned}(a \vee b) \wedge (\neg b \vee c) &\equiv (\neg a \rightarrow b) \wedge (b \rightarrow c) \\ &\equiv \neg(\neg(\neg a \rightarrow b) \vee \neg(b \rightarrow c)) \\ &\equiv \neg((\neg a \rightarrow b) \rightarrow \neg(b \rightarrow c)).\end{aligned}$$

Supplementary Explanation

Answer. $\neg((\neg a \rightarrow b) \rightarrow \neg(b \rightarrow c))$.

Derivation/proof. Set X=(¬a→b), equivalent to a∨b, and Y=(b→c), equivalent to ¬b∨c. Their conjunction satisfies X∧Y≡¬(¬X∨¬Y)≡¬(X→¬Y). Substitute the two subexpressions to obtain the answer. Expanding its outer negated implication recovers X∧Y and then the original two clauses.

The full intermediate identity is $X \wedge Y \equiv \neg(\neg X \vee \neg Y) \equiv \neg(X \rightarrow \neg Y)$.

L048

Propositional Logic

2020 Midterm | Question 2 | Source page 1 | Difficulty 3 | 10 marks
Q-2020-MID-P02 | Answer provenance: official

> Determine whether the following two propositions are logically equivalent.
>
> (i) $((\neg p \wedge q) \rightarrow (p \vee s)) \vee ((\neg(\neg q \wedge p) \wedge q) \rightarrow (s \vee r))$,
>
> (ii) $q \rightarrow (p \vee s \vee r)$.
>
> If they are, prove it by a series of logical equivalences. If they are not, give a counterexample.

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

$$
\begin{aligned}
&((\neg p \wedge q) \rightarrow (p \vee s)) \vee ((\neg(\neg q \wedge p) \wedge q) \rightarrow (s \vee r)) \\
&\equiv ((\neg p \wedge q) \rightarrow (p \vee s)) \vee (((q \vee \neg p) \wedge q) \rightarrow (s \vee r)) \\
&\equiv ((\neg p \wedge q) \rightarrow (p \vee s)) \vee (q \rightarrow (s \vee r)) \\
&\equiv (\neg(\neg p \wedge q) \vee (p \vee s)) \vee (\neg q \vee (s \vee r)) \\
&\equiv ((p \vee \neg q) \vee (p \vee s)) \vee (\neg q \vee (s \vee r)) \\
&\equiv \neg q \vee p \vee s \vee r \\
&\equiv q \rightarrow (p \vee s \vee r).
\end{aligned}
$$

Supplementary Explanation

Answer. They are logically equivalent.

Derivation/proof. By De Morgan, $\neg(\neg q \wedge p) \wedge q = (q \vee \neg p) \wedge q \equiv q$ by absorption. The first implication is $\neg(\neg p \wedge q) \vee p \vee s \equiv p \vee \neg q \vee s$. The second is $\neg q \vee s \vee r$. Their disjunction is $\neg q \vee p \vee s \vee r$ by associativity, commutativity and idempotence, which is exactly $q \rightarrow (p \vee s \vee r)$.

All transformations are equivalences, so the expressions agree on all sixteen truth assignments.

L049

Propositional Logic

2018 Midterm | Question 1(a) | Source page 3 | Difficulty 3
Q-2018-MID-P01-a | Answer provenance: official

> There are only two kinds of inhabitants on an island: knights, who always tell the truth; and knaves, who always lie. You encounter two of the island's inhabitants, Alice and Bob. Alice tells you, "Bob is a knave." Bob clarifies, "Actually, Alice and I are both knaves."
>
> (a) Let p be the proposition "Alice is a knight", and q be the proposition "Bob is a knight". Write a single compound proposition constructed from p and q which contains all the information given or implied in the question. (Hint: Consider that Alice could be a knight or a knave, and Bob could be a knight or a knave.)

Official Answer

Source: physical PDF page 3 (2018-midterm.pdf).

$((p \wedge \neg q) \vee (\neg p \wedge q)) \wedge ((q \wedge (\neg q \wedge \neg p)) \vee (\neg q \wedge (p \vee q)))$. (XOR can be used to connect Alice's two cases and Bob's two cases. Some other answers are possible. Using implication is generally not correct.)

Supplementary Explanation

Answer. $(p \leftrightarrow \neg q) \wedge (q \leftrightarrow (\neg p \wedge \neg q))$.

Derivation/proof. Alice's statement has truth value $\neg q$. Being a knight means that statement is true, and being a knave means it is false, giving $p \leftrightarrow \neg q$. Bob's whole statement is $\neg p \wedge \neg q$, giving $q \leftrightarrow (\neg p \wedge \neg q)$. Both conversations occurred, so conjoin the constraints. Expanding each biconditional produces the source's two-case expression.

Expanded explicitly, the first constraint is $(p \wedge \neg q) \vee (\neg p \wedge q)$ and the second is $(q \wedge \neg p \wedge \neg q) \vee (\neg q \wedge (p \vee q))$. Parenthesis balancing in the official display above is a typography repair; its two-case method is unchanged.

L050

Propositional Logic

2016 Midterm | Question 4(b) | Source page 7 | Difficulty 3 | 6 marks
Q-2016-MID-P04-b | Answer provenance: official

> (b) Are the following two statements logically equivalent? If yes, prove that these two statements are logically equivalent. Otherwise, give a counter example to show the statements are not logically equivalent.
>
> (i) $(p \Rightarrow q) \Rightarrow r$
>
> (ii) $(p \vee r) \wedge (q \Rightarrow r)$

Official Answer

Source: PDF physical page 7 (2016-midterm.pdf).

(b) Yes. They are logically equivalent.

$$
\begin{gathered}
(p \Rightarrow q) \Rightarrow r \\
(\neg p \vee q) \Rightarrow r \\
\neg(\neg p \vee q) \vee r \\
(p \wedge \neg q) \vee r \\
(p \vee r) \wedge (\neg q \vee r) \\
(p \vee r) \wedge (q \Rightarrow r)
\end{gathered}
$$

Supplementary Explanation

Answer. Yes, they are equivalent.

Derivation/proof. $(p \to q) \to r \equiv \neg(\neg p \vee q) \vee r \equiv (p \wedge \neg q) \vee r \equiv (p \vee r) \wedge (\neg q \vee r) \equiv (p \vee r) \wedge (q \to r)$. The steps use implication elimination twice, De Morgan and double negation, distribution, and implication introduction.

L051

Predicate Logic

2025 Fall Midterm | Question 2(a)(i) | Source page 6 | Difficulty 1 | 2 marks
Q-2025F-MID-P02-a-i | Answer provenance: official

> (a) [4 points, 2 pts each] Consider the set of Natural numbers $\mathbb{N}$ to be the domain. Express the following sentences using only $\forall, \exists, \wedge, \vee, \rightarrow, \leftrightarrow, \neg, =, \neq, <, >, +, \times$.
>
> (i) x is a multiple of 5.

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Midterm.pdf).

> (a) (i) $\exists y$ s.t. $5 \cdot y = x$.

Supplementary Explanation

Answer. $\exists y(5 \times y = x)$.

Derivation/proof. If x is a multiple of five, some natural number y satisfies $x = 5y$, establishing the formula. Conversely, take its witness y; the equality directly says that x is a multiple of five. The quantifier, multiplication, equality, and constant five are permitted, and all variables remain in the natural-number domain.

L052

Predicate Logic

2025 Fall Midterm | Question 2(a)(ii) | Source page 6 | Difficulty 1 | 2 marks
Q-2025F-MID-P02-a-ii | Answer provenance: official

> (a) [4 points, 2 pts each] Consider the set of Natural numbers $\mathbb{N}$ to be the domain. Express the following sentences using only $\forall, \exists, \wedge, \vee, \rightarrow, \leftrightarrow, \neg, =, \neq, <, >, +, \times$.
>
> (ii) x is a perfect square number. (a perfect square number x can be written as $x = y^2$ for an integer y)

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Midterm.pdf).

> (ii) $\exists y$ s.t. $y \cdot y = x$.

Supplementary Explanation

Answer. $\exists y(y \times y = x)$.

Derivation/proof. If the natural number x is the square of an integer z, take $y = |z|$. This root is nonnegative and $y \times y = x$. If the convention starts the natural numbers at one, $x > 0$ also makes the root positive. Conversely, a natural-number witness satisfying this equation is itself an integer root. Thus the formula is equivalent to the definition and uses only the permitted multiplication operation.

L053

Predicate Logic

2025 Fall Midterm | Question 2(c) | Source page 7 | Difficulty 1
Q-2025F-MID-P02-c-premise1 | Answer provenance: official

> (c) [5 points] Consider the following predicates.
>
> $S(x)$ be "x is a satisfactory"
>
> $C(x)$ be "x is a clear explanation"
>
> $E(x)$ be "x is an excuse"
>
> Assume the following premises:
>
> Premise 1: All clear explanations are satisfactory
>
> Premise 2: Some excuses are unsatisfactory
>
> First, write each premise as a logical form using quantifiers and the above predicates
>
> Premise 1: All clear explanations are satisfactory

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Midterm.pdf).

> (c) The premises turn into: $\forall x(C(x) \rightarrow S(x))$

Supplementary Explanation

Answer. $\forall x(C(x) \rightarrow S(x))$.

Derivation/proof. Fix any object x. If C(x) is true, the premise requires S(x), which is exactly $C(x) \rightarrow S(x)$. If C(x) is false, the English sentence imposes no requirement, matching the implication's vacuous truth. Universally quantify x. The premise does not require every satisfactory object to be a clear explanation.

L054

Predicate Logic

2024 Fall Midterm | Question 2(a) | Source page 5 | Difficulty 1 | 3 marks
Q-2024F-MID-P02-a | Answer provenance: official

> Consider the following predicates: $Prime(x)$: x is prime $Even(x)$: x is even The domain is all natural numbers $(0, 1, 2, \ldots)$.
>
> $$\forall x((Even(x) \land x > 2) \rightarrow \exists y \exists z(Prime(y) \land Prime(z) \land (y + z = x)))$$
>
> (a) Express the following statement in English:
>
> $$\forall x((Even(x) \land x > 2) \rightarrow \exists y \exists z(Prime(y) \land Prime(z) \land (y + z = x)))$$

Official Answer

Source: physical PDF page 6 (COMP2711_F2024_Midterm.pdf).

> Answer: (a) “Every even natural number greater than 2 is the sum of two prime numbers.” In reality this is a famous open problem called Goldbach's conjecture! We still have no proof for it and we don't know if its really correct =)

Supplementary Explanation

Answer. Every even natural number greater than 2 is the sum of two prime numbers.

Derivation/proof. $\forall x$ ranges over all natural numbers, and $Even(x) \land x > 2$ restricts which ones require the consequent. The quantifiers $\exists y \exists z$ choose two natural numbers. $Prime(y) \land Prime(z)$ makes both prime, and $y + z = x$ makes their sum the chosen even number. Thus: Every even natural number greater than 2 is the sum of two prime numbers. There is no $y \neq z$ condition, so adding “distinct” would change the claim.

L055

Predicate Logic

2024 Spring Midterm | Question 3(a) | Source page 5 | Difficulty 1 | 1 mark
Q-2024S-MID-P03-a | Answer provenance: official

> Let $P(x, y)$ be a propositional function. For each part below, state whether or not it is a tautology. No explanation is needed.
>
> (a) $\exists x \forall y P(x, y) \rightarrow \forall y \exists x P(x, y)$.

Official Answer

Source: physical PDF page 5 (COMP2711_S2024_Midterm.pdf).

> Answer. (a) Tautology

Supplementary Explanation

Answer. Tautology.

Derivation/proof. If the antecedent holds, there is a fixed x_0 such that $P(x_0, y)$ holds for every y. For an arbitrary y, the same x_0 witnesses $\exists x P(x, y)$, proving the consequent. When the antecedent is false, the implication is true automatically; an empty domain also makes the antecedent false. The implication therefore holds in every interpretation.

L056

Predicate Logic

2022 Final Part 2 | Question 5(a)(ii) | Source page 3 | Difficulty 1
Q-2022-FINAL2-P05-a-ii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> For every vertex in V_1, there are edges that connect it with all vertices in V_2

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (ii) $\forall u \in V_1 \forall v \in V_2 (E(u, v))$.

Supplementary Explanation

Answer. $\forall u \in V_1 \forall v \in V_2 E(u, v)$.

Derivation/proof. For each u in the first part, assert E(u,v) for every v in the second. This is exactly two restricted universal quantifiers. Undirectedness supplies the reverse orientation automatically. Since neither quantifier ranges over two vertices in the same part, the formula says nothing about internal edges.

Internal edges have not been forbidden.

L057

Predicate Logic

2022 Midterm | Question 2(a) | Source page 1 | Difficulty 1
Q-2022-MID-P02-a | Answer provenance: official

> Let $W(x, y)$ denote the predicate that x has visited website y, where the domain for x consists of people in the world and the domain for y consists of all websites. Express each of the following statements using predicate logic.
>
> There is a website that both Alice and Bob have visited.

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (a) $\exists y(W(\text{Alice}, y) \wedge W(\text{Bob}, y))$

Supplementary Explanation

Answer. $\exists y(W(\text{Alice}, y) \wedge W(\text{Bob}, y))$.

Derivation/proof. The existential variable y ranges over websites. The two W predicates share that same y, requiring one common site visited by Alice and Bob. Alice and Bob are named people rather than quantified websites. Conjunction expresses both visits; separate website witnesses would not require their visited sets to overlap.

The two predicates use the same y.

L058

Predicate Logic

2022 Midterm | Question 3(1) | Source page 1 | Difficulty 1
Q-2022-MID-P03-1 | Answer provenance: official

> Let $P(x), Q(x), R(x), S(x)$ be the following predicates:
>
> $P(x)$: "x is a hummingbird." $Q(x)$: "x is large." $R(x)$: "x lives on honey." $S(x)$: "x is richly colored."
>
> Let the domain be all the birds. First translate the following statements into predicate logic (you may assume that "small" is the same as "not large" and that "dull in color" is the same as "not richly colored"):
>
> (1) "All hummingbirds are richly colored." (2) "No large birds live on honey." (3) "Birds that do not live on honey are dull in color." (4) "Hummingbirds are small."
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> "All hummingbirds are richly colored."

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (1) $\forall x(P(x) \to S(x))$.

Supplementary Explanation

Answer. $\forall x(P(x) \to S(x))$.

Derivation/proof. The domain already includes only birds. For any such x, the sentence makes being a hummingbird sufficient for rich color. Thus P is the antecedent and S the consequent under ∀x. It places no color requirement on birds that are not hummingbirds and does not assert every richly colored bird is a hummingbird.

For a non-hummingbird the antecedent is false.

L059

Predicate Logic

2022 Midterm | Question 3(2) | Source page 2 | Difficulty 1
Q-2022-MID-P03-2 | Answer provenance: official

> Let $P(x), Q(x), R(x), S(x)$ be the following predicates:
>
> $P(x)$: "x is a hummingbird." $Q(x)$: "x is large." $R(x)$: "x lives on honey." $S(x)$: "x is richly colored."
>
> Let the domain be all the birds. First translate the following statements into predicate logic (you may assume that "small" is the same as "not large" and that "dull in color" is the same as "not richly colored"):
>
> (1) "All hummingbirds are richly colored." (2) "No large birds live on honey." (3) "Birds that do not live on honey are dull in color." (4) "Hummingbirds are small."
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> "No large birds live on honey."

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (2) $\neg\exists x(Q(x) \wedge R(x))$.

Supplementary Explanation

Answer. $\neg\exists x(Q(x) \wedge R(x))$.

Derivation/proof. A counterexample to "no large birds live on honey" would be a bird satisfying both Q and R. Denying the existence of such a bird gives the displayed formula. Equivalently ∀x(Q(x)→¬R(x)), by quantifier negation and De Morgan. It does not prohibit small honey-fed birds.

Equivalently, $\forall x(\neg Q(x) \vee \neg R(x))$, or $\forall x(Q(x) \rightarrow \neg R(x))$.

L060

Predicate Logic

2022 Midterm | Question 3(4) | Source page 2 | Difficulty 1
Q-2022-MID-P03-4 | Answer provenance: official

> Let $P(x), Q(x), R(x), S(x)$ be the following predicates:
>
> $P(x)$: "x is a hummingbird." $Q(x)$: "x is large." $R(x)$: "x lives on honey." $S(x)$: "x is richly colored."
>
> Let the domain be all the birds. First translate the following statements into predicate logic (you may assume that "small" is the same as "not large" and that "dull in color" is the same as "not richly colored"):
>
> (1) "All hummingbirds are richly colored." (2) "No large birds live on honey." (3) "Birds that do not live on honey are dull in color." (4) "Hummingbirds are small."
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> "Hummingbirds are small."

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (4) $\forall x(P(x) \to \neg Q(x))$.

Supplementary Explanation

Answer. $\forall x(P(x) \to \neg Q(x))$.

Derivation/proof. A bird being a hummingbird is P(x), and the source's explicit convention makes small equal to ¬Q(x). The statement applies to every bird satisfying P, so universally quantify an implication. No existence of hummingbirds is asserted; if there were none, the universal implication would still be true.

The formula does not additionally assert that hummingbirds exist.

L061

Predicate Logic

2021 Midterm | Question 1(a) | Source page 1 | Difficulty 1 | 3 marks
Q-2021-MID-P01-a | Answer provenance: official

> Determine the truth value of each of the following statements. The domain for all variables are all integers. Justify your answer.
>
> $\exists x \forall y (y = x^2 + 2x + 1)$

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> (a) False. For every x, there is always an integer $y = x^2 + 2x + 2 > x^2 + 2x + 1$.

Supplementary Explanation

Answer. False.

Derivation/proof. For each integer x, choose the integer $y = x^2 + 2x + 2$. It differs from $x^2 + 2x + 1$ by one, so the inner universal statement fails. Therefore no existential x works. The order requires a single x to serve every integer y, unlike choosing x after seeing y.

The difference is $1 \neq 0$. The refuting y may depend on x, exactly as allowed by the negation $\forall x \exists y$.

L062

Predicate Logic

2021 Midterm | Question 2 | Source page 1 | Difficulty 1
Q-2021-MID-P02-translation-course | Answer provenance: official

> Consider the following predicates.
>
> $c(x)$: "x studied COMP2711". $g(x)$: "x knows how to compute the gcd of two integers". $h(x)$: "x can get a high paying job".
>
> "Sarah, a student who studied COMP2711, knows how to compute the gcd of two integers. Everyone who knows how to compute the gcd of two integers can get a high-paying job. Therefore, at least one student who studied COMP2711 can get a high-paying job."
>
> Please first translate the premises and conclusion into predicate logic sentences. Then show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> Sarah, a student who studied COMP2711

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> We are given premises $c(\text{Sarah})$, $g(\text{Sarah})$, and $\forall x(g(x) \to h(x))$, and we want to conclude $\exists x(c(x) \land h(x))$.

Supplementary Explanation

Answer. $c(\text{Sarah})$.

Derivation/proof. The source identifies Sarah as a student who studied COMP2711. The defined predicate c expresses exactly that course fact, with Sarah as its named argument. This premise is distinct from her ability to compute gcds, so both must be retained when translating the full sentence.

This is an independent course-attendance fact.

L063

Predicate Logic

2021 Midterm | Question 2 | Source page 1 | Difficulty 1
Q-2021-MID-P02-translation-gcd | Answer provenance: official

> Consider the following predicates.
>
> $c(x)$: "x studied COMP2711". $g(x)$: "x knows how to compute the gcd of two integers". $h(x)$: "x can get a high paying job".
>
> "Sarah, a student who studied COMP2711, knows how to compute the gcd of two integers. Everyone who knows how to compute the gcd of two integers can get a high-paying job. Therefore, at least one student who studied COMP2711 can get a high-paying job."
>
> Please first translate the premises and conclusion into predicate logic sentences. Then show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> knows how to compute the gcd of two integers.

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> We are given premises $c(\text{Sarah})$, $g(\text{Sarah})$, and $\forall x(g(x) \to h(x))$, and we want to conclude $\exists x(c(x) \wedge h(x))$.

Supplementary Explanation

Answer. $g(\text{Sarah})$.

Derivation/proof. The grammatical subject remains Sarah. Predicate g already means knowing how to compute the gcd of two integers, so apply g to that named person. The integers mentioned are internal to the given predicate meaning; the translation does not need new number quantifiers or assert that every student has the skill.

In the second clause, the subject remains Sarah.

L064

Predicate Logic

2021 Midterm | Question 2 | Source page 1 | Difficulty 1
Q-2021-MID-P02-translation-rule | Answer provenance: official

> Consider the following predicates.
>
> $c(x)$: "x studied COMP2711". $g(x)$: "x knows how to compute the gcd of two integers". $h(x)$: "x can get a high paying job".
>
> "Sarah, a student who studied COMP2711, knows how to compute the gcd of two integers. Everyone who knows how to compute the gcd of two integers can get a high-paying job. Therefore, at least one student who studied COMP2711 can get a high-paying job."
>
> Please first translate the premises and conclusion into predicate logic sentences. Then show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> Everyone who knows how to compute the gcd of two integers can get a high-paying job.

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> We are given premises $c(\text{Sarah})$, $g(\text{Sarah})$, and $\forall x(g(x) \to h(x))$, and we want to conclude $\exists x(c(x) \wedge h(x))$.

Supplementary Explanation

Answer. $\forall x(g(x) \to h(x))$.

Derivation/proof. The word everyone calls for a universal quantifier. Having gcd-computation skill is the qualifying condition g(x); being able to obtain a high-paying job is h(x). An implication makes the latter necessary whenever the former holds, while leaving people without the skill unconstrained.

Job eligibility is a necessary condition for the skill under this implication; the skill is not asserted to be a necessary condition for job eligibility.

L065

Predicate Logic

2020 Final Part 1 | Question 1(a)(iii) | Source page 1 | Difficulty 1
Q-2020-FINAL1-P01-a-iii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "For every vertex in V_1, there are edges that connect it with all vertices in V_2".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$\forall u \in V_1 \forall v \in V_2 (E(u, v))$.

Supplementary Explanation

Answer. $\forall u \in V_1 \forall v \in V_2 E(u, v)$.

Derivation/proof. The phrase "every vertex" in the first part supplies the first universal quantifier; "all vertices" in the other part supplies the second. For each resulting pair assert adjacency. This places no restriction on edges whose two endpoints are both in V1 or both in V2.

The formula $\forall u \in V_1 \forall v \in V_2 E(u, v)$ requires all cross-part edges. By itself it does not prohibit edges within a part; that is a separate restriction in the preceding property.

L066

Predicate Logic

2020 Midterm | Question 1(a) | Source page 1 | Difficulty 1
Q-2020-MID-P01-a | Answer provenance: official

> Let $Z(x), D(x), F(x)$ and $C(x)$ be the following predicates:
>
> $Z(x)$: "x attended the Zoom meeting of midterm dry run". $D(x)$: "x gets some marks deducted for the midterm exam". $F(x)$: "x submitted a file for the midterm dry run". $C(x)$: "x is cheating".
>
> Express the following statements using quantifiers, logical connectives, and the predicates above, where the domain consists of all students in COMP2711.
>
> Any student absent from the Zoom meeting of midterm dry run gets some marks deducted for the midterm exam.

Official Answer

Source: physical PDF page 1 (2020-midterm.pdf).

$\forall x(\neg Z(x) \rightarrow D(x))$.

Supplementary Explanation

Answer. $\forall x(\neg Z(x) \rightarrow D(x))$.

Derivation/proof. "Any student" ranges over every member of the given student domain. Being absent is $\neg Z(x)$; the statement requires $D(x)$ whenever that condition holds. Thus use implication inside the universal quantifier. A student who attended is unconstrained by this sentence.

The universal conditional constrains only absent students. For a student who attended, its antecedent is false and the formula is true regardless of the deduction predicate.

L067

Predicate Logic

2020 Midterm | Question 1(c) | Source page 1 | Difficulty 1
Q-2020-MID-P01-c | Answer provenance: official

> Let $Z(x), D(x), F(x)$ and $C(x)$ be the following predicates:
>
> $Z(x)$: "x attended the Zoom meeting of midterm dry run". $D(x)$: "x gets some marks deducted for the midterm exam". $F(x)$: "x submitted a file for the midterm dry run". $C(x)$: "x is cheating".
>
> Express the following statements using quantifiers, logical connectives, and the predicates above, where the domain consists of all students in COMP2711.
>
> Some students were absent from the Zoom meeting of midterm dry run but did not get marks deducted for the midterm exam.

Official Answer

Source: physical PDF page 1 (2020-midterm.pdf).

$\exists x(\neg Z(x) \wedge \neg D(x))$.

Supplementary Explanation

Answer. $\exists x(\neg Z(x) \wedge \neg D(x))$.

Derivation/proof. The existential quantifier supplies at least one student. That same witness is absent and has no deduction, so both predicates are negated and conjoined. An existential implication would be too weak: a student who attended could satisfy it vacuously without demonstrating the requested exception.

L068

Predicate Logic

2020 Midterm | Question 3(a) | Source page 2 | Difficulty 1
Q-2020-MID-P03-a | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\forall x(|x| \cdot x \geq x)$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

False. When $x = -2$, we have $2 \cdot (-2) = -4 < -2$.

Supplementary Explanation

Answer. False.

Derivation/proof. Take the real value $x = -2$. Then $|x|x = 2(-2) = -4$, which is less than -2. Hence this one member of the quantified domain violates the inequality and refutes the universal statement. Absolute value does not make the product nonnegative because the second factor remains negative.

The counterexample calculation is $2 \cdot (-2) = -4 < -2 = x$. Although the absolute-value factor is positive, its multiplication by the negative second factor refutes the proposed inequality.

L069

Predicate Logic

2020 Midterm | Question 3(c) | Source page 2 | Difficulty 1
Q-2020-MID-P03-c | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\forall x \exists y (x = 2y + 3)$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

True. There is always a $y = (x - 3)/2$.

Supplementary Explanation

Answer. True.

Derivation/proof. For arbitrary real x choose $y = (x - 3)/2$, also real. Substitution gives $2y + 3 = x - 3 + 3 = x$. Since this construction works for every x, the universal-existential statement holds. The witness is allowed to vary with x because its existential quantifier occurs inside the universal one.

The formula $\forall x \exists y$ allows a different witness for each x. Choosing $y = (x - 3)/2$ gives $2((x - 3)/2) + 3 = x$ for every real x.

L070

Predicate Logic

2020 Midterm | Question 3(d) | Source page 2 | Difficulty 1
Q-2020-MID-P03-d | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\exists y \forall x (x = 2y + 3)$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

False. For every y, there exists an $x \neq 2y + 3$.

Supplementary Explanation

Answer. False.

Derivation/proof. For any proposed real y, choose $x = 2y + 4$. This is a real number but differs from $2y + 3$ by one. Thus no fixed y can satisfy the inner universal condition. Equivalently x=0 and x=1 would force the same expression $2y + 3$ to have two distinct values.

For every candidate y, choose $x = 2y + 4$. Then $x \neq 2y + 3$, so each candidate fails the universal requirement; a single fixed y cannot serve all x.

L071

Predicate Logic

2019 Final | Question 1(b) | Source page 3 | Difficulty 1 | 2 marks
Q-2019-FINAL-P01-b | Answer provenance: official

> For each of the following statements, determine whether it is true or false, and justify your answer. The domain is all integers.
>
> (b) $\forall x \forall y((x-y)^2 > 0)$

Official Answer

Source: physical PDF page 3 (2019-final.pdf).

False. Counter example: $x = y = 0$.

Supplementary Explanation

Answer. False; $x = y = 0$ is a counterexample.

Derivation/proof. The two universal quantifiers allow equal inputs. At $x = y = 0$, $(x-y)^2 = 0^2 = 0$, and $0 > 0$ is false. Although every real square is nonnegative, strict positivity fails whenever its base is zero, so the universally strict claim is false.

L072

Predicate Logic

2019 Final | Question 1(c) | Source page 3 | Difficulty 1 | 2 marks
Q-2019-FINAL-P01-c | Answer provenance: official

> For each of the following statements, determine whether it is true or false, and justify your answer. The domain is all integers.
>
> (c) $\forall x \exists y(2x - y \leq 1)$

Official Answer

Source: physical PDF page 3 (2019-final.pdf).

True. For any x, setting $y = 2x$ makes the proposition true.

Supplementary Explanation

Answer. True.

Derivation/proof. Let x be an arbitrary integer and choose $y = 2x$, which is also an integer. Then $2x - y = 0 \leq 1$. This constructs a valid witness for every allowed x, proving the quantified statement. The witness is selected after x, as required by the quantifier order.

For the order $\forall x \exists y$, the witness may depend on x. Taking $y = 2x$ yields $2x - 2x = 0 \leq 1$, which validates that quantifier order directly.

L073

Predicate Logic

2019 Midterm | Question 2(a) | Source page 4 | Difficulty 1 | 2 marks
Q-2019-MID-P02-a | Answer provenance: official

> Let $P(x)$ be the statement "x is even" and $Q(x)$ be the statement "x is an integer". Express each of the following statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain is all real numbers.
>
> (a) Some real numbers are not positive.

Official Answer

Source: physical PDF page 4 (2019-midterm.pdf).

$\exists x(x \leq 0)$.

Supplementary Explanation

Answer. $\exists x(x \leq 0)$.

Derivation/proof. "Some" introduces an existential quantifier. Since the domain is already all real numbers, no separate real-number predicate is needed. The negation of $x > 0$ is $x \leq 0$, so the witness is required to satisfy that inequality; for example zero does.

The allowed choice $x = 0$ must be considered when deciding the quantified claim; a universal statement over the reals cannot silently exclude zero.

L074

Predicate Logic

2019 Midterm | Question 2(b) | Source page 4 | Difficulty 1 | 2 marks
Q-2019-MID-P02-b | Answer provenance: official

> Let $P(x)$ be the statement "x is even" and $Q(x)$ be the statement "x is an integer". Express each of the following statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain is all real numbers.
>
> (b) Every integer is even.

Official Answer

Source: physical PDF page 4 (2019-midterm.pdf).

$\forall x(Q(x) \rightarrow P(x))$.

Supplementary Explanation

Answer. $\forall x(Q(x) \rightarrow P(x))$.

Derivation/proof. For every real x, only the case in which $Q(x)$ holds is constrained. In that case the sentence requires $P(x)$. Thus the correct translation is a universally quantified implication. Translating a false English statement does not license changing it into a true one.

A formula can be a proposition although the statement is false: a proposition needs a definite truth value, not necessarily truth.

L075

Predicate Logic

2017 Midterm | Question 2(1) | Source page 4 | Difficulty 1
Q-2017-MID-P02-1 | Answer provenance: official

> Consider the following predicates:
>
> $C(x)$: x is a child. $K(x)$: x is kind. $N(x, y)$: x and y are neighbours. $L(x)$: x learned.
>
> Write the following sentences using the above predicates, adding quantifiers when necessary:
>
> 1. All children are kind.

Official Answer

Source: physical PDF page 4 (2017-midterm.pdf).

$\forall x(C(x) \to K(x))$.

Supplementary Explanation

Answer. $\forall x(C(x) \to K(x))$.

Derivation/proof. For each person x, being a child is a sufficient condition for being kind. Quantifying this implication over all people captures "all children." For a nonchild, the implication is automatically true, which is correct because the sentence says nothing about that person.

L076

Predicate Logic

2017 Midterm | Question 2(3) | Source page 4 | Difficulty 1
Q-2017-MID-P02-3 | Answer provenance: official

> Consider the following predicates:
>
> $C(x)$: x is a child. $K(x)$: x is kind. $N(x, y)$: x and y are neighbours. $L(x)$: x learned.
>
> Write the following sentences using the above predicates, adding quantifiers when necessary:
>
> 3. People who are not learned are unkind.

Official Answer

Source: physical PDF page 4 (2017-midterm.pdf).

$\forall x(\neg L(x) \to \neg K(x))$.

Supplementary Explanation

Answer. $\forall x(\neg L(x) \to \neg K(x))$.

Derivation/proof. The antecedent is "not learned," namely $\neg L(x)$; the consequent is "unkind," namely $\neg K(x)$. The statement applies to every person. Its contrapositive $\forall x(K(x) \to L(x))$ is equivalent, but the converse need not hold.

L077

Predicate Logic

2013 Final | Question 6(a)(i) | Source page 9 | Difficulty 1
Q-2013-FINAL-P06-a-i | Answer provenance: official

> Consider the following three statements:
>
> - (i) All rich people have famous-brand products and don't have part-time jobs.
> - (ii) Some students with part-time jobs have famous-brand products.
> - (iii) All students with part-time jobs are not rich.
>
> Let U be the universe of all people and define the following predicates:
>
> - $R(x)$: x is a rich person
> - $F(x)$: x has famous-brand products
> - $S(x)$: x is a student
> - $J(x)$: x has part-time jobs.
>
> (a) Write down the logic statements for the three sentences above.
>
> (i) All rich people have famous-brand products and don't have part-time jobs.

Official Answer

Source: PDF physical page 9 (2013-final.pdf).

(a) (i) $\forall x \in U(R(x) \Rightarrow (F(x) \wedge \neg J(x)))$

Supplementary Explanation

Answer. $\forall x \in U\,[R(x) \Rightarrow (F(x) \wedge \neg J(x))]$.

Derivation/proof. The phrase ‘all rich people’ sets a universal implication with antecedent $R(x)$; the two claimed properties are conjoined in the consequent.

People who are not rich are not constrained by this statement.

L078

Predicate Logic

2013 Final | Question 6(a)(ii) | Source page 9 | Difficulty 1
Q-2013-FINAL-P06-a-ii | Answer provenance: official

> Consider the following three statements:
>
> - (i) All rich people have famous-brand products and don't have part-time jobs.
> - (ii) Some students with part-time jobs have famous-brand products.
> - (iii) All students with part-time jobs are not rich.
>
> Let U be the universe of all people and define the following predicates:
>
> - $R(x)$: x is a rich person
> - $F(x)$: x has famous-brand products
> - $S(x)$: x is a student
> - $J(x)$: x has part-time jobs.
>
> (a) Write down the logic statements for the three sentences above.
>
> (ii) Some students with part-time jobs have famous-brand products.

Official Answer

Source: PDF physical page 9 (2013-final.pdf).

(a) (ii) $\exists x \in U(S(x) \wedge J(x) \wedge F(x))$

Supplementary Explanation

Answer. $\exists x \in U\,[S(x) \wedge J(x) \wedge F(x)]$.

Derivation/proof. 'Some' requires an existential witness, and that same witness must be a student, have a part-time job, and have the products.

L079

Predicate Logic

2013 Final | Question 6(a)(iii) | Source page 9 | Difficulty 1
Q-2013-FINAL-P06-a-iii | Answer provenance: official

Consider the following three statements:

- (i) All rich people have famous-brand products and don't have part-time jobs.
- (ii) Some students with part-time jobs have famous-brand products.
- (iii) All students with part-time jobs are not rich.

Let U be the universe of all people and define the following predicates:

- $R(x)$: x is a rich person
- $F(x)$: x has famous-brand products
- $S(x)$: x is a student
- $J(x)$: x has part-time jobs.

(a) Write down the logic statements for the three sentences above.

(iii) All students with part-time jobs are not rich.

Official Answer

Source: PDF physical page 9 (2013-final.pdf).

(a) (iii) $\forall x \in U(S(x) \wedge J(x) \Rightarrow \neg R(x))$

Supplementary Explanation

Answer. $\forall x \in U\,[(S(x) \wedge J(x)) \Rightarrow \neg R(x)]$.

Derivation/proof. The restricted class is people who are both students and have part-time jobs; the asserted property is not-rich.

L080

Predicate Logic

2025 Fall Final | Question 1(c)(i) | Source page 3 | Difficulty 2 | 2 marks
Q-2025F-FINAL-P01-c-i | Answer provenance: official

> (c) [4 points, 2 pts each] Lets the domain, be the students of COMP2711. Consider the following predicates:
>
> $s(x)$: Student x is a senior. $m(x)$: Student x is a mathematics major. $c(x)$: Student x is a comp major $Compare(x, y)$: Student x gets a higher grade than student y
>
> Write each of the following statement in terms of quantifiers, logical operators and the predicates above.
>
> (i) Every senior in the class is majoring either in comp or mathematics.

Official Answer

Source: physical PDF page 4 (COMP2711_F2025_Final.pdf).

(c) (i) $\forall x\big(s(x) \rightarrow (c(x) \vee m(x))\big)$.

Supplementary Explanation

Answer. $\forall x(s(x) \rightarrow (c(x) \vee m(x)))$.

Derivation/proof. Take any student x in the domain. If $s(x)$ holds, the student must have at least one of the two majors, expressed by $c(x) \vee m(x)$. Therefore use $s(x) \rightarrow (c(x) \vee m(x))$ and universally quantify over all students. The statement does not prohibit a double major, and the official formula uses inclusive disjunction: a senior who satisfies both $c(x)$ and $m(x)$ still qualifies. Nonseniors are not constrained by this sentence.

L081

Predicate Logic

2025 Fall Midterm | Question 2(c) | Source page 7 | Difficulty 2
Q-2025F-MID-P02-c-premise2 | Answer provenance: official

> (c) [5 points] Consider the following predicates.
>
> $S(x)$ be "x is a satisfactory"
>
> $C(x)$ be "x is a clear explanation"
>
> $E(x)$ be "x is an excuse"
>
> Assume the following premises:
>
> Premise 1: All clear explanations are satisfactory
>
> Premise 2: Some excuses are unsatisfactory
>
> First, write each premise as a logical form using quantifiers and the above predicates
>
> Premise 2: Some excuses are unsatisfactory

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Midterm.pdf).

> $\exists x\, E(x) \land \neg S(x)$.

Supplementary Explanation

Answer. $\exists x(E(x) \land \neg S(x))$.

Derivation/proof. E(x) says the object is an excuse, and $\neg S(x)$ says it is unsatisfactory. Both describe the same object, so first form $E(x) \land \neg S(x)$ and place the entire conjunction in the scope of one existential quantifier. Every occurrence is bound. If only E(x) were within the quantifier's scope, the external occurrence in S(x) would be free and would not express a complete premise.

L082

Predicate Logic

2025 Spring Final | Question 1(c) | Source page 2 | Difficulty 2 | 2 marks
Q-2025S-FINAL-P01-c | Answer provenance: official

> (c) [2 pts] Assume A and B are two sets with elements in $\mathbb{N}$. Express the following predicate using only $\forall, \exists, \wedge, \vee, \rightarrow, \leftrightarrow, \neg, \in, \notin$, parentheses, variables and constants. Take the natural numbers $\mathbb{N}$ as the domain. You are not allowed to use any other predicates.
>
> Set A and set B are disjoint

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(c) Possible Solutions:

- $\forall x(x \in A \rightarrow x \notin B) \wedge \forall x(x \in B \rightarrow x \notin A)$
- $\forall x(x \notin A \vee x \notin B)$
- $\neg \exists x(x \in A \wedge x \in B)$

Supplementary Explanation

Answer. $\forall x(x \notin A \vee x \notin B)$.

Derivation/proof. Belonging to both sets is $x \in A \wedge x \in B$. Excluding any such witness gives $\neg \exists x(x \in A \wedge x \in B)$. Quantifier negation and De Morgan's law give $\forall x(\neg(x \in A) \vee \neg(x \in B))$, or $\forall x(x \notin A \vee x \notin B)$. The natural-number domain contains all elements of both sets, making this necessary and sufficient and using only permitted symbols.

L083

Predicate Logic

2025 Spring Final | Question 1(d) | Source page 2 | Difficulty 2 | 2 marks
Q-2025S-FINAL-P01-d | Answer provenance: official

> (d) [2 pts] Express the negation of the statement in (c) so that all negation symbols, if any, must immediately precede predicates.

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Final.pdf).

(d) $\exists x(x \in A \wedge x \in B)$

Supplementary Explanation

Answer. $\exists x(x \in A \wedge x \in B)$.

Derivation/proof. Negate the statement $\forall x(x \notin A \vee x \notin B)$ from part (c). Quantifier negation gives $\exists x \neg(x \notin A \vee x \notin B)$. De Morgan's law gives $\exists x(\neg(x \notin A) \wedge \neg(x \notin B))$. Removing double negations yields $\exists x(x \in A \wedge x \in B)$. The same witness x belongs to both sets, and no negation remains before a compound expression.

L084

Predicate Logic

2024 Fall Final | Question 1(b) | Source page 4 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P01-b | Answer provenance: official

> (b) Assume the domain to be all integers, $Prime(x)$ is a predicates that means x is prime. Express the following statement in terms of quantifiers, variable(s), predicates, and the inequality symbols $<$ or $>$. For every integer n bigger than 1, there is a prime strictly between n and $2n$. (Strictly between n and $2n$ means it can not be equal to n an it can not be equal to $2n$)

Official Answer

Source: physical PDF page 4 (COMP2711_F2024_Final.pdf).

> (b) $\forall n((n > 1) \rightarrow \exists p(Prime(p) \wedge (n < p < 2n)))$

Supplementary Explanation

Answer. $\forall n((n > 1) \rightarrow \exists p(Prime(p) \wedge n < p \wedge p < 2n))$.

Derivation/proof. Use $\forall n$ over the integer domain. Since the requirement applies only when $n > 1$, put that condition in the implication antecedent. The consequent $\exists p$ allows the prime to depend on n. Require $Prime(p)$, $n < p$ and $p < 2n$ simultaneously, excluding both endpoints. For $n \leq 1$ the implication makes no additional demand. This translates the supplied sentence without needing to prove its number-theoretic truth.

L085

Predicate Logic

2024 Fall Final | Question 1(c) | Source page 4 | Difficulty 2 | 2 marks
Q-2024F-FINAL-P01-c | Answer provenance: official

> Assume the domain to be all integers, $Prime(x)$ is a predicates that means x is prime. For every integer n bigger than 1, there is a prime strictly between n and $2n$. (Strictly between n and $2n$ means it can not be equal to n an it can not be equal to $2n$)
>
> (c) Negate the statement in (b) so that all negation symbols immediately precede predicates.

Official Answer

Source: physical PDF page 4 (COMP2711_F2024_Final.pdf).

> (c) $\exists n((n > 1) \wedge \forall p(\neg Prime(p) \vee (p \leq n \vee p \geq 2n)))$

Supplementary Explanation

Answer. $\exists n((n > 1) \wedge \forall p(\neg Prime(p) \vee p \leq n \vee p \geq 2n))$.

Derivation/proof. Negating the original universal implication first gives $\exists n((n > 1) \wedge \neg\exists p(Prime(p) \wedge n < p \wedge p < 2n))$. Negate the existential quantifier to obtain a universal one, then apply De Morgan to get $\neg Prime(p) \vee \neg(n < p) \vee \neg(p < 2n)$. The integer linear order makes the last two conditions $p \leq n$ and $p \geq 2n$. This gives the answer, with the sole negation symbol immediately preceding the prime predicate.

L086

Predicate Logic

2024 Fall Midterm | Question 2(b) | Source page 5 | Difficulty 2 | 4 marks
Q-2024F-MID-P02-b | Answer provenance: official

> Consider the following predicates: $Prime(x)$: x is prime $Even(x)$: x is even The domain is all natural numbers $(0, 1, 2, \ldots)$.
>
> $$\forall x((Even(x) \wedge x > 2) \rightarrow \exists y \exists z(Prime(y) \wedge Prime(z) \wedge (y + z = x)))$$
>
> (b) Negate the statement in (a) so that all negation symbols immediately precede predicates.

Official Answer

Source: physical PDF page 6 (COMP2711_F2024_Midterm.pdf).

> (b) The statement in part (a) can be also written like this:
>
> $$\forall x(\neg(Even(x) \wedge x > 2) \vee \exists y \exists z(Prime(y) \wedge Prime(z) \wedge y + z = x)).$$
>
> Here is the negation of this statement:
>
> $$\exists x((Even(x) \wedge x > 2) \wedge \forall y \forall z \neg(Prime(y) \wedge Prime(z) \wedge y + z = x)).$$
>
> Which also can be written as:
>
> $$\exists x((Even(x) \wedge x > 2) \wedge \forall y \forall z(\neg Prime(y) \vee \neg Prime(z) \vee y + z \neq x)).$$

Supplementary Explanation

Answer. $\exists x((Even(x) \wedge x > 2) \wedge \forall y \forall z(\neg Prime(y) \vee \neg Prime(z) \vee y + z \neq x))$.

Derivation/proof. Write the original as $\forall x(A(x) \rightarrow \exists y \exists z B(x, y, z))$. Its negation is $\exists x(A(x) \wedge \forall y \forall z \neg B(x, y, z))$. Substitute $A = Even(x) \wedge x > 2$. Applying De Morgan to $B = Prime(y) \wedge Prime(z) \wedge (y + z = x)$ gives $\neg Prime(y) \vee \neg Prime(z) \vee (y + z \neq x)$. This yields the displayed answer: negation signs immediately precede the $Prime$ predicates, and $\neq$ expresses the negated equality.

L087

Predicate Logic

2024 Spring Midterm | Question 3(b) | Source page 5 | Difficulty 2 | 1 mark
Q-2024S-MID-P03-b | Answer provenance: official

> Let $P(x, y)$ be a propositional function. For each part below, state whether or not it is a tautology. No explanation is needed.
>
> (b) $\forall y \exists x P(x, y) \rightarrow \exists x \forall y P(x, y)$.

Official Answer

Source: physical PDF page 5 (COMP2711_S2024_Midterm.pdf).

> Answer. (b) Not a tautology

Supplementary Explanation

Answer. Not a tautology.

Derivation/proof. Use domain $\{0, 1\}$ and let $P(x, y)$ mean $x = y$. For each y choose $x = y$, so the antecedent holds. But $x = 0$ fails for $y = 1$, and $x = 1$ fails for $y = 0$, so no fixed x works for every y. The consequent is false and the implication is false in this countermodel.

L088

Predicate Logic

2024 Spring Midterm | Question 3(c) | Source page 5 | Difficulty 2 | 1 mark
Q-2024S-MID-P03-c | Answer provenance: official

> Let $P(x, y)$ be a propositional function. For each part below, state whether or not it is a tautology. No explanation is needed.
>
> (c) $\exists x \forall y P(x, y) \rightarrow \forall x \exists y P(x, y)$.

Official Answer

Source: physical PDF page 5 (COMP2711_S2024_Midterm.pdf).

> Answer. (c) Not a tautology

Supplementary Explanation

Answer. Not a tautology.

Derivation/proof. Use domain $\{0, 1\}$ and let $P(x, y)$ mean $x = 0$. Choosing $x = 0$ makes it true for every y, so the antecedent is true. For $x = 1$, however, $P(1, y)$ is false for every y, making the consequent $\forall x \exists y P(x, y)$ false. Thus this interpretation falsifies the implication.

L089

Predicate Logic

2024 Spring Midterm | Question 3(d) | Source page 5 | Difficulty 2 | 1 mark
Q-2024S-MID-P03-d | Answer provenance: official

> Let $P(x, y)$ be a propositional function. For each part below, state whether or not it is a tautology. No explanation is needed.
>
> (d) $\forall x \exists y P(x, y) \rightarrow \exists x \forall y P(x, y)$.

Official Answer

Source: physical PDF page 5 (COMP2711_S2024_Midterm.pdf).

> Answer. (d) Not a tautology

Supplementary Explanation

Answer. Not a tautology.

Derivation/proof. Use domain $\{0, 1\}$ and let $P(x, y)$ mean $x = y$. For each x choose $y = x$, making the antecedent true. For any fixed x, the other value of y makes $P(x, y)$ false, so no row is entirely true. The consequent is false, proving the implication is not a tautology.

L090

Predicate Logic

2024 Spring Midterm | Question 4(a) | Source page 6 | Difficulty 2
Q-2024S-MID-P04-a | Answer provenance: official

> (a) Consider the domain to be the set of all integers. Express the following statement using quantifiers: "Every integer multiple of 4 can be expressed as the difference of two perfect squares." You are only allowed to use $\forall, \exists, \wedge, \vee, \neg, \rightarrow, \leftrightarrow$, equality $(=)$, power (e.g. (a^b)), addition (+,-), multiplication$(\times)$ and divisibility $(|)$.

Official Answer

Source: physical PDF page 6 (COMP2711_S2024_Midterm.pdf).

> Answer. (a) possible solution 1: $\forall x \exists a \exists b (4x = a^2 - b^2)$; possible solution 2: $\forall x (4 \mid x \rightarrow \exists a \exists b\, x = a^2 - b^2)$

Supplementary Explanation

Answer. $\forall x (4 \mid x \rightarrow \exists a \exists b (x = a^2 - b^2))$.

Derivation/proof. The domain is already the integers, so no integer predicate is needed. The antecedent $4 \mid x$ selects multiples of four, while $x = a^2 - b^2$ expresses the required difference of integer squares. The quantifiers $\exists a \exists b$ choose bases for each qualifying x, and $\forall x$ covers all integers. This yields $\forall x (4 \mid x \rightarrow \exists a \exists b (x = a^2 - b^2))$. The other official form, $\forall x \exists a \exists b (4x = a^2 - b^2)$, directly enumerates multiples as $4x$ and is equivalent.

L091

Predicate Logic

2024 Spring Midterm | Question 4(b) | Source page 6 | Difficulty 2
Q-2024S-MID-P04-b | Answer provenance: official

> (b) Express the negation of your statement in (a), such that the symbol $\neg$ does not appear in your statement. (As an example if you want to say $\neg(a = b)$ you must use $a \neq b$, and if you want to say $\neg(a \mid b)$ you must use $a \nmid b$)

Official Answer

Source: physical PDF page 6 (COMP2711_S2024_Midterm.pdf).

> (b) possible solution 1: $\exists x \forall a \forall b (4x \neq a^2 - b^2)$; possible solution 2: $\exists x (4 \mid x \wedge \forall a \forall b (x \neq a^2 - b^2))$

Supplementary Explanation

Answer. $\exists x (4 \mid x \wedge \forall a \forall b (x \neq a^2 - b^2))$.

Derivation/proof. Start from $\neg \forall x (4 \mid x \to \exists a \exists b (x = a^2 - b^2))$. Negating the universal quantifier gives an existential one. Using $\neg(A \to B) \equiv A \wedge \neg B$ then gives $\exists x (4 \mid x \wedge \neg \exists a \exists b (x = a^2 - b^2))$. Negating the two existential quantifiers produces two universal quantifiers; replace $\neg(x = a^2 - b^2)$ with $x \neq a^2 - b^2$. The resulting answer contains no negation symbol, as required.

L092

Predicate Logic

2023 Fall Final | Question 1(a)(i) | Source page 3 | Difficulty 2 | 1 mark
Q-2023F-FINAL-P01-a-i | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (i) $\forall x \forall a \forall b: \ x \mid ab \rightarrow (x \mid a \vee x \mid b)$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (a) (i) False

Supplementary Explanation

Answer. False.

Derivation/proof. Take $x = 6$, $a = 2$, and $b = 3$. Then $ab = 6$, so $6 \mid ab$ is true. But $6 \nmid 2$ and $6 \nmid 3$, making the disjunction false. Thus the implication is false for this one integer triple, which falsifies the universally quantified expression. The counterexample also shows precisely why a primality assumption would matter.

Take $x = 6, a = 2, b = 3$. The implication is $T \rightarrow F$, since $6 \mid 6$ but $6 \nmid 2$ and $6 \nmid 3$. The divisor is not restricted to primes.

L093

Predicate Logic

2023 Fall Final | Question 1(a)(iii) | Source page 3 | Difficulty 2 | 1 mark
Q-2023F-FINAL-P01-a-iii | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (iii) $\forall x \forall a \forall b : \ (x \mid ab \rightarrow (x \mid a \vee x \mid b)) \rightarrow P(x)$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (iii) False

Supplementary Explanation

Answer. False.

Derivation/proof. Let $x = 1$, and for concreteness take $a = b = 1$. Both $1 \mid ab$ and $1 \mid a \vee 1 \mid b$ are true, so their implication is true. But $P(1)$ is false because 1 is not prime. The outer implication is therefore $T \rightarrow F = F$, falsifying the universal formula. This respects the actual printed placement of the a,b quantifiers outside the whole implication.

For $x = 1, a = b = 1$, the divisibility implication is true but $P(1) = F$. The outer implication therefore has value $T \rightarrow F = F$.

L094

Predicate Logic

2023 Spring Midterm | Question 2(a) | Source page 5 | Difficulty 2 | 3 marks
Q-2023S-MID-P02-a | Answer provenance: official

> **Shared instructions for (a)–(b); editorial grouping:**
>
> Determine the truth value of each of these statements and prove your answer. Let the domain be $\mathbb{R}$.
>
> **Shared instructions for (c)–(d); editorial grouping:**
>
> Are the following statements equivalent? If yes, give a proof. If not, give a counter example specifying two predicates $P(x)$ and $Q(x)$ and a domain that show the statements are not logically equivalent:
>
> (a) $\forall x \forall y (x^2 + y^2 \geq 2xy)$

Official Answer

Source: physical PDF page 5 (COMP2711_S2023_Midterm.pdf).

> (a) True. $x^2 + y^2 \geq 2xy \leftrightarrow x^2 - 2xy + y^2 \geq 0 \leftrightarrow (x - y)^2 \geq 0$. Since $(x - y)^2 \geq 0$, the statement is true for all x and y.

Supplementary Explanation

Answer. True.

Derivation/proof. Let $x, y \in \mathbb{R}$ be arbitrary. Since a real square is nonnegative, $(x - y)^2 \geq 0$. Expanding gives $x^2 - 2xy + y^2 \geq 0$, which rearranges to $x^2 + y^2 \geq 2xy$. No restriction was imposed on either chosen real number, so universal generalization proves the statement. Equality occurs exactly when $x = y$, confirming the boundary is included.

Equality holds if and only if $x = y$.

L095

Predicate Logic

2023 Spring Midterm | Question 2(b) | Source page 5 | Difficulty 2 | 3 marks
Q-2023S-MID-P02-b | Answer provenance: official

> **Shared instructions for (a)–(b); editorial grouping:**
>
> Determine the truth value of each of these statements and prove your answer. Let the domain be $\mathbb{R}$.
>
> **Shared instructions for (c)–(d); editorial grouping:**
>
> Are the following statements equivalent? If yes, give a proof. If not, give a counter example specifying two predicates $P(x)$ and $Q(x)$ and a domain that show the statements are not logically equivalent:
>
> (b) $\exists x \forall y (x \geq y^2)$

Official Answer

Source: physical PDF page 5 (COMP2711_S2023_Midterm.pdf).

> (b) False. For every x, there exists a y, such as $y = \sqrt{|x| + 1}$, that $x < y^2$.

Supplementary Explanation

Answer. False.

Derivation/proof. Negating the assertion gives $\forall x \exists y (x < y^2)$. For an arbitrary real x, choose $y = \sqrt{|x| + 1}$. This is real and satisfies $y^2 = |x| + 1 > x$, whether x is negative or nonnegative. Therefore every proposed x fails for a suitable y, so no real number is an upper bound for all real squares.

L096

Predicate Logic

2023 Spring Midterm | Question 2(c) | Source page 5 | Difficulty 2 | 3 marks
Q-2023S-MID-P02-c | Answer provenance: official

> **Shared instructions for (a)–(b); editorial grouping:**
>
> Determine the truth value of each of these statements and prove your answer. Let the domain be $\mathbb{R}$.
>
> **Shared instructions for (c)–(d); editorial grouping:**
>
> Are the following statements equivalent? If yes, give a proof. If not, give a counter example specifying two predicates $P(x)$ and $Q(x)$ and a domain that show the statements are not logically equivalent:
>
> (c) $\neg(\exists x(x > 0 \wedge P(x)))$ and $\forall x(x \leq 0 \vee \neg P(x))$

Official Answer

Source: physical PDF page 5 (COMP2711_S2023_Midterm.pdf).

> (c) Equivalent. $\neg(\exists x(x > 0 \wedge p(x))) \equiv \forall x(\neg(x > 0) \vee \neg p(x)) \equiv \forall x(x \leq 0 \vee \neg p(x))$. (By De Morgan's Law)

Supplementary Explanation

Answer. Equivalent.

Derivation/proof. Apply quantifier negation and De Morgan in order:

$$\neg\exists x(x > 0 \wedge P(x)) \equiv \forall x\neg(x > 0 \wedge P(x)) \equiv \forall x(\neg(x > 0) \vee \neg P(x)) \equiv \forall x(x \leq 0 \vee \neg P(x)).$$

These are equivalences for every real x and every interpretation of P. The predicate Q mentioned in the general instructions is not used by either formula, so its interpretation is immaterial.

L097

Predicate Logic

2022 Final Part 2 | Question 5(a)(i) | Source page 3 | Difficulty 2
Q-2022-FINAL2-P05-a-i | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. You can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (i) Every edge connects a vertex in V_1 and a vertex in V_2 (ii) For every vertex in V_1, there are edges that connect it with all vertices in V_2
>
> Every edge connects a vertex in V_1 and a vertex in V_2

Official Answer

Source: physical PDF page 3 (2022-final-part-2.pdf).

> (a) (i) $(\forall u \in V_1 \forall v \in V_1 (\neg E(u, v))) \wedge (\forall u \in V_2 \forall v \in V_2 (\neg E(u, v)))$ or $\forall u, v \in V (E(u, v) \rightarrow ((u \in V_1 \wedge v \in V_2) \vee (u \in V_2 \wedge v \in V_1)))$

Supplementary Explanation

Answer. $(\forall u \in V_1 \forall v \in V_1 \neg E(u, v)) \wedge (\forall u \in V_2 \forall v \in V_2 \neg E(u, v))$.

Derivation/proof. Because V1,V2 partition all vertices, any pair either lies within one part or across the two. The formula excludes both within-part possibilities, leaving only cross-part pairs as potential edges. Conversely every cross-part-only graph satisfies both universal nonadjacency clauses. It places no requirement that any particular cross edge exist.

The formula does not require every cross-part pair to be adjacent.

L098

Predicate Logic

2022 Midterm | Question 2(b) | Source page 1 | Difficulty 2
Q-2022-MID-P02-b | Answer provenance: official

> Let $W(x, y)$ denote the predicate that x has visited website y, where the domain for x consists of people in the world and the domain for y consists of all websites. Express each of the following statements using predicate logic.
>
> There is a person other than David who has visited all the websites David has visited.

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (b) $\exists y \forall z (y \neq \text{David} \wedge (W(\text{David}, z) \rightarrow W(y, z)))$

Supplementary Explanation

Answer. $\exists x (x \neq \text{David} \wedge \forall y (W(\text{David}, y) \rightarrow W(x, y)))$.

Derivation/proof. The person x must be different from David, so place that condition alongside the quantified website requirement. For every website y, if David visited it then x did too. The implication asserts inclusion of David's visited set in x's visited set and permits x to have visited additional websites. The person is chosen once before ranging over sites.

The person cannot change from one website to another. In the supplementary formula, x denotes the person and y the website; the official formula uses y, z respectively.

L099

Predicate Logic

2022 Midterm | Question 3(3) | Source page 2 | Difficulty 2
Q-2022-MID-P03-3 | Answer provenance: official

> Let $P(x), Q(x), R(x), S(x)$ be the following predicates:
>
> $P(x)$: "x is a hummingbird." $Q(x)$: "x is large." $R(x)$: "x lives on honey." $S(x)$: "x is richly colored."
>
> Let the domain be all the birds. First translate the following statements into predicate logic (you may assume that "small" is the same as "not large" and that "dull in color" is the same as "not richly colored"):
>
> (1) "All hummingbirds are richly colored." (2) "No large birds live on honey." (3) "Birds that do not live on honey are dull in color." (4) "Hummingbirds are small."
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> "Birds that do not live on honey are dull in color."

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

> (3) $\forall x(\neg R(x) \to \neg S(x))$.

Supplementary Explanation

Answer. $\forall x(\neg R(x) \to \neg S(x))$.

Derivation/proof. The restrictive clause selects birds not living on honey, ¬R. The source explicitly equates dull color with not richly colored, ¬S. Universally imply the latter from the former. Its contrapositive S→R is equivalent, but the converse ¬S→¬R is not entailed.

The equivalent contrapositive is $\forall x(S(x) \to R(x))$.

L100

Predicate Logic

2021 Midterm | Question 1(b) | Source page 1 | Difficulty 2 | 3 marks
Q-2021-MID-P01-b | Answer provenance: official

> Determine the truth value of each of the following statements. The domain for all variables are all integers. Justify your answer.
>
> $\forall y \exists x (y = x^2 + 2x + 1)$

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> (b) False. Note that $x^2 + 2x + 1 = (x + 1)^2$. When $y = 3$, there is no integer x so that $(x + 1)^2 = 3$.

Supplementary Explanation

Answer. False.

Derivation/proof. The right side equals $(x + 1)^2$, a nonnegative integer square. Choose y=−1, an allowed integer. No integer square equals −1, so the required x does not exist for this y. The official example y=3 also works because 3 is not an integer square.

The choice $y = -1$ is impossible, and the official choice 3 is not an integer square.

L101

Predicate Logic

2021 Midterm | Question 1(c) | Source page 1 | Difficulty 2 | 3 marks
Q-2021-MID-P01-c | Answer provenance: official

> Determine the truth value of each of the following statements. The domain for all variables are all integers. Justify your answer.
>
> $\exists x \forall y \exists z (2y - z = 4x)$

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> (c) True. Let $x = 0$. Then for any y, set $z = 2y$.

Supplementary Explanation

Answer. True.

Derivation/proof. Choose x=0 once. For any integer y choose z=2y, which is an integer and satisfies $2y - z = 2y - 2y = 0 = 4x$. The z witness may vary with y because its quantifier is inside ∀y. This directly meets the quantifier order without needing to solve for a common z.

L102

Predicate Logic

2021 Midterm | Question 1(d) | Source page 1 | Difficulty 2 | 3 marks
Q-2021-MID-P01-d | Answer provenance: official

> Determine the truth value of each of the following statements. The domain for all variables are all integers. Justify your answer.
>
> $\exists x \exists z \forall y (2y - z = 4x)$

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> (d) False. For every x and z, there is an integer $y = 4|x| + |z| + 1$, and thus $2y = 8|x| + 2|z| + 2 > 4x + z$.

Supplementary Explanation

Answer. False.

Derivation/proof. Suppose fixed integers x,z made the equation true for every y. At y=0 it gives −z=4x; at y=1 it gives 2−z=4x. Subtract the equations to obtain 2=0, a contradiction. Hence no fixed pair x,z can satisfy the universal condition.

The conflicting equations are $-z = 4x$ and $2 - z = 4x$; subtraction gives $2 = 0$.

L103

Predicate Logic

2021 Midterm | Question 2 | Source page 1 | Difficulty 2
Q-2021-MID-P02-proof | Answer provenance: official

> Consider the following predicates.
>
> $c(x)$: "x studied COMP2711". $g(x)$: "x knows how to compute the gcd of two integers". $h(x)$: "x can get a high paying job".
>
> "Sarah, a student who studied COMP2711, knows how to compute the gcd of two integers. Everyone who knows how to compute the gcd of two integers can get a high-paying job. Therefore, at least one student who studied COMP2711 can get a high-paying job."
>
> Please first translate the premises and conclusion into predicate logic sentences. Then show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> Then show a step-by-step proof using inference rules.

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf), physical PDF page 2 (2021-midterm.pdf).

> We are given premises $c(\text{Sarah})$, $g(\text{Sarah})$, and $\forall x(g(x) \to h(x))$, and we want to conclude $\exists x(c(x) \wedge h(x))$.

Step	Reason
1. $\forall x(g(x) \to h(x))$	Premise
2. $g(\text{Sarah}) \to h(\text{Sarah})$	Universal instantiation using (1)
3. $g(\text{Sarah})$	Premise
4. $h(\text{Sarah})$	Modus ponens using (2) and (3)
5. $c(\text{Sarah})$	Premise
6. $c(\text{Sarah}) \wedge h(\text{Sarah})$	Conjunction using (4) and (5)
7. $\exists x(c(x) \wedge h(x))$	Existential generalization using (6)

Supplementary Explanation

Answer. The conclusion follows from the three premises.

Derivation/proof. 1. $\forall x(g(x) \to h(x))$ (premise). 2. $g(\text{Sarah}) \to h(\text{Sarah})$ (1, universal instantiation). 3. $g(\text{Sarah})$ (premise). 4. $h(\text{Sarah})$ (2,3, modus ponens). 5. $c(\text{Sarah})$ (premise). 6. $c(\text{Sarah}) \wedge h(\text{Sarah})$ (5,4). 7. $\exists x(c(x) \wedge h(x))$ (6, existential generalization). The same named individual witnesses both conjuncts.

Step 4 follows from (2) and (3).

L104

Predicate Logic

2021 Midterm | Question 2 | Source page 1 | Difficulty 2
Q-2021-MID-P02-translation-conclusion | Answer provenance: official

> Consider the following predicates.
>
> $c(x)$: "x studied COMP2711". $g(x)$: "x knows how to compute the gcd of two integers". $h(x)$: "x can get a high paying job".
>
> "Sarah, a student who studied COMP2711, knows how to compute the gcd of two integers. Everyone who knows how to compute the gcd of two integers can get a high-paying job. Therefore, at least one student who studied COMP2711 can get a high-paying job."
>
> Please first translate the premises and conclusion into predicate logic sentences. Then show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> Therefore, at least one student who studied COMP2711 can get a high-paying job.

Official Answer

Source: physical PDF page 1 (2021-midterm.pdf).

> We are given premises $c(\text{Sarah})$, $g(\text{Sarah})$, and $\forall x(g(x) \to h(x))$, and we want to conclude $\exists x(c(x) \land h(x))$.

Supplementary Explanation

Answer. $\exists x(c(x) \land h(x))$.

Derivation/proof. The conclusion asserts at least one witness, so use an existential quantifier. That same person must satisfy course participation c(x) and job eligibility h(x), hence their conjunction. An existential implication could be true merely because someone did not study the course, which would not establish the stated conclusion.

To express both properties of one witness, a conjunction is required inside the existential quantifier.

L105

Predicate Logic

2020 Final Part 1 | Question 1(a)(ii) | Source page 1 | Difficulty 2
Q-2020-FINAL1-P01-a-ii | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V (E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "Every edge connects a vertex in V_1 and a vertex in V_2".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$(\forall u \in V_1 \forall v \in V_1 (\neg E(u, v))) \wedge (\forall u \in V_2 \forall v \in V_2 (\neg E(u, v)))$.

Supplementary Explanation

Answer. $\forall u, v \in V (E(u, v) \rightarrow ((u \in V_1 \wedge v \in V_2) \vee (u \in V_2 \wedge v \in V_1)))$.

Derivation/proof. Universally consider both endpoints of an edge. Because the graph is undirected either orientation of the two parts is possible, so the consequent is a disjunction of the two opposite-part assignments. The given partition covers V and is disjoint, making this equivalent to forbidding edges within each part.

The two within-part restrictions are $\forall u \in V_1 \forall v \in V_1 \neg E(u, v)$ and $\forall u \in V_2 \forall v \in V_2 \neg E(u, v)$. These forbid every internal edge because the edge is undirected; this holds regardless of which endpoint is written first.

L106

Predicate Logic

2020 Midterm | Question 1(b) | Source page 1 | Difficulty 2
Q-2020-MID-P01-b | Answer provenance: official

> Let $Z(x), D(x), F(x)$ and $C(x)$ be the following predicates:
>
> $Z(x)$: "x attended the Zoom meeting of midterm dry run". $D(x)$: "x gets some marks deducted for the midterm exam". $F(x)$: "x submitted a file for the midterm dry run". $C(x)$: "x is cheating".
>
> Express the following statements using quantifiers, logical connectives, and the predicates above, where the domain consists of all students in COMP2711.
>
> If a student attended the Zoom meeting of midterm dry run but got some marks deducted for the midterm exam, then he/she must have not submitted a file for the midterm dry run.

Official Answer

Source: physical PDF page 1 (2020-midterm.pdf).

$\forall x((Z(x) \land D(x)) \rightarrow \neg F(x))$.

Supplementary Explanation

Answer. $\forall x((Z(x) \land D(x)) \rightarrow \neg F(x))$.

Derivation/proof. "But" joins the two factual conditions with $\land$: the same student attended and lost marks. Their conjunction is the complete if-clause. The then-clause negates submission, not attendance. Universally quantify x because the assertion concerns any student satisfying those conditions.

L107

Predicate Logic

2020 Midterm | Question 1(d) | Source page 1 | Difficulty 2
Q-2020-MID-P01-d | Answer provenance: official

> Let $Z(x), D(x), F(x)$ and $C(x)$ be the following predicates:
>
> $Z(x)$: "x attended the Zoom meeting of midterm dry run". $D(x)$: "x gets some marks deducted for the midterm exam". $F(x)$: "x submitted a file for the midterm dry run". $C(x)$: "x is cheating".
>
> Express the following statements using quantifiers, logical connectives, and the predicates above, where the domain consists of all students in COMP2711.
>
> Any student who submitted a file for the midterm dry run but were absent from the Zoom meeting of midterm dry run is considered as cheating.

Official Answer

Source: physical PDF page 1 (2020-midterm.pdf).

$\forall x((F(x) \land \neg Z(x)) \rightarrow C(x))$.

Supplementary Explanation

Answer. $\forall x((F(x) \land \neg Z(x)) \rightarrow C(x))$.

Derivation/proof. The restrictive clause describes a student who submitted a file and did not attend. Combine these as $F(x) \land \neg Z(x)$. The predicate asserted of every such student is cheating, so C is the consequent. Nothing in the wording asserts the converse that every cheater has this particular pattern.

The source gives one sufficient condition for cheating, so do not strengthen the implication to a biconditional. Students outside the stated attendance-and-submission pattern are not classified by this sentence.

L108

Predicate Logic

2020 Midterm | Question 3(b) | Source page 2 | Difficulty 2
Q-2020-MID-P03-b | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\forall x \forall y((x > 2 \wedge y > 2) \rightarrow (xy > x + y))$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

True. When $y > 2$, $y/(y-1) < 2$. So, $x > y/(y-1)$. This implies $x(y-1) > y$, and thus $xy > x + y$.

Supplementary Explanation

Answer. True.

Derivation/proof. If either hypothesis fails the implication is true. Otherwise $x - 1 > 1$ and $y - 1 > 1$, so $(x-1)(y-1) > 1$. Expanding gives $xy - x - y + 1 > 1$, hence $xy > x + y$. The strict inequalities ensure a strict final result, including values arbitrarily close to 2.

Under the two hypotheses, $x - 1 > 1, y - 1 > 1$. Their product is greater than one, and expansion and subtraction then give $xy > x + y$.

L109

Predicate Logic

2020 Midterm | Question 3(e) | Source page 2 | Difficulty 2
Q-2020-MID-P03-e | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\forall x((x > 1) \to (x^2 > x)) \leftrightarrow \neg\exists x((x > 1) \to (x^2 \leq x))$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

False.

$$\begin{aligned}\neg\exists x((x > 1) \to (x^2 \leq x)) &\equiv \forall x\neg((x > 1) \to (x^2 \leq x)) \\ &\equiv \forall x\neg(\neg(x > 1) \vee (x^2 \leq x)) \\ &\equiv \forall x((x > 1) \wedge (x^2 > x)),\end{aligned}$$

which is false when $x \leq 1$. However, $\forall x((x > 1) \to (x^2 > x))$ is true.

Supplementary Explanation

Answer. False.

Derivation/proof. The left side is true: for x>1, multiplying x>1 by positive x gives $x^2 > x$. On the right, x=0 makes the antecedent of the existential implication false, so that implication is true. Hence its existential quantification is true and its negation false. The biconditional is therefore $T \leftrightarrow F$.

The left side is true, while the negated existential on the right is false because its existential has the witness zero. Therefore $T \leftrightarrow F = F$.

L110

Predicate Logic

2020 Midterm | Question 3(f) | Source page 2 | Difficulty 2
Q-2020-MID-P03-f | Answer provenance: official

> For each of the following statement, determine it is true or false. Justification is not required. The domain is the set of real numbers.
>
> $\forall x((x > 1) \to (x^2 > x)) \leftrightarrow \neg\exists x((x > 1) \wedge (x^2 \leq x))$

Official Answer

Source: physical PDF page 2 (2020-midterm.pdf).

True.

$$\begin{aligned}\neg\exists x((x > 1) \wedge (x^2 \leq x)) &\equiv \forall x\neg((x > 1) \wedge (x^2 \leq x)) \\ &\equiv \forall x((x \leq 1) \vee (x^2 > x)) \\ &\equiv \forall x((x > 1) \to (x^2 > x)).\end{aligned}$$

Supplementary Explanation

Answer. True.

Derivation/proof. Negate the existential: $\neg\exists x((x > 1) \wedge (x^2 \leq x)) \equiv \forall x\neg((x > 1) \wedge (x^2 \leq x))$. De Morgan gives $\forall x((x \leq 1) \vee (x^2 > x))$, which is $\forall x((x > 1) \to (x^2 > x))$. Thus both sides are identical by logical laws, regardless of separately verifying the inequality.

The two sides are two logically equivalent statements: negating an existential counterexample to the inequality gives exactly the universal implication. Their biconditional is therefore true.

L111

Predicate Logic

2019 Final | Question 1(a) | Source page 3 | Difficulty 2 | 2 marks
Q-2019-FINAL-P01-a | Answer provenance: official

> For each of the following statements, determine whether it is true or false, and justify your answer. The domain is all integers.
>
> (a) $\exists x\forall y(2x - y > 1)$

Official Answer

Source: physical PDF page 3 (2019-final.pdf).

False. This is the negation of (c), which is true (shown below).

Supplementary Explanation

Answer. False.

Derivation/proof. For any integer x proposed as the existential witness, choose the integer $y = 2x$. Then $2x - y = 0$, which is not greater than 1. Thus no x works for all y. This proves the negation $\forall x\exists y(2x - y \leq 1)$, rather than merely testing one candidate x.

With the chosen witness $y = 2x$, the expression is $2x - y = 0 \leq 1$. This confirms the inequality for every real universal choice of x.

L112

Predicate Logic

2019 Final | Question 1(d) | Source page 3 | Difficulty 2 | 2 marks
Q-2019-FINAL-P01-d | Answer provenance: official

> For each of the following statements, determine whether it is true or false, and justify your answer. The domain is all integers.
>
> (d) $\exists x \forall y \exists z (x + y + z = 0)$

Official Answer

Source: physical PDF page 3 (2019-final.pdf).

True. Set $x = 0$. Then for any y, setting $z = -y$ makes the proposition true.

Supplementary Explanation

Answer. True.

Derivation/proof. Choose the fixed integer $x = 0$ before y is given. For each integer y, choose $z = -y$. Then z is an integer and $x + y + z = 0 + y - y = 0$. This respects the quantifier order: x is independent of y, whereas z may depend on y.

L113

Predicate Logic

2019 Midterm | Question 2(c) | Source page 4 | Difficulty 2 | 2 marks
Q-2019-MID-P02-c | Answer provenance: official

> Let $P(x)$ be the statement "x is even" and $Q(x)$ be the statement "x is an integer". Express each of the following statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain is all real numbers.
>
> (c) Some integers are odd.

Official Answer

Source: physical PDF page 4 (2019-midterm.pdf).

$\exists x (Q(x) \wedge \neg P(x))$.

Supplementary Explanation

Answer. $\exists x (Q(x) \wedge \neg P(x))$.

Derivation/proof. Over the real domain, not even alone does not establish that a value is an odd integer. The same witness must be an integer, $Q(x)$, and not even, $\neg P(x)$. For integers, those two conditions describe oddness, and "some" requires their conjunction to hold for at least one x.

L114

Predicate Logic

2019 Midterm | Question 2(d) | Source page 4 | Difficulty 2 | 3 marks
Q-2019-MID-P02-d | Answer provenance: official

> Let $P(x)$ be the statement "x is even" and $Q(x)$ be the statement "x is an integer". Express each of the following statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain is all real numbers.
>
> (d) If $x < y$, then x is not equal to y.

Official Answer

Source: physical PDF page 4 (2019-midterm.pdf).

$\forall x \forall y((x < y) \rightarrow (x \neq y))$.

Supplementary Explanation

Answer. $\forall x \forall y((x < y) \rightarrow (x \neq y))$.

Derivation/proof. The claim is a general relation between arbitrary real numbers x, y, so quantify both universally. The comparison $x < y$ is the antecedent and $x \neq y$ its consequence. The implication does not assert that every pair satisfies $x < y$; for equal or reversed pairs it is vacuously true.

Quantification must bind both variables to produce a proposition; leaving either free leaves an open predicate whose truth can depend on an unspecified value.

L115

Predicate Logic

2019 Midterm | Question 2(e) | Source page 4 | Difficulty 2 | 3 marks
Q-2019-MID-P02-e | Answer provenance: official

> Let $P(x)$ be the statement "x is even" and $Q(x)$ be the statement "x is an integer". Express each of the following statements using predicates, quantifiers, logical connectives, and mathematical operators where the domain is all real numbers.
>
> (e) There is no largest real number.

Official Answer

Source: physical PDF page 4 (2019-midterm.pdf).

$\forall x \exists y(y > x)$.

Supplementary Explanation

Answer. $\forall x \exists y(y > x)$.

Derivation/proof. For any candidate x to be a largest real, there must be a real y larger than it. Thus every x has a suitable possibly different y, giving $\forall x \exists y(y > x)$. The witness $y = x + 1$ checks the intended meaning. Reversing the quantifiers would require one number larger than all reals.

For any x choose $y = x + 1$. An equivalent form is $\neg \exists x \forall y(y \leq x)$.

L116

Predicate Logic

2018 Final | Question 1(b) | Source page 3 | Difficulty 2
Q-2018-FINAL-P01-b | Answer provenance: official

> For each of the following predicates, decide whether it is true or false. If it is true, prove it. If it is false, state it is false; you do not have to prove it.
>
> (b) (Prime number p; x, y, z in $\mathbb{Z}_p$; $y \neq 0$) $\forall x \forall y \exists z (x + yz \equiv 0 \pmod{p})$

Official Answer

Source: physical PDF page 3 (2018-final.pdf).

True. The inverse of y must exist in $\mathbb{Z}_p$, so we can set $z \equiv -y^{-1}x \pmod{p}$ to satisfy the equation for any x and y.

Supplementary Explanation

Answer. True; choose $z \equiv -y^{-1}x \pmod{p}$.

Derivation/proof. Because p is prime and $y \not\equiv 0 \pmod{p}$, $\gcd(y, p) = 1$, so a multiplicative inverse y^{-1} exists. Choose the representative of $-y^{-1}x$ in $\mathbb{Z}_p$. Then $x + yz \equiv x - yy^{-1}x \equiv x - x \equiv 0$. The witness may depend on both universally chosen values.

Choosing the second variable equal to the first makes the difference $x - x = 0$, independent of the value of x. This supplies the existential witness for every universal choice.

L117

Predicate Logic

2018 Midterm | Question 2(c) | Source page 4 | Difficulty 2
Q-2018-MID-P02-c | Answer provenance: official

> Some of the following propositions and predicates are tautologies. For each statement that is not a tautology, prove it by providing a counterexample. For each statement that is a tautology, just state it is a tautology; no proof is required.
>
> (c) $(\forall x P(x)) \to (\exists x (P(x) \to Q(x)))$

Official Answer

Source: physical PDF page 4 (2018-midterm.pdf).

Not a tautology. It is false if $Q(x)$ is always false and $P(x)$ is always true.

$$\begin{aligned}(\forall x P(x)) \to (\exists x (P(x) \to Q(x))) &\equiv T \to (\exists x (T \to F)) \\ &\equiv T \to F \\ &\equiv F.\end{aligned}$$

Supplementary Explanation

Answer. Not a tautology. On $D = \{a\}$, let $P(a) = T, Q(a) = F$.

Derivation/proof. The universal antecedent is true in this domain. Its only candidate for the existential consequent gives $P(a) \to Q(a) = T \to F = F$, so that consequent is false. Thus the full implication is false, proving failure of logical validity.

L118

Predicate Logic

2018 Midterm | Question 2(d) | Source page 4 | Difficulty 2
Q-2018-MID-P02-d | Answer provenance: official

> Some of the following propositions and predicates are tautologies. For each statement that is not a tautology, prove it by providing a counterexample. For each statement that is a tautology, just state it is a tautology; no proof is required.
>
> (d) In the domain of real numbers: $(\exists x \forall y(x > 2 \to x^2 + y^2 > 10)) \wedge (\forall x \exists y(x > 2 \to x^2 + y^2 > 10))$

Official Answer

Source: physical PDF page 4 (2018-midterm.pdf).

Tautology.

Supplementary Explanation

Answer. It is true over the real domain.

Derivation/proof. For the first conjunct choose $x = 0$: the antecedent $x > 2$ is false for every y, so every implication is true. For the second, for any x choose $y = 4$. Then $x^2 + y^2 \geq 16 > 10$, so the implication is true irrespective of x. Both conjuncts hold.

For the displayed choice of the other variable, $x^2 + 16 \geq 16 > 10$ for every real x. Nonnegativity of a square is enough to establish the universal claim.

L119

Predicate Logic

2017 Midterm | Question 1(c) | Source page 3 | Difficulty 2
Q-2017-MID-P01-c | Answer provenance: official

> Some of the following propositions and predicates are tautologies. Give a proof for each of the statements that are tautologies. You do not need to disprove the ones that are not tautologies.
>
> (c) $(\forall x P(x)) \to (\exists x(P(x) \to Q(x)))$

Official Answer

Source: physical PDF page 3 (2017-midterm.pdf).

Not a tautology. It is false if $Q(x)$ is always false. (No counter example required)

$$\begin{aligned}(\forall x P(x)) \to \exists x(P(x) \to Q(x)) &\equiv \exists x \neg P(x) \vee \exists x(\neg P(x) \vee Q(x)) \\ &\equiv \exists x \exists y(\neg P(x) \vee \neg P(y) \vee Q(y)).\end{aligned}$$

Supplementary Explanation

Answer. It is not a tautology.

Derivation/proof. Use the one-element domain $D = \{a\}$ with $P(a) = T$ and $Q(a) = F$. Then $\forall x P(x)$ is true, but $P(a) \to Q(a)$ is false, so the existential consequent is false and the outer implication is false. Merely setting every Q false is insufficient unless every P is also true.

The implication fails only with a true antecedent and false consequent: $T \to F = F$. Thus the stated condition on Q alone does not settle the implication; the antecedent must also be checked.

L120

Predicate Logic

2017 Midterm | Question 1(d) | Source page 3 | Difficulty 2
Q-2017-MID-P01-d | Answer provenance: official

> Some of the following propositions and predicates are tautologies. Give a proof for each of the statements that are tautologies. You do not need to disprove the ones that are not tautologies.
>
> (d) For the domain of real numbers, $\exists y(\exists x(x^2 = 2) \vee \exists x(x > 2 \wedge x^2 + y^2 = 2))$

Official Answer

Source: physical PDF page 3 (2017-midterm.pdf).

Tautology.

$$\begin{aligned}\exists y(\exists x(x^2 = 2) \vee \exists x(x > 2 \wedge x^2 + y^2 = 2)) &\equiv \exists y(T \vee \exists x(x > 2 \wedge x^2 + y^2 = 2)) \\ &\equiv \exists y(T \vee F) \\ &\equiv T.\end{aligned}$$

Supplementary Explanation

Answer. It is true over the stated real domain.

Derivation/proof. Choose $y = 0$ and choose $x = \sqrt{2}$ for the first existential statement. Then $x^2 = 2$, so the first disjunct is true and therefore the disjunction and outer existential are true. The second disjunct is actually impossible because $x > 2$ implies $x^2 + y^2 > 4$.

For the second disjunct, $x > 2$ gives $x^2 > 4$ and $y^2 \geq 0$, hence $x^2 + y^2 > 4 > 2$. Thus that disjunct has no witness, while the first disjunct still makes the full statement true.

L121

Predicate Logic

2017 Midterm | Question 2(2) | Source page 4 | Difficulty 2
Q-2017-MID-P02-2 | Answer provenance: official

> Consider the following predicates:
>
> $C(x)$: x is a child. $K(x)$: x is kind. $N(x,y)$: x and y are neighbours. $L(x)$: x learned.
>
> Write the following sentences using the above predicates, adding quantifiers when necessary:
>
> 2. People who are unkind can only have unkind neighbours.

Official Answer

Source: physical PDF page 4 (2017-midterm.pdf).

$\forall x(\neg K(x) \rightarrow \forall y(N(x,y) \rightarrow \neg K(y)))$, or equivalently, $\forall x \forall y((\neg K(x) \wedge N(x,y)) \rightarrow \neg K(y))$.

Supplementary Explanation

Answer. $\forall x \forall y((\neg K(x) \wedge N(x,y)) \rightarrow \neg K(y))$.

Derivation/proof. Fix any people x, y. Only when x is unkind and y is a neighbor of x does the sentence constrain y, requiring $\neg K(y)$. Universal quantification covers all such pairs. Equivalently, $\forall x(\neg K(x) \rightarrow \forall y(N(x,y) \rightarrow \neg K(y)))$.

The universal conditional does not require a neighbour to exist: if nobody satisfies the neighbour predicate, every instance is true vacuously.

L122

Predicate Logic

2017 Midterm | Question 2(4) | Source page 4 | Difficulty 2
Q-2017-MID-P02-4 | Answer provenance: official

> Consider the following predicates:
>
> $C(x)$: x is a child. $K(x)$: x is kind. $N(x,y)$: x and y are neighbours. $L(x)$: x learned.
>
> Write the following sentences using the above predicates, adding quantifiers when necessary:
>
> 4. Mencius, a child, has unkind neighbours.

Official Answer

Source: physical PDF page 4 (2017-midterm.pdf).

$C(\text{Mencius}) \wedge \exists x(N(x,\text{Mencius}) \wedge \neg K(x))$.

Supplementary Explanation

Answer. $C(\text{Mencius}) \wedge \exists x(N(x,\text{Mencius}) \wedge \neg K(x))$.

Derivation/proof. The apposition “a child” asserts $C(\text{Mencius})$. “Has unkind neighbours” supplies an existential witness who is both a neighbor and unkind. The source uses the usual existential reading, not a universal claim that all of Mencius's neighbors are unkind.

Both predicates must refer to the same witness inside one existential quantifier; separate existential witnesses could name different people and would not express the requested statement.

L123

Predicate Logic

2016 Midterm | Question 6(a) | Source page 10 | Difficulty 2 | 3 marks
Q-2016-MID-P06-a | Answer provenance: official

> Let $q(x)$ be "x is even". Let Z be a set of all integers. Let E be a set of all even integers.
>
> For each of the following pairs, say whether (i) is equivalent to (ii). If they are equivalent, all you have to do is to say that they are equivalent. If they are not equivalent, give a counter example. A counter example should involve a specification of $p(x)$ and an explanation as to why the resulting statement is false.
>
> (a) (i) $\forall x \in E(p(x))$
>
> (ii) $\forall x \in Z(q(x) \Rightarrow p(x))$

Official Answer

Source: PDF physical page 10 (2016-midterm.pdf).

(a) Yes.

Supplementary Explanation

Answer. Equivalent.

Derivation/proof. If (i) holds, every even integer satisfies p, and every odd integer satisfies $q \to p$ vacuously, proving (ii). Conversely, apply (ii) to an arbitrary even integer: q is true, so modus ponens yields p, proving (i).

For odd x, $q(x)$ is false, making the implication true. Conversely, assume (ii); an even integer has true antecedent, and modus ponens gives $p(x)$.

L124

Predicate Logic

2016 Midterm | Question 6(b) | Source page 10 | Difficulty 2 | 4 marks
Q-2016-MID-P06-b | Answer provenance: official

> Let $q(x)$ be "x is even". Let Z be a set of all integers. Let E be a set of all even integers.
>
> For each of the following pairs, say whether (i) is equivalent to (ii). If they are equivalent, all you have to do is to say that they are equivalent. If they are not equivalent, give a counter example. A counter example should involve a specification of $p(x)$ and an explanation as to why the resulting statement is false.
>
> (b) (i) $\forall x \in E(p(x))$
>
> (ii) $\forall x \in Z(q(x) \wedge p(x))$

Official Answer

Source: PDF physical page 10 (2016-midterm.pdf).

(b) No.

Let $p(x)$ be "x is even".

"$\forall x \in E(p(x))$" is true but "$\forall x \in Z(q(x) \wedge p(x))$" is false.

Supplementary Explanation

Answer. Not equivalent; take $p(x)$ to mean "x is even."

Derivation/proof. For this predicate, (i) says all even integers are even and is true. In (ii), set $x = 1$: $q(1)$ is false, so $q(1) \wedge p(1)$ is false. Therefore (ii) is false and the truth values differ.

The concrete counterexample has $q(1) = F$ and $q(1) \wedge p(1) = F$.

L125

Predicate Logic

2016 Midterm | Question 6(d) | Source page 10 | Difficulty 2 | 3 marks
Q-2016-MID-P06-d | Answer provenance: official

> Let $q(x)$ be "x is even". Let Z be a set of all integers. Let E be a set of all even integers.
>
> For each of the following pairs, say whether (i) is equivalent to (ii). If they are equivalent, all you have to do is to say that they are equivalent. If they are not equivalent, give a counter example. A counter example should involve a specification of $p(x)$ and an explanation as to why the resulting statement is false.
>
> (d) (i) $\exists x \in E(p(x))$
>
> (ii) $\exists x \in Z(q(x) \wedge p(x))$

Official Answer

Source: PDF physical page 10 (2016-midterm.pdf).

(d) Yes.

Supplementary Explanation

Answer. Equivalent.

Derivation/proof. A witness $x \in E$ for (i) is an integer with $q(x)$ and $p(x)$, hence witnesses (ii). Conversely, a witness of $q(x) \wedge p(x)$ is even by $q(x)$, so belongs to E and witnesses (i).

If (i) has witness u, then $q(u) \wedge p(u)$ witnesses (ii). Conversely, conjunction elimination for witness v yields $q(v)$ and $p(v)$, so $v \in E$ and it witnesses (i).

L126

Predicate Logic

2015 Midterm | Question 3(a) | Source page 7 | Difficulty 2 | 2 marks
Q-2015-MID-P03-statement-a | Answer provenance: official

> Consider the following three statements:
>
> (a) A rich and famous person is a successful person.
>
> (b) A rich person is successful if she/he is famous.
>
> (c) A famous person is successful if she/he is rich.
>
> Convert the statements into logic statements using the domain U (set of all persons), and predicates $rich(x)$ (person x is rich), $famous(x)$ (person x is famous), $successful(x)$ (person x is successful).
>
> Are the three statements logically equivalent? If not, explain why. If yes, give a proof. Show all steps of your proof, and state clearly the assumption or the inference rule used in each step.
>
> (a) A rich and famous person is a successful person.

Official Answer

Source: PDF physical page 8 (2015-midterm.pdf).

(a) $\forall x \in U((rich(x) \land famous(x)) \rightarrow successful(x))$.

Supplementary Explanation

Answer. $\forall x \in U((rich(x) \land famous(x)) \rightarrow successful(x))$.

Derivation/proof. Choose an arbitrary person x. The claim applies when both $rich(x)$ and $famous(x)$ hold, and then requires $successful(x)$. Put these conditions into an implication and universally quantify x .

People who do not satisfy both conditions are not constrained by this assertion of success.

L127

Predicate Logic

2015 Midterm | Question 3(b) | Source page 7 | Difficulty 2 | 2 marks
Q-2015-MID-P03-statement-b | Answer provenance: official

> Consider the following three statements:
>
> (a) A rich and famous person is a successful person.
>
> (b) A rich person is successful if she/he is famous.
>
> (c) A famous person is successful if she/he is rich.
>
> Convert the statements into logic statements using the domain U (set of all persons), and predicates $rich(x)$ (person x is rich), $famous(x)$ (person x is famous), $successful(x)$ (person x is successful).
>
> Are the three statements logically equivalent? If not, explain why. If yes, give a proof. Show all steps of your proof, and state clearly the assumption or the inference rule used in each step.
>
> (b) A rich person is successful if she/he is famous.

Official Answer

Source: PDF physical page 8 (2015-midterm.pdf).

(b) $\forall x \in U(rich(x) \rightarrow (famous(x) \rightarrow successful(x)))$.

Supplementary Explanation

Answer. $\forall x \in U(rich(x) \rightarrow (famous(x) \rightarrow successful(x)))$.

Derivation/proof. For each person x, being rich requires the conditional assertion "if famous, then successful." Thus $rich(x)$ is the outer antecedent and $famous(x) \rightarrow successful(x)$ its consequent.

L128

Predicate Logic

2015 Midterm | Question 3(c) | Source page 7 | Difficulty 2 | 2 marks
Q-2015-MID-P03-statement-c | Answer provenance: official

> Consider the following three statements:
>
> (a) A rich and famous person is a successful person.
>
> (b) A rich person is successful if she/he is famous.
>
> (c) A famous person is successful if she/he is rich.
>
> Convert the statements into logic statements using the domain U (set of all persons), and predicates $rich(x)$ (person x is rich), $famous(x)$ (person x is famous), $successful(x)$ (person x is successful).
>
> Are the three statements logically equivalent? If not, explain why. If yes, give a proof. Show all steps of your proof, and state clearly the assumption or the inference rule used in each step.
>
> (c) A famous person is successful if she/he is rich.

Official Answer

Source: PDF physical page 8 (2015-midterm.pdf).

(c) $\forall x \in U(famous(x) \rightarrow (rich(x) \rightarrow successful(x)))$.

Supplementary Explanation

Answer. $\forall x \in U(famous(x) \rightarrow (rich(x) \rightarrow successful(x)))$.

Derivation/proof. For arbitrary x, assume $famous(x)$. What must then hold is the conditional $rich(x) \rightarrow successful(x)$. Universal quantification gives the displayed formula.

L129

Predicate Logic

2013 Final | Question 6(b) | Source page 9 | Difficulty 2
Q-2013-FINAL-P06-b | Answer provenance: official

> Consider the following three statements:
>
> - (i) All rich people have famous-brand products and don't have part-time jobs.
> - (ii) Some students with part-time jobs have famous-brand products.
> - (iii) All students with part-time jobs are not rich.
>
> Let U be the universe of all people and define the following predicates:
>
> - $R(x)$: x is a rich person
> - $F(x)$: x has famous-brand products
> - $S(x)$: x is a student
> - $J(x)$: x has part-time jobs.
>
> (b) Does (i) logically imply (iii)? If yes, give a proof. Clearly state the logic inference rule that you use at each step. If no, explain why not.

Official Answer

Source: PDF physical page 9 (2013-final.pdf).

(b) Yes, (i) does imply (iii).

Proof: Statement (i) can be broken up into the following two statements:

$$\forall x \in U(R(x) \Rightarrow F(x))$$
$$\forall x \in U(R(x) \Rightarrow \neg J(x))$$

The contrapositive of the second rule is:

$$\forall x \in U(J(x) \Rightarrow \neg R(x))$$

Let x be a generic element in U. We have

$$S(x) \wedge J(x) \Rightarrow J(x)$$
$$J(x) \Rightarrow \neg R(x)$$

By transitivity of logic implication, we get:

$$S(x) \wedge J(x) \Rightarrow \neg R(x)$$

By the rule of Universal Generalization, we obtain:

$$\forall x \in U(S(x) \wedge J(x) \Rightarrow \neg R(x)).$$

Q.E.D

Supplementary Explanation

Answer. Yes, (i) implies (iii).

Derivation/proof. From (i), simplify to $\forall x(R(x) \Rightarrow \neg J(x))$. By contraposition, $\forall x(J(x) \Rightarrow \neg R(x))$. For arbitrary x, $(S(x) \wedge J(x)) \Rightarrow J(x)$; transitivity yields $(S(x) \wedge J(x)) \Rightarrow \neg R(x)$. Universally generalize.

Universal instantiation gives the premise for an arbitrary x; conjunction elimination supplies $J(x)$, and modus ponens supplies $\neg R(x)$. Since x was arbitrary, universal generalization is valid.

L130

Predicate Logic

2025 Fall Final | Question 1(c)(ii) | Source page 3 | Difficulty 3 | 2 marks
Q-2025F-FINAL-P01-c-ii | Answer provenance: official

> (c) [4 points, 2 pts each] Lets the domain, be the students of COMP2711. Consider the following predicates:
>
> $s(x)$: Student x is a senior. $m(x)$: Student x is a mathematics major. $c(x)$: Student x is a comp major $Compare(x, y)$: Student x gets a higher grade than student y
>
> Write each of the following statement in terms of quantifiers, logical operators and the predicates above.
>
> (ii) The highest grade belongs to a comp major student.

Official Answer

Source: physical PDF page 4 (COMP2711_F2025_Final.pdf).

(ii) $\exists y\big(c(y) \land \forall x(x \neq y \rightarrow \neg Compare(x, y))\big)$.

Supplementary Explanation

Answer. $\exists y(c(y) \land \forall x(x \neq y \rightarrow \neg Compare(x, y)))$.

Derivation/proof. First choose a student y with $\exists y$ and require $c(y)$. For each other student x, having a grade higher than y is forbidden. This is $x \neq y \rightarrow \neg Compare(x, y)$, under $\forall x$. Conjoin the major requirement and the universal condition inside the scope of $\exists y$. A student scoring above y falsifies the formula; tied students with nobody scoring higher do not. Thus at least one highest-grade holder is a comp major, without requiring a unique highest grade.

L131

Predicate Logic

2025 Fall Midterm | Question 2(b) | Source page 6 | Difficulty 3 | 4 marks
Q-2025F-MID-P02-b | Answer provenance: official

(b) [4 points] Express the negations of each of these statements so that all negation symbols immediately precede predicates.

$$\forall x\big(p(x) \vee \exists y(p(y) \to \forall z\, q(x,z))\big)$$

Official Answer

Source: physical PDF page 7 (COMP2711_F2025_Midterm.pdf), physical PDF page 8 (COMP2711_F2025_Midterm.pdf).

(b)

$$\neg\forall x(p(x) \vee \exists y(p(y) \to \forall z\, q(x,z))) = \exists x\neg(p(x) \vee \exists y(p(y) \to \forall z\, q(x,z))).$$

Apply De Morgan's law to the disjunction:

$$\neg(p(x) \vee \exists y(p(y) \to \forall z\, q(x,z))) = \neg p(x) \wedge \neg\exists y(p(y) \to \forall z\, q(x,z)).$$

Negate the existential quantifier:

$$\neg\exists y(p(y) \to \forall z\, q(x,z)) = \forall y\neg(p(y) \to \forall z\, q(x,z)).$$

Rewrite the implication $p(y) \to \forall z\, q(x,z) = \neg p(y) \vee \forall z\, q(x,z)$, so:

$$\neg(p(y) \to \forall z\, q(x,z)) = \neg(\neg p(y) \vee \forall z\, q(x,z)) = p(y) \wedge \neg\forall z\, q(x,z).$$

Negate the universal quantifier:

$$\neg\forall z\, q(x,z) = \exists z\neg q(x,z).$$

Thus:

$$\neg(p(y) \to \forall z\, q(x,z)) = p(y) \wedge \exists z\neg q(x,z).$$

So:

$$\neg\exists y(p(y) \to \forall z\, q(x,z)) = \forall y(p(y) \wedge \exists z\neg q(x,z)).$$

Combine:

$$\exists x(\neg p(x) \wedge \forall y(p(y) \wedge \exists z\neg q(x,z))).$$

This expression has all negation symbols immediately preceding predicates ($\neg p(x)$ and $\neg q(x,z)$).

$$\boxed{\exists x(\neg p(x) \wedge \forall y(p(y) \wedge \exists z\neg q(x,z)))}.$$

Supplementary Explanation

Answer. $\exists x(\neg p(x) \wedge \forall y(p(y) \wedge \exists z\neg q(x,z)))$.

Derivation/proof. First negate the outer universal to obtain $\exists x\neg(p(x) \vee \exists y(p(y) \to \forall z q(x,z)))$. De Morgan's law and quantifier negation give $\exists x(\neg p(x) \wedge \forall y\neg(p(y) \to \forall z q(x,z)))$. Use $\neg(A \to B) \equiv A \wedge \neg B$ to get $\exists x(\neg p(x) \wedge \forall y(p(y) \wedge \neg\forall z q(x,z)))$. Finally replace the last negated universal by $\exists z\neg q(x,z)$. The witness z may depend on x and y, and every negation now immediately precedes p or q.

L132

Predicate Logic

2025 Spring Midterm | Question 2(b)(v) | Source page 4 | Difficulty 3
Q-2025S-MID-P02-b-v | Answer provenance: derived

> (b) Express the following statements using the predicates defined above (i.e., you can use mod and gcd) and $prime(x)$, which returns true iff x is prime. You are not allowed to use other predicates. You can also use $\forall$, $\exists$, $\wedge$, $\vee$, $\rightarrow$, $\leftrightarrow$, $\neg$, $+$, $=$, $\neq$, $<$, $>$, multiplication, parentheses, variables, and constants.
>
> (v) For any prime number p, every non-zero element in $\mathbb{Z}_p$ has a multiplicative inverse.

Official Answer

Source: physical PDF page 4 (COMP2711_S2025_Midterm.pdf).

(v) $\forall p(prime(p) \rightarrow \forall x(x < p \rightarrow \exists y(y < p \wedge xy \equiv 1 \pmod{p})))$

Supplementary Explanation

Answer. $\forall p(prime(p) \rightarrow \forall x((0 < x \wedge x < p) \rightarrow \exists y(y < p \wedge xy \equiv 1 \pmod{p})))$.

Derivation/proof. The outer implication restricts the modulus to primes. A nonzero representative must satisfy $0 < x \wedge x < p$ because the natural-number domain already guarantees nonnegativity. Its inverse is expressed by $\exists y(y < p \wedge xy \equiv 1 \pmod{p})$. A product congruent to one excludes $y = 0$ automatically. The printed condition $x < p$ also includes $p = 2, x = 0$, which would require $0y \equiv 1 \pmod{2}$, an impossibility. Thus the additional nonzero restriction is essential.

L133

Predicate Logic

2024 Fall Midterm | Question 2(c) | Source page 5 | Difficulty 3 | 5 marks
Q-2024F-MID-P02-c | Answer provenance: official

> (c) Either prove that the two statements below are logically equivalent or give a counterexample to show that they are not logically equivalent.
>
> $$\exists y[P(y) \rightarrow \forall x Q(x)]$$
>
> $$\exists y P(y) \rightarrow \forall x Q(x)$$

Official Answer

Source: physical PDF page 6 (COMP2711_F2024_Midterm.pdf).

> (c) These two statements are obviously not equivalent. Consider this scenario, Imagine having $\{a, b\}$ in our domain. Assume $P(a) = T, P(b) = F, Q(a) = T, Q(b) = F$. The top statement is True. Because $P(b) = F$ which makes the top statement True. However the bottom statement becomes False.

Supplementary Explanation

Answer. Not equivalent.

Derivation/proof. Use domain $\{a, b\}$ with $P(a) = T, P(b) = F, Q(a) = T, Q(b) = F$. Since $Q(b) = F$, $\forall x Q(x) = F$. For the first expression choose $y = b$, giving $P(b) \rightarrow \forall x Q(x) = F \rightarrow F = T$, so its existential claim is true. In the second, $P(a) = T$ makes $\exists y P(y) = T$, and the implication is $T \rightarrow F = F$. One interpretation gives different truth values, proving non-equivalence.

L134

Predicate Logic

2023 Fall Final | Question 1(a)(ii) | Source page 3 | Difficulty 3 | 1 mark
Q-2023F-FINAL-P01-a-ii | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> (ii) $\forall x \forall a \forall b : \ P(x) \rightarrow (x \mid ab \rightarrow (x \mid a \vee x \mid b))$

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (ii) True

Supplementary Explanation

Answer. True.

Derivation/proof. If $P(x)$ is false, the outer implication is true. Suppose x is prime and $x \mid ab$. If $x \mid a$, the desired disjunction holds. Otherwise $\gcd(x, a) = 1$, so integers u,v satisfy $ux + va = 1$. Multiply by b to get $uxb + vab = b$. Both terms on the left are divisible by x, since $x \mid ab$; therefore $x \mid b$. This proves the assertion for all integer a and b.

When x is prime and $x \nmid a$, Bézout gives $ux + va = 1$. Multiplication gives $uxb + vab = b$, so $x \mid b$ if $x \mid ab$. If x is not prime, or if $x \nmid ab$, the corresponding implication is vacuously true.

L135

Predicate Logic

2023 Fall Final | Question 1(b) | Source page 3 | Difficulty 3 | 2 marks
Q-2023F-FINAL-P01-b | Answer provenance: official

> In this question the domain is the set of all integers. Consider the following predicates:
>
> $P(x)$: x is prime.
>
> $Q(x, y)$: x, y are relatively prime (co-prime).
>
> $E(x, y, z)$: x, y are congruent mod z (i.e., $x \equiv y \pmod{z}$).
>
> (a) Determine whether the following logical expressions are True or False. No justification is needed. ($a \mid b$ means b is divisible by a)
>
> Write down Fermat's little theorem using quantifiers and the predicates given above. You are also allowed to use power. As an example $E(x^2, y, z)$ is acceptable.
>
> (Fermat's little theorem: If p is prime and a is any integer not divisible by p, then $a^{p-1} \equiv 1 \pmod{p}$.)

Official Answer

Source: physical PDF page 3 (COMP2711_F2023_Final.pdf).

> (b) $\forall x\forall y:(P(x)\land Q(y,x))\implies E(y^{x-1},1,x).

Supplementary Explanation

Answer. $\forall p \forall a \big((P(p) \land Q(a, p)) \to E(a^{p-1}, 1, p) \big)$.

Derivation/proof. The quantified variables p and a range over integers. $P(p)$ expresses that p is prime. Under this hypothesis, $p \nmid a$ is equivalent to $\gcd(a, p) = 1$, namely $Q(a, p)$, since a prime has only the positive divisors 1 and p. The predicate $E(a^{p-1}, 1, p)$ encodes exactly the congruence in Fermat's theorem. Connecting the two hypotheses to that conclusion with implication yields the stated quantified translation.

Using p, a instead of x, y, the two necessary hypotheses are $P(p)$ and $Q(a, p)$; the conclusion is $E(a^{p-1}, 1, p)$. For prime p, coprimality is equivalent to $p \nmid a$.

L136

Predicate Logic

2023 Fall Midterm | Question 2(a) | Source page 5 | Difficulty 3 | 4 marks
Q-2023F-MID-P02-a | Answer provenance: official

> Express the negation of the following statement so that all negation operators immediately precede the predicates:
>
> $$\exists x(P(x) \to \forall y(Q(y) \to (R(y) \vee S(y)))).$$

Official Answer

Source: physical PDF page 5 (COMP2711_F2023_Midterm.pdf).

> (a)
>
> $$\begin{aligned}
> &\neg\exists x(P(x) \to \forall y(Q(y) \to (R(y) \vee S(y)))) \\
> &\equiv \forall x\neg(\neg P(x) \vee \forall y(\neg Q(y) \vee (R(y) \vee S(y)))) \\
> &\equiv \forall x(P(x) \wedge \neg\forall y(\neg Q(y) \vee R(y) \vee S(y))) \\
> &\equiv \forall x(P(x) \wedge \exists y\neg(\neg Q(y) \vee R(y) \vee S(y))) \\
> &\equiv \forall x(P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y))) \\
> &\equiv \forall x P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y)).
> \end{aligned}$$

Supplementary Explanation

Answer. $\forall x\big(P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y))\big)$.

Derivation/proof. Negate the whole statement, rather than merely its final disjunction:

$$\begin{aligned}
\neg\exists x(P(x) \to \forall y(Q(y) \to (R(y) \vee S(y)))) &\equiv \forall x(P(x) \wedge \neg\forall y(Q(y) \to (R(y) \vee S(y)))) \\
&\equiv \forall x(P(x) \wedge \exists y(Q(y) \wedge \neg(R(y) \vee S(y)))) \\
&\equiv \forall x(P(x) \wedge \exists y(Q(y) \wedge \neg R(y) \wedge \neg S(y))).
\end{aligned}$$

All remaining negations are immediately before R or S. Keeping the quantifier scopes explicit also avoids any unnecessary domain-dependent extraction of a quantifier.

The quantifier-negation step is $\neg\forall y F(y) \equiv \exists y\neg F(y)$. Keeping the final existential inside its original scope avoids relying on an unstated nonempty-domain convention when extracting it.

L137

Predicate Logic

2023 Fall Midterm | Question 2(b) | Source page 5 | Difficulty 3 | 4 marks
Q-2023F-MID-P02-b | Answer provenance: official

> Let the domain consist of all integers. $CoPrime(a, b)$ is a predicate which means a and b are co-prime (you do not need to know the meaning of 'co-prime' in order to answer this question). Express the following statement using quantifiers:
>
> Two numbers a and b are co-prime if and only if any integer can be expressed as a linear combination of a and b.
>
> (A linear combination is an expression constructed from a set of terms by multiplying each term with a constant and summing the results (e.g., a linear combination of x and y is any expression of the form $ax + by$, where a and b are constants).)
>
> Note: you are only allowed to use quantifiers, addition, multiplication, equality, and the $CoPrime$ predicate.

Official Answer

Source: physical PDF page 5 (COMP2711_F2023_Midterm.pdf).

> (b) $\forall a \forall b (CoPrime(a, b) \iff \forall c \exists x \exists y (c = ax + by))$

Supplementary Explanation

Answer. $\forall a \forall b \big(CoPrime(a, b) \leftrightarrow \forall c \exists x \exists y (c = ax + by)\big)$.

Derivation/proof. The outer $\forall a \forall b$ covers every pair of integers named in the sentence. “If and only if” becomes the biconditional. “Any integer” introduces $\forall c$; “can be expressed” then introduces integer coefficients $\exists x \exists y$. The equation $c = ax + by$ uses only the allowed arithmetic operations and equality. The order permits the coefficients to depend on a, b, and the target c, exactly as required.

The coefficients may depend on a, b, c.

L138

Predicate Logic

2023 Spring Midterm | Question 2(d) | Source page 5 | Difficulty 3 | 3 marks
Q-2023S-MID-P02-d | Answer provenance: official

> **Shared instructions for (a)–(b); editorial grouping:**
>
> Determine the truth value of each of these statements and prove your answer. Let the domain be $\mathbb{R}$.
>
> **Shared instructions for (c)–(d); editorial grouping:**
>
> Are the following statements equivalent? If yes, give a proof. If not, give a counter example specifying two predicates $P(x)$ and $Q(x)$ and a domain that show the statements are not logically equivalent:
>
> (d) $\forall x(\neg P(x) \wedge Q(x))$ and $\neg\forall x\exists y(P(x) \vee \neg Q(y))$

Official Answer

Source: physical PDF page 5 (COMP2711_S2023_Midterm.pdf).

> (d) Not equivalent. Counterexample: $P(x) : x > 0$, $Q(x) : x^2 \geq 0$. L.H.S is false but R.H.S is true.

Supplementary Explanation

Answer. Not equivalent. Take domain $\mathbb{R}$, $P(x) : x > 0$, and $Q(x) : x^2 \geq 0$.

Derivation/proof. On this domain Q is true for every element, whereas $P(1)$ is true and $P(0)$ is false. The left formula is false at $x = 1$. For the right formula, at $x = 0$ the inner expression $P(0) \vee \neg Q(y)$ is false for every y; thus $\forall x\exists y(P(x) \vee \neg Q(y))$ is false and its negation is true. The two formulas therefore disagree under the same interpretation.

Here $P(1)$ is true and $P(0)$ is false. The left and right formulas are false and true, respectively, under one common interpretation.

L139

Predicate Logic

2022 Midterm | Question 2(c) | Source page 1 | Difficulty 3
Q-2022-MID-P02-c | Answer provenance: official

> Let $W(x, y)$ denote the predicate that x has visited website y, where the domain for x consists of people in the world and the domain for y consists of all websites. Express each of the following statements using predicate logic.
>
> There are two distinct people who have visited exactly the same sites.

Official Answer

Source: physical PDF page 1 (2022-midterm.pdf).

> (c) $\exists x \exists y \forall z(((x \neq y) \wedge (W(x, z) \leftrightarrow W(y, z))))$

Supplementary Explanation

Answer. $\exists x \exists x'(x \neq x' \wedge \forall y(W(x, y) \leftrightarrow W(x', y)))$.

Derivation/proof. Quantify two people x,x′ and explicitly require inequality. For each website y, the biconditional says either both visited y or neither did, which makes their visited-site sets identical. A one-way implication would express only subset containment. This formula also correctly allows both visited sets to be empty, since the statement does not require a visit.

The biconditional enforces both directions of containment.

L140

Predicate Logic

2022 Midterm | Question 3 | Source page 2 | Difficulty 3
Q-2022-MID-P03-proof | Answer provenance: official

> Let $P(x), Q(x), R(x), S(x)$ be the following predicates:
>
> $P(x)$: "x is a hummingbird." $Q(x)$: "x is large." $R(x)$: "x lives on honey." $S(x)$: "x is richly colored."
>
> Let the domain be all the birds. First translate the following statements into predicate logic (you may assume that "small" is the same as "not large" and that "dull in color" is the same as "not richly colored"):
>
> (1) "All hummingbirds are richly colored." (2) "No large birds live on honey." (3) "Birds that do not live on honey are dull in color." (4) "Hummingbirds are small."
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules. You don't have to write down the name of the rule used; instead, you can just write down from which statement(s) a new statement is derived.
>
> Let the first three statements be premises and the fourth be the conclusion. Show a step-by-step proof using inference rules.

Official Answer

Source: physical PDF page 2 (2022-midterm.pdf).

(1) $\forall x(P(x) \to S(x))$. (2) $\neg\exists x(Q(x) \land R(x))$. (3) $\forall x(\neg R(x) \to \neg S(x))$. (4) $\forall x(P(x) \to \neg Q(x))$.

Step	Reason
(5) $\forall x(\neg Q(x) \lor \neg R(x))$	(2) equivalence
(6) $\forall x(\neg P(x) \lor S(x))$	(1) equivalence
(7) $\forall x(R(x) \lor \neg S(x))$	(3) equivalence
(8) $R(a) \lor \neg S(a)$ for arbitrary a	Universal instantiation using (7)
(9) $\neg P(a) \lor S(a)$	Universal instantiation using (6)
(10) $\neg P(a) \lor R(a)$	Resolution using (8) and (9)
(11) $\neg Q(a) \lor \neg R(a)$	Universal instantiation using (5)
(12) $\neg P(a) \lor \neg Q(a)$	Resolution using (10) and (11)
(13) $P(a) \to \neg Q(a)$	(12) equivalence
(14) $\forall x(P(x) \to \neg Q(x))$	Universal generalization using (13)

Supplementary Explanation

Answer. $\forall x(P(x) \to \neg Q(x))$ follows.

Derivation/proof. 1. ∀x(P(x)→S(x)) (premise). 2. ∀x(Q(x)→¬R(x)) (premise 2 by equivalence). 3. ∀x(¬R(x)→¬S(x)) (premise). 4. Take arbitrary bird a and assume P(a). 5. S(a) (1,4). 6. R(a) (3 instantiated at a, then contrapositive and 5). 7. ¬Q(a) (2 instantiated, contrapositive with 6). 8. P(a)→¬Q(a) (4–7, discharge assumption). 9. ∀x(P(x)→¬Q(x)) (8, universal generalization). The arbitrary bird a has no extra assumptions after step8.

In the supplementary proof, instantiate (1) at a and use the assumption $P(a)$ to obtain $S(a)$. The next two implications are $S(a) \to R(a)$ and $R(a) \to \neg Q(a)$. After discharging the temporary assumption, apply universal generalization to (8). No undischarged special assumption about a remains.

L141

Predicate Logic

2020 Final Part 1 | Question 1(a)(i) | Source page 1 | Difficulty 3
Q-2020-FINAL1-P01-a-i | Answer provenance: official

> Let G be a simple undirected graph with a set of vertices V. Let V_1 and V_2 be subsets of V so that $V_1 \cup V_2 = V$ and $V_1 \cap V_2 = \varnothing$. Let $E(x, y)$ be the predicate representing that there is an edge from x to y. Note that the graph being undirected means that $\forall u \in V \forall v \in V(E(u, v) \Leftrightarrow E(v, u))$.
>
> (a) Express each of the following properties in predicate logic. Your can only use $V, V_1, V_2, E(\cdot, \cdot)$, logical and mathematical operators.
>
> (ii) "Every edge connects a vertex in V_1 and a vertex in V_2".
>
> (iii) "For every vertex in V_1, there are edges that connect it with all vertices in V_2".
>
> "Every vertex in the graph has degree exactly 1".

Official Answer

Source: physical PDF page 1 (2020-final-part-1.pdf).

$\forall u \in V(\exists v \in V(E(u, v) \wedge \forall w \in V(E(u, w) \rightarrow v = w)))$.

Supplementary Explanation

Answer. $\forall u \in V \exists v \in V(E(u, v) \wedge \forall w \in V(E(u, w) \rightarrow w = v))$.

Derivation/proof. For each vertex u, the existential v supplies an adjacent vertex. The universal w clause says that any neighbor of u equals v, excluding a second neighbor. Together they mean degree exactly one. Simplicity excludes loops and multiple edges, so counting distinct neighbors equals degree.

The existence conjunct excludes isolated vertices, while uniqueness excludes a second distinct neighbour.

After requiring an adjacent vertex v, the clause $\forall w \in V(E(u, w) \rightarrow v = w)$ says that every neighbour equals that same v. Together they express existence and uniqueness, excluding isolated vertices as well as vertices with two neighbours.

L142

Predicate Logic

2020 Midterm | Question 4 | Source page 2 | Difficulty 3 | 10 marks
Q-2020-MID-P04 | Answer provenance: official

> Recall that we can express unique existence as
>
> (1) $\exists x(P(x) \wedge \forall y(P(y) \rightarrow x = y))$.
>
> In many unique existence proofs, instead of proving (1), we prove the following:
>
> (2) $\exists x P(x)$
>
> (3) $\forall x \forall y(P(x) \wedge P(y) \rightarrow x = y)$.
>
> Your task here is to prove (1) from (2) and (3) using the rules of inference for propositional and predicate logic.

Official Answer

Source: physical PDF page 3 (2020-midterm.pdf).

(4) $P(c)$ for some c — (2), existential instantiation (5) $\forall y(P(c) \wedge P(y) \rightarrow c = y)$ — (3), universal instantiation (6) $\forall y(\neg P(c) \vee \neg P(y) \vee c = y)$ — (5), equivalence (7) $\forall y(\neg P(y) \vee c = y)$ — (4), (6), resolution (8) $\forall y(P(y) \rightarrow c = y)$ — (7), equivalence (9) $\forall y(P(c) \wedge (P(y) \rightarrow c = y))$ — (4), (8), conjunction (10) $\exists x \forall y(P(x) \wedge (P(y) \rightarrow x = y))$ — (9), existential generalization (11) $\exists x(P(x) \wedge \forall y(P(y) \rightarrow x = y))$ — (10), null qualification

Supplementary Explanation

Answer. (2) and (3) entail (1).

Derivation/proof. Use existence to choose one fresh witness, and uniqueness to show that every satisfying element equals it.

4. From (2), introduce a fresh witness c with $P(c)$ by existential instantiation.
5. Let y be arbitrary. Universal instantiation of (3) gives $(P(c) \wedge P(y)) \rightarrow c = y$.
6. Temporarily assume $P(y)$.
7. Conjunction from (4) and (6) gives $P(c) \wedge P(y)$; modus ponens with (5) yields $c = y$.
8. Discharge the temporary assumption (6) to obtain $P(y) \rightarrow c = y$ by conditional proof.
9. Since y is arbitrary and is not free in an undischarged assumption, universal generalization yields $\forall y(P(y) \rightarrow c = y)$.
10. Conjoin (4) and (9) to obtain $P(c) \wedge \forall y(P(y) \rightarrow c = y)$.
11. Existential generalization gives $\exists x(P(x) \wedge \forall y(P(y) \rightarrow x = y))$, which is (1). The conclusion has no free occurrence of c, so the fresh-witness subproof can close.

Both premises are essential: uniqueness alone permits no satisfying element, whereas existence alone permits several. The fresh witness and the arbitrary y play different roles, so the generalization in step 9 does not assert that every element satisfies P.

L143

Predicate Logic

2019 Midterm | Question 4 | Source page 6 | Difficulty 3 | 8 marks
Q-2019-MID-P04 | Answer provenance: official

> Consider the following argument:
>
> Premise 1: $\exists x P(x)$.
>
> Premise 2: $\forall x(P(x) \to Q(x))$.
>
> Premise 3: $\forall x(P(x) \to R(x))$.
>
> Conclusion: $\exists x(Q(x) \wedge R(x))$.
>
> Show that the argument is valid. Number your steps and refer to those numbers in the reason you give for each step, but it is not necessary to give the name of the inference rule.

Official Answer

Source: physical PDF page 6 (2019-midterm.pdf).

Step — Reason

1. $\exists x P(x)$ — Hypothesis
2. $P(c)$ — (1), Existential instantiation
3. $\forall x(P(x) \to Q(x))$ — Hypothesis
4. $P(c) \to Q(c)$ — (3), Universal instantiation
5. $\forall x(P(x) \to R(x))$ — Hypothesis
6. $P(c) \to R(c)$ — (5), Universal instantiation
7. $Q(c)$ — (2) and (4), Modus ponens
8. $R(c)$ — (2) and (6), Modus ponens
9. $Q(c) \wedge R(c)$ — (7) and (8), Conjunction
10. $\exists x(Q(x) \wedge R(w))$ — (9), Existential generalization

Supplementary Explanation

Answer. The argument is valid.

Derivation/proof. Use one fresh witness throughout the numbered inference.

1. $\exists x P(x)$ (premise).
2. Let c be a fresh witness with $P(c)$ (1).
3. $\forall x(P(x) \to Q(x))$ (premise).
4. $P(c) \to Q(c)$ (3, instantiate at c).
5. $\forall x(P(x) \to R(x))$ (premise).
6. $P(c) \to R(c)$ (5, instantiate at c).
7. $Q(c)$ (2,4).
8. $R(c)$ (2,6).
9. $Q(c) \wedge R(c)$ (7,8).
10. $\exists x(Q(x) \wedge R(x))$ (9, existential generalization; discharge the fresh witness).

The modus ponens step yielding $Q(c)$ follows from (2) and (4), and the one yielding $R(c)$ follows from (2) and (6). Both conclusions concern the same fresh witness, which is why conjunction and then existential generalization are valid.

L144

Predicate Logic

2019 Midterm | Question 5 | Source page 7 | Difficulty 3 | 6 marks
Q-2019-MID-P05 | Answer provenance: official

> Express the negations of $\forall x(P(x) \land \exists y \forall z(Q(x,y) \to R(y,z)))$ so that all negation symbols immediately precede predicates.

Official Answer

Source: physical PDF page 7 (2019-midterm.pdf).

$$\begin{aligned}
&\neg\forall x(P(x) \land \exists y \forall z(Q(x,y) \to R(y,z))) \\
&\equiv \exists x \neg(P(x) \land \exists y \forall z(Q(x,y) \to R(y,z))) \\
&\equiv \exists x(\neg P(x) \lor \neg\exists y \forall z(Q(x,y) \to R(y,z))) \\
&\equiv \exists x(\neg P(x) \lor \forall y \neg\forall z(Q(x,y) \to R(y,z))) \\
&\equiv \exists x(\neg P(x) \lor \forall y \exists z \neg(Q(x,y) \to R(y,z))) \\
&\equiv \exists x(\neg P(x) \lor \forall y \exists z(Q(x,y) \land \neg R(y,z))).
\end{aligned}$$

Supplementary Explanation

Answer. $\exists x(\neg P(x) \lor \forall y \exists z(Q(x,y) \land \neg R(y,z)))$.

Derivation/proof. Move the outer negation through $\forall x$ to get $\exists x \neg(P(x) \land \exists y \forall z(Q \to R))$. De Morgan gives $\exists x(\neg P(x) \lor \neg\exists y \forall z(Q \to R))$. Negating the inner quantifiers gives $\forall y \exists z \neg(Q \to R)$. Finally $\neg(Q \to R) \equiv Q \land \neg R$. Restoring all arguments yields the answer with negations only before P, R.

For the contrapositive argument, expand the negation of an implication as $\neg(A \to B) \equiv A \land \neg B$. This identifies the precise case ruled out by the proof instead of replacing an implication by its converse.

L145

Predicate Logic

2016 Midterm | Question 6(c) | Source page 10 | Difficulty 3 | 5 marks
Q-2016-MID-P06-c | Answer provenance: official

> Let $q(x)$ be "x is even". Let Z be a set of all integers. Let E be a set of all even integers.
>
> For each of the following pairs, say whether (i) is equivalent to (ii). If they are equivalent, all you have to do is to say that they are equivalent. If they are not equivalent, give a counter example. A counter example should involve a specification of $p(x)$ and an explanation as to why the resulting statement is false.
>
> (c) (i) $\exists x \in E(p(x))$
>
> (ii) $\exists x \in Z(q(x) \Rightarrow p(x))$

Official Answer

Source: PDF physical page 10 (2016-midterm.pdf).

(c) No.

Let $p(x)$ be "x is odd".

"$\exists x \in E(p(x))$" is false but "$\exists x \in Z(q(x) \Rightarrow p(x))$" is true (because when $x = 1$ (where $x \in Z$), $q(x)$ is false and $p(x)$ is true (which means that "$q(x) \Rightarrow p(x)$" is true)).

Supplementary Explanation

Answer. Not equivalent; take $p(x)$ to mean "x is odd."

Derivation/proof. No even integer is odd, so (i) is false. With witness $x = 1$, $q(1)$ is false and $p(1)$ true, making $q(1) \to p(1)$ true. Thus (ii) is true, disproving equivalence.

At $x = 1$, the implication is $F \to T = T$.

L146

Predicate Logic

2015 Midterm | Question 1(c) | Source page 3 | Difficulty 3 | 4 marks
Q-2015-MID-P01-c | Answer provenance: official

> For each of the following parts, circle the correct answer or answers. There is no need to explain how your answers are derived.
>
> (c) Let U be a domain (a.k.a. a universe), and $P(x)$ and $Q(x)$ be two predicates. Which of the following statements is/are equivalent to $\forall x \in U(P(x) \vee Q(x))$?
>
> 1. $\exists x \in U(\neg P(x) \wedge \neg Q(x))$
> 2. $\forall y \in U(\neg P(y) \rightarrow Q(y))$
> 3. $\neg(\forall x \in U(\neg P(x)) \vee \forall x \in U(\neg Q(x)))$
> 4. $\neg(\exists x \in U(\neg P(x)) \wedge \exists x \in U(\neg Q(x)))$
> 5. None of above.

Official Answer

Source: PDF physical page 4 (2015-midterm.pdf).

(c) 2

Supplementary Explanation

Answer. 2.

Derivation/proof. For each y, $\neg P(y) \rightarrow Q(y) \equiv P(y) \vee Q(y)$, proving option 2. Option 1 is the target's negation. Option 3 is $(\exists x P(x)) \wedge (\exists x Q(x))$; with $U = \{a, b\}$ and both predicates true only at a, it is true but the target fails at b. Option 4 equals $(\forall x P(x)) \vee (\forall x Q(x))$; let P hold only at a and Q only at b. Then the target is true but option 4 false.

For option 1, a one-element domain where P is true makes the target true and option 1 false.

L147

Predicate Logic

2015 Midterm | Question 1(d) | Source page 3 | Difficulty 3 | 4 marks
Q-2015-MID-P01-d | Answer provenance: official

> For each of the following parts, circle the correct answer or answers. There is no need to explain how your answers are derived.
>
> (d) Let U_1 and U_2 be two domains, and let $P(x)$ and $R(x, y)$ be two predicates. Which of the following statements logically implies/imply $\exists x \in U_2 \forall y \in U_1 (P(x) \to R(x, y))$?
>
> 1. $\forall y \in U_1 \exists x \in U_2 (P(x) \to R(x, y))$
> 2. $\exists x \in U_2 \forall y \in U_1 (P(x))$
> 3. $\exists x \in U_2 \forall y \in U_1 (R(x, y))$
> 4. $\forall x \in U_2 \forall y \in U_1 (P(x) \to R(x, y))$
> 5. None of above.

Official Answer

Source: PDF physical page 4 (2015-midterm.pdf).

(d) 3,4

Supplementary Explanation

Answer. 3, 4, under the usual nonempty-domain convention used by the official answer.

Derivation/proof. Option 3 supplies a witness for which every consequent is true, making all implications true. For option 4 choose any $x \in U_2$; universal instantiation gives the target. To refute option 1 take $U_1 = U_2 = \{0, 1\}$, P always true, and $R(x, y)$ meaning $x = y$: a witness exists for each y, but no common witness exists. For option 2 take singleton domains, P true, and R false. If empty U_2 were permitted, option 4 would need that extra nonemptiness assumption.

L148

Predicate Logic

2015 Midterm | Question 3 | Source page 7 | Difficulty 3
Q-2015-MID-P03-equivalence | Answer provenance: official

> Consider the following three statements:
>
> (a) A rich and famous person is a successful person.
>
> (b) A rich person is successful if she/he is famous.
>
> (c) A famous person is successful if she/he is rich.
>
> Convert the statements into logic statements using the domain U (set of all persons), and predicates $rich(x)$ (person x is rich), $famous(x)$ (person x is famous), $successful(x)$ (person x is successful).
>
> Are the three statements logically equivalent? If not, explain why. If yes, give a proof. Show all steps of your proof, and state clearly the assumption or the inference rule used in each step.
>
> Are the three statements logically equivalent? If not, explain why. If yes, give a proof. Show all steps of your proof, and state clearly the assumption or the inference rule used in each step.

Official Answer

Source: PDF physical page 8 (2015-midterm.pdf).

The three statements are equivalent because they are all equivalent to the same fourth statement, as shown below:

$$\begin{aligned}
&\forall x \in U((rich(x) \land famous(x)) \to successful(x)) \\
&\equiv \forall x \in U(\neg(rich(x) \land famous(x)) \lor successful(x)) && p \to q \equiv \neg p \lor q \\
&\equiv \forall x \in U(\neg rich(x) \lor \neg famous(x) \lor successful(x)) && \text{DeMorgan's \& Associative Law}
\end{aligned}$$

$$\begin{aligned}
&\forall x \in U(rich(x) \to (famous(x) \to successful(x))) \\
&\equiv \forall x \in U(rich(x) \to (\neg famous(x) \lor successful(x))) && p \to q \equiv \neg p \lor q \\
&\equiv \forall x \in U(\neg rich(x) \lor (\neg famous(x) \lor successful(x))) && p \to q \equiv \neg p \lor q \\
&\equiv \forall x \in U(\neg rich(x) \lor \neg famous(x) \lor successful(x)) && \text{Associative Law}
\end{aligned}$$

$$\begin{aligned}
&\forall x \in U(famous(x) \to (rich(x) \to successful(x))) \\
&\equiv \forall x \in U(famous(x) \to (\neg rich(x) \lor successful(x))) && p \to q \equiv \neg p \lor q \\
&\equiv \forall x \in U(\neg famous(x) \lor (\neg rich(x) \lor successful(x))) && p \to q \equiv \neg p \lor q \\
&\equiv \forall x \in U(\neg rich(x) \lor \neg famous(x) \lor successful(x)) && \text{Associative \& Commutative Law}
\end{aligned}$$

Supplementary Explanation

Answer. Yes; all three statements are logically equivalent.

Derivation/proof. For fixed x, abbreviate the predicates by R, F, S.

$(R \land F) \to S \equiv \neg(R \land F) \lor S \equiv \neg R \lor \neg F \lor S$ (implication, De Morgan, associativity).

$R \to (F \to S) \equiv R \to (\neg F \lor S) \equiv \neg R \lor \neg F \lor S$ (implication twice, associativity).

$F \to (R \to S) \equiv F \to (\neg R \lor S) \equiv \neg F \lor \neg R \lor S \equiv \neg R \lor \neg F \lor S$ (implication twice, commutativity).

Each identity holds pointwise for every x, so putting the same $\forall x \in U$ around each preserves equivalence.

L149

Predicate Logic

2014 Final | Question 5(b) | Source page 9 | Difficulty 3
Q-2014-FINAL-P05-b | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (b) $\exists x \in U\ p(x) \Rightarrow \exists x \in U\ q(x)$ and $\exists x \in U (p(x) \Rightarrow q(x))$

Official Answer

Source: PDF physical page 9 (2014-final.pdf).

(b) Not equivalent. Counterexample:

- U: All animals; $p(x)$: x can fly; $q(x)$: x can fly airplane.
- LHS is false because there are animals that can fly, but no animal can fly airplane.
- RHS is true because there are animals that cannot fly and hence $p(x) \Rightarrow q(x)$ is true.

Supplementary Explanation

Answer. No.

Derivation/proof. Let U be animals, $p(x)$ mean 'x can fly', and $q(x)$ mean 'x can fly an airplane'. The left implication is false (some animal flies but none flies an airplane), while the right is true because any non-flying animal makes $p(x) \Rightarrow q(x)$ true.

To remove ambiguity in the informal animal example, take $U = \{u, v\}$, $p(u) = T, p(v) = F$, and $q(u) = q(v) = F$. On the left, $\exists x\, p(x)$ is true and $\exists x\, q(x)$ is false. On the right, v witnesses $p(v) \Rightarrow q(v) = F \Rightarrow F = T$.

L150

Predicate Logic

2014 Final | Question 5(c) | Source page 9 | Difficulty 3
Q-2014-FINAL-P05-c | Answer provenance: official

> For each of the following pairs of logic statements, either prove that the two statements are logically equivalent, or give a counterexample. In your proof, you may use either a truth table or logic laws. A counterexample should consist of a truth setting of the variables and the truth values of the statements under the setting.
>
> (c) $\exists x \in U\ p(x) \Rightarrow \forall y \in V\ q(y)$ and $\forall x \in U \forall y \in V (p(x) \Rightarrow q(y))$

Official Answer

Source: PDF physical page 9 (2014-final.pdf).

(c) Equivalent.

$$\begin{aligned} \exists x \in U\, p(x) \Rightarrow \forall y \in V\, q(y) &\equiv \neg(\exists x \in U\, p(x)) \vee (\forall y \in V\, q(y)) \\ &\equiv (\forall x \in U\, \neg p(x)) \vee (\forall y \in V\, q(y)) \\ &\equiv \forall x \in U \forall y \in V (\neg p(x) \vee q(y)) \\ &\equiv \forall x \in U \forall y \in V (p(x) \Rightarrow q(y)) \end{aligned}$$

Supplementary Explanation

Answer. Yes.

Derivation/proof. The left formula is $\neg\exists x, p(x) \vee \forall y, q(y) = \forall x \neg p(x) \vee \forall y q(y)$. Distributing universal quantifiers over this two-variable disjunction gives $\forall x \forall y (\neg p(x) \vee q(y))$, the right formula.

Conversely, if the right side holds and $p(u)$ holds for some u, instantiate at (u, y) for arbitrary y to derive $q(y)$. If there is no such u, the left antecedent is false. These cases also cover empty domains.

If no object satisfies p, every implication on the right has a false antecedent. If some object satisfies p, the left side forces $q(y)$ for every y, again proving the right side.

L151

Inference and Proofs

2019 Midterm | Question 6(b) | Source page 8 | Difficulty 1 | 3 marks
Q-2019-MID-P06-b | Answer provenance: official

> (b) For all integers a, b, if $a \mid b$ and $b \mid a$, then $a = b$.

Official Answer

Source: physical PDF page 8 (2019-midterm.pdf).

False. Set $a = 1$ and $b = -1$.

Supplementary Explanation

Answer. False; $a = 1, b = -1$.

Derivation/proof. Here $-1 = 1 \cdot (-1)$, so $1 \mid -1$. Also $1 = (-1)(-1)$, so $-1 \mid 1$. Yet $1 \neq -1$. The hypotheses determine absolute values for nonzero inputs, not their signs, and the integer domain allows this counterexample.

Mutual divisibility forces equal absolute values, not necessarily equal signs.

The example $1 = (-1) \cdot (-1)$ shows why a product equal to one need not have both integer factors equal to positive one; the negative unit is the missing case.

L152

Inference and Proofs

2017 Midterm | Question 4(a) | Source page 6 | Difficulty 1
Q-2017-MID-P04-a | Answer provenance: official

> Prove or disprove the following statements. Assume that a, b, c, d, and m are integers and $m > 1$.
>
> (a) If $a \equiv b \pmod{m}$ and $a \equiv c \pmod{m}$, then $a \equiv b + c \pmod{m}$.

Official Answer

Source: physical PDF page 6 (2017-midterm.pdf).

False. Set $a = 1, b = 3, c = 5, m = 2$ as a counter example. We have $1 \equiv 3 \pmod{2}$, $1 \equiv 5 \pmod{2}$, but $1 \not\equiv 5 \pmod{2}$.

Supplementary Explanation

Answer. False; $a = 1, b = 3, c = 5, m = 2$ is a counterexample.

Derivation/proof. Here $1 \equiv 3 \equiv 5 \pmod{2}$, so both hypotheses hold. But $b + c = 8 \equiv 0 \pmod{2}$, while $a \equiv 1 \pmod{2}$. The two residues differ, so the conclusion fails. In general the hypotheses give $b + c \equiv 2a$, not a.

L153

Inference and Proofs

2017 Midterm | Question 4(b) | Source page 6 | Difficulty 1
Q-2017-MID-P04-b | Answer provenance: official

> Prove or disprove the following statements. Assume that a, b, c, d, and m are integers and $m > 1$.
>
> (b) If $a \equiv b \pmod{m}$, then $2a \equiv 2b \pmod{2m}$.

Official Answer

Source: physical PDF page 6 (2017-midterm.pdf).

True. $a \equiv b \pmod{m}$ implies $m \mid a - b$, i.e., $a - b = km$ for some integer k. Multiplying both sides by 2 gives $2a - 2b = 2mk$. Therefore $2a \equiv 2b \pmod{2m}$.

Supplementary Explanation

Answer. True.

Derivation/proof. The hypothesis means $a - b = km$ for some integer k. Multiplying the equality by two gives $2a - 2b = k(2m)$. Thus $2m$ divides $2a - 2b$, exactly the definition of the required congruence.

Starting from $m \mid (a - b)$, multiplication by two yields $2m \mid (2a - 2b)$, which is precisely the congruence modulo $2m$. Conversely, cancel two from the integer multiple identity to recover the original divisibility.

L154

Inference and Proofs

2025 Fall Final | Question 1(b) | Source page 3 | Difficulty 2 | 3 marks
Q-2025F-FINAL-P01-b | Answer provenance: official

> (b) [3 points] Let p, q, r, s, t denote five propositions. You are given the following premises:
>
> 1. $p \wedge q$
> 2. $p \rightarrow (r \wedge q)$
> 3. $r \rightarrow (s \vee t)$
> 4. $\neg s$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion "t" is true.

Official Answer

Source: physical PDF page 4 (COMP2711_F2025_Final.pdf).

(b)

1. $p \wedge q$ (given)
2. p (1, simplification)
3. $p \rightarrow (r \wedge q)$ (given)
4. $r \wedge q$ (2, 3, modus ponens)
5. r (4, simplification)
6. $r \rightarrow (s \vee t)$ (given)
7. $s \vee t$ (5, 6, modus ponens)
8. $\neg s$ (given)
9. t (7, 8, disjunctive syllogism)

Thus, the premises logically imply t.

Supplementary Explanation

Answer. t.

Derivation/proof.

1. Simplification of premise 1, $p \wedge q$, gives p.
2. Modus ponens with premise 2 gives $r \wedge q$.
3. Simplification of step 2 gives r.
4. Modus ponens with premise 3 gives $s \vee t$.
5. Premise 4 is $\neg s$. Combining it with step 4 by disjunctive syllogism gives t.

Every step uses only a given premise or an earlier result. Thus, when all premises are true, the conclusion must be true.

L155

Inference and Proofs

2023 Fall Midterm | Question 3(a) | Source page 7 | Difficulty 2 | 4 marks
Q-2023F-MID-P03-a | Answer provenance: official

> Let S and T denote two sets. Prove the following by contradiction:
>
> If $S \cap T = \varnothing$ and $S \cup T = T$ then $S = \varnothing$.

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Midterm.pdf).

> (a) Assume that S is not an empty set. This implies that $\exists x$ s.t. $x \in S$. If $x \in S$, then $x \in S \cup T$ or $x \in T$ (because $S \cup T = T$). However, $x \in S \cap T$ contradicts with the fact that $S \cap T = \varnothing$. Consequently, we must have $S = \varnothing$.

Supplementary Explanation

Answer. $S = \varnothing$.

Derivation/proof. Assume $S \cap T = \varnothing$ and $S \cup T = T$, but suppose $S \neq \varnothing$. Then there is an element $x \in S$. It follows that $x \in S \cup T$, and the union equality implies $x \in T$. Hence $x \in S \cap T$, contradicting the premise that this intersection is empty. The assumption $S \neq \varnothing$ must therefore be false, proving the conclusion by contradiction.

L156

Inference and Proofs

2022 Midterm | Question 6(a) | Source page 3 | Difficulty 2 | 8 marks
Q-2022-MID-P06-a | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that $p \nmid n$.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (a) We use proof by contradiction. Suppose $n = kp$ for some integer k. Then $4n^2 + 1 = 4(kp)^2 + 1 = 4k^2p^2 + 1 = pk'$ for some integer k', i.e., $p(k' - 4k^2p) = 1$, which is impossible for prime p.

Supplementary Explanation

Answer. p does not divide n.

Derivation/proof. Assume p|n. Then n=pt for an integer t, so p divides $4n^2=4p^2t^2$. By hypothesis p also divides $4n^2+1$. Subtracting, p divides1, which is impossible because a prime p≥2. Thus the assumption fails and p∤n. This directly proves the coprimality needed in the following Fermat argument.

Writing $n = pt$ gives $p \mid (4n^2 + 1) - 4n^2 = 1$, impossible, so $p \nmid n$.

L157

Inference and Proofs

2020 Midterm | Question 8 | Source page 4 | Difficulty 2 | 10 marks
Q-2020-MID-P08 | Answer provenance: official

> Show that $\gcd(21n+4, 14n+3) = 1$, for any $n \in \mathbb{N}$.

Official Answer

Source: physical PDF page 4 (2020-midterm.pdf).

Using the Euclid's algorithm, we have $\gcd(21n+4, 14n+3) = \gcd(14n+3, 7n+1) = \gcd(7n+1, 1) = 1$.

Supplementary Explanation

Answer. The gcd equals 1 for every permitted n.

Derivation/proof. Subtract the second argument from the first: $(21n+4) - (14n+3) = 7n+1$. Subtract twice this remainder from the second: $(14n+3) - 2(7n+1) = 1$. Any common divisor therefore divides 1, so it is 1. Equivalently $3(14n+3) - 2(21n+4) = 1$ is a Bezout certificate valid for every integer n, including the requested naturals.

The independent Bezout certificate is $3(14n+3) - 2(21n+4) = 1$, also valid at $n = 0$.

The Euclidean reductions end in $\gcd(7n+1, 1) = 1$. This final gcd is one for every allowed n, so no exceptional natural-number case remains.

L158

Inference and Proofs

2019 Midterm | Question 6(a) | Source page 8 | Difficulty 2 | 3 marks
Q-2019-MID-P06-a | Answer provenance: official

> (a) For all integers a, b, c, if $a \mid c$ and $b \mid c$, then $ab \mid c^2$.

Official Answer

Source: physical PDF page 8 (2019-midterm.pdf).

True. If $c = ak$ and $c = bl$, then $c^2 = ab(kl)$, so $ab \mid c^2$.

Supplementary Explanation

Answer. True.

Derivation/proof. The premises give integers k, l with $c = ak$ and $c = bl$. Multiplying these equalities produces $c^2 = (ak)(bl) = ab(kl)$. Since kl is an integer, this is exactly the divisibility witness for $ab \mid c^2$. No coprimality assumption is required.

No division is used, so zero values are also covered under the divisibility convention permitting $0 \mid 0$.

There is no division by a possibly zero integer in the product argument: expand the integer factors directly. Thus the reasoning remains valid even when one of the integers is zero.

L159

Inference and Proofs

2019 Midterm | Question 8 | Source page 10 | Difficulty 2 | 8 marks
Q-2019-MID-P08 | Answer provenance: official

> Prove the following statement by contradiction.
>
> "For any non-zero real number x, if $x + \frac{1}{x} < 2$, then $x < 0$".

Official Answer

Source: physical PDF page 10 (2019-midterm.pdf).

Assume $x + 1/x < 2$ and $x > 0$. We have

$$\begin{aligned} x + \frac{1}{x} < 2 &\to \frac{x^2+1}{x} < 2 \\ &\to x^2 + 1 < 2x \quad \text{since } x > 0 \\ &\to x^2 - 2x + 1 < 0 \\ &\to (x-1)^2 < 0. \end{aligned}$$

This contradicts with the fact that $(x-1)^2 \geq 0$, completing the proof.

Supplementary Explanation

Answer. The implication is true.

Derivation/proof. Assume $x \neq 0$ and $x + 1/x < 2$, but deny the conclusion: $x \geq 0$. Nonzero then gives $x > 0$. Multiplication by this positive x yields $x^2 + 1 < 2x$, so $(x-1)^2 = x^2 - 2x + 1 < 0$. A real square is nonnegative, a contradiction. Therefore $x < 0$.

L160

Inference and Proofs

2018 Final | Question 5(a) | Source page 7 | Difficulty 2
Q-2018-FINAL-P05-a | Answer provenance: official

> (a) For what values of n will a complete graph K_n be bipartite? Explain your answers.

Official Answer

Source: physical PDF page 7 (2018-final.pdf).

Only when $n = 2$. If $n \geq 3$, then consider any vertex colored blue; all of its neighbors will be colored red, and they are neighbors of each other, violating two-coloring. (We do not penalize saying $n = 1$ is bipartite because it is an unclear edge case.)

Supplementary Explanation

Answer. $n = 1$ or $n = 2$, for $n \geq 1$.

Derivation/proof. For K_1, put the lone vertex in one part and let the other part be empty; no edge violates bipartition. For K_2, separate its endpoints. For $n \geq 3$, three vertices form a triangle. If one is blue, both neighbors must be red, but their mutual edge then violates two-coloring. Hence there are no other cases.

The small cases $n = 1, 2$ are both bipartite: a one-vertex graph has the other part empty, and the two-vertex graph places its endpoints in different parts. Here the grading note explicitly accepts the singleton case, which must not be lost when copying the printed main answer.

L161

Inference and Proofs

2018 Final | Question 6(a) | Source page 8 | Difficulty 2
Q-2018-FINAL-P06-a | Answer provenance: official

> In this question, we consider simple graphs $G = (V, E)$ with $|V| \geq 3$ where all vertices satisfy $deg(v) \geq 2$. Let us call them two-dense graphs.
>
> Consider the following statement:
>
> All two-dense graphs have a triangle subgraph (a cycle of length 3).
>
> Here is an inductive proof of the statement.
>
> Proof. Let the proposition $P(n)$ be the above statement for $|V| = n$. We will prove $P(n)$ is true for $n \geq 3$.
>
> Base case ($P(3)$): For the two-dense graph G with three vertices, each vertex has a degree of 2, so G is simply a triangle and has itself as a subgraph.
>
> Inductive step: Suppose $P(k)$ is true, i.e. any two-dense graph with k vertices has a triangle subgraph. Extend any such graph G to obtain $G' = (V', E')$ by adding one vertex and at least two new edges from the new vertex to any different vertices in $|V|$. G' is a simple graph with $k + 1$ vertices and all vertices satisfy $deg(v) \geq 2$, so G' is two-dense. G' contains G as a subgraph, so using the inductive assumption, G' also has a triangle subgraph. Therefore, $P(k + 1)$ is true.
>
> (a) Give a counterexample to show that the statement is wrong.

Official Answer

Source: physical PDF page 9 (2018-final.pdf).

Any cycle C_n with $n \geq 4$.

Supplementary Explanation

Answer. C_4 is a counterexample.

Derivation/proof. Take vertices $1, 2, 3, 4$ and edges $12, 23, 34, 41$. This graph is simple, has four vertices, and each vertex has degree two, satisfying the two-dense definition. Any three selected vertices miss at least one edge because there are no diagonals, so there is no triangle. Thus every hypothesis holds while the conclusion fails.

The example lacks edges 13 or 24; every vertex has degree two in its four-cycle. Degree at least two therefore does not force completeness, since nonadjacent pairs remain.

L162

Inference and Proofs

2018 Midterm | Question 3(b) | Source page 5 | Difficulty 2 | 3 marks
Q-2018-MID-P03-b | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (b) For any subset P satisfying the above criterion, we can write every $p_i \in P$ as $c_i \cdot 2^{k_i}$, where c_i is odd. Prove that the following is true: $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow c_i \neq c_j)$.

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

WLOG consider $p_i < p_j \in P$. If $i \neq j$, then $p_i \nmid p_j$. However, if $c_i = c_j$, then $p_i = c_i \cdot 2^{k_i} \mid c_i \cdot 2^{k_i + (k_j - k_i)} = c_i \cdot 2^{k_j} = p_j$. This is a contradiction, so $c_i \neq c_j$. (3 marks)

Supplementary Explanation

Answer. Distinct elements of P have distinct odd parts.

Derivation/proof. Suppose $c_i = c_j = c$. Since the elements differ, their exponents differ; assume $k_i < k_j$. Then $p_j = c2^{k_j} = p_i 2^{k_j - k_i}$, so $p_i \mid p_j$, contradicting the defining criterion. Thus equal odd parts are impossible.

The thin vertical stroke in the printed chain is divisibility $\mid$, not equality.

L163

Inference and Proofs

2018 Midterm | Question 4(a) | Source page 6 | Difficulty 2
Q-2018-MID-P04-a-proof | Answer provenance: official

> Consider finite sets S and T, and a function $f : S \to T$.
>
> "If f is injective, then $|T| \geq |S|$."
>
> Then, prove the contrapositive directly.

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

The contrapositive is "If $|T| < |S|$, then f is not injective". We can prove it by the pigeonhole principle. f maps pigeons in S to holes in T. Since $|T| < |S|$, there must be two pigeons that share a same hole, say $a, b \in S$. Then $a \neq b$ but $f(a) = f(b)$, so f is not injective.

Supplementary Explanation

Answer. If $|T| < |S|$, the function f is not injective.

Derivation/proof. Assume $|S| > |T|$. Regard the elements of S as pigeons and the elements of T as holes, sending s to hole $f(s)$. By pigeonhole principle, two distinct elements $a, b \in S$ share an image: $f(a) = f(b)$. This is precisely failure of injectivity.

To show failure of injectivity, the two inputs must satisfy $a \neq b$ while their outputs agree; equality of outputs for the same input would prove nothing.

L164

Inference and Proofs

2018 Midterm | Question 4(b) | Source page 6 | Difficulty 2
Q-2018-MID-P04-b | Answer provenance: official

> (b) For functions on finite sets $f : S \to T, g : T \to U$, if f is injective, and g is surjective, is it true that $g \circ f$ must be injective? Prove it.

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

False. Students can provide any counterexample such that f is injective, g is surjective, and $g \circ f$ is not injective. For example: $S = T = \{a, b\}, U = \{0\}, f(x) = x, g(y) = 0$, then $g(f(x)) = 0$. It is not injective because $g(f(a)) = g(f(b)) = 0$.

Supplementary Explanation

Answer. No. Let $S = T = \{a, b\}, U = \{0\}$, f be the identity and g be constant zero.

Derivation/proof. The identity sends distinct inputs to distinct elements, so f is injective. The image of g is all of the singleton U, so g is surjective. Yet $g(f(a)) = 0 = g(f(b))$ with $a \neq b$, hence the composite is not injective. Every required premise is satisfied by this counterexample.

L165

Inference and Proofs

2018 Midterm | Question 5(b) | Source page 7 | Difficulty 2
Q-2018-MID-P05-b | Answer provenance: official

> (b) Let x, y be positive integers and p be a positive prime integer. Show that, if $p \mid xy$, then $p \mid x$ or $p \mid y$. (Note: You may use the result from (a) even if you did not solve that part.)

Official Answer

Source: physical PDF page 7 (2018-midterm.pdf).

Since p is prime, it follows that $\gcd(p, x)$ is either p or 1, because these are the only divisors of p. If $\gcd(p, x) = p$, then x is a multiple of p, that is, $p \mid x$. Else, $\gcd(p, x) = 1$, that is, x, p are relatively prime. In this case, using (a), we get that $p \mid y$.

Supplementary Explanation

Answer. At least one of x, y is divisible by p.

Derivation/proof. If $p \mid x$, the conclusion already holds. Otherwise, because the only positive divisors of p are $1, p$, we have $\gcd(p, x) = 1$. Applying part (a) to $p \mid xy$ with the coprime factor x gives $p \mid y$. These two cases exhaust all possibilities.

Primality is essential to the coprime case; the conclusion does not extend to arbitrary composite divisors.

Because p is prime, $p \nmid x$ forces $\gcd(p, x) = 1$. The preceding coprime divisor result then transfers the divisibility of the product to its other factor.

L166

Inference and Proofs

2017 Final | Question 13(a) | Source page 17 | Difficulty 2 | 2 marks
Q-2017-FINAL-P13-a | Answer provenance: derived

> In class we learned about several special types of graphs: complete graphs K_n, cycles C_n, and complete bipartite graphs $K_{m,n}$. Recall the definitions:
>
> K_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 1$), there is exactly one edge between every pair of vertices in V. K_1 is a single vertex and K_2 is two vertices connected by an edge.
>
> C_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 3$), there is exactly one edge between v_i and v_{i+1} for all $1 \leq i \leq n$, plus exactly one edge from v_n to v_1.
>
> $K_{m,n}$: For every vertex $u_i \in U = \{u_1, u_2, \ldots, u_m\}$ and $v_j \in V = \{v_1, v_2, \ldots, v_n\}$ ($m \geq 1, n \geq 1$), there is exactly one edge connecting u_i and v_j. There are no edges between two vertices in U, and no edges between two vertices in V.
>
> (a) Can a complete graph K_n be bipartite? Explain what conditions n must satisfy if it is possible.

Official Answer

Source: physical PDF page 17 (2017-final.pdf).

Yes, only if $n = 2$: the graph with two vertices and one edge. Consider the colouring method to determine if a graph is bipartite. If $n > 2$, then if we mark a vertex as red, all other vertices will be marked as blue, and they will be neighbours of each other, so the colouring contradicts the requirement for bipartite graphs.

Supplementary Explanation

Answer. Yes, exactly for $n = 1, 2$ under the standard bipartite definition.

Derivation/proof. For $n = 1$, put the sole vertex in one part and leave the other empty; there are no edges violating the condition. For $n = 2$, put the endpoints in opposite parts. If $n \geq 3$, any three vertices form a triangle. Starting with one color, its two neighbors must both receive the other color, but they are adjacent, which is impossible. A direct enumeration of the small cases agrees.

For the boundary graph K_1, take the unique vertex in one bipartition and the other part empty. It is bipartite despite not belonging to the printed two-vertex-only description.

L167

Inference and Proofs

2017 Final | Question 13(b) | Source page 17 | Difficulty 2 | 2 marks
Q-2017-FINAL-P13-b | Answer provenance: official

> In class we learned about several special types of graphs: complete graphs K_n, cycles C_n, and complete bipartite graphs $K_{m,n}$. Recall the definitions:
>
> K_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 1$), there is exactly one edge between every pair of vertices in V. K_1 is a single vertex and K_2 is two vertices connected by an edge.
>
> C_n: For $V = \{v_1, v_2, \ldots, v_n\}$ ($n \geq 3$), there is exactly one edge between v_i and v_{i+1} for all $1 \leq i \leq n$, plus exactly one edge from v_n to v_1.
>
> $K_{m,n}$: For every vertex $u_i \in U = \{u_1, u_2, \ldots, u_m\}$ and $v_j \in V = \{v_1, v_2, \ldots, v_n\}$ ($m \geq 1, n \geq 1$), there is exactly one edge connecting u_i and v_j. There are no edges between two vertices in U, and no edges between two vertices in V.
>
> (b) Can a cycle C_n be bipartite? Explain what conditions n must satisfy if it is possible.

Official Answer

Source: physical PDF page 17 (2017-final.pdf).

Yes, when n is even. Using the colouring method, if vertex 1 is red, then 2 is blue, 3 is red, and so on. Vertex n is blue if n is even, in which case there is no contradiction (all red vertices have blue neighbours, all blue vertices have red neighbours). Vertex n is red if n is odd, but it is a neighbour of vertex 1, so there is a contradiction; no colouring is possible.

Supplementary Explanation

Answer. Exactly when n is even ($n \geq 4$ in this family).

Derivation/proof. Color v_1 red and alternate along $v_1 v_2 \cdots v_n$. If n is even, v_n is blue, so its closing edge to red v_1 is valid and every edge crosses colors. If n is odd, v_n is red, making the closing edge monochromatic. The alternating requirement is forced by every preceding edge, so no other two-coloring can rescue an odd cycle.

Color the cycle alternately: odd-numbered vertices are red and even-numbered vertices are blue. The closing edge has differently colored ends exactly when the length is even.

L168

Inference and Proofs

2017 Midterm | Question 3(a) | Source page 5 | Difficulty 2
Q-2017-MID-P03-a | Answer provenance: official

> Assuming that any number that is not rational is an irrational number, prove the following:
>
> (a) The sum of any irrational number and any rational number must be an irrational number.

Official Answer

Source: physical PDF page 5 (2017-midterm.pdf).

We use a proof by contradiction: assume that the sum of some rational number x and some irrational number y is a rational number z. $x = \frac{a}{b}$, and $z = \frac{c}{d}$ for some integers c and d. We have $y = z - x = \frac{c}{d} - \frac{a}{b} = \frac{cb-ad}{db}$. Since $cb - ad$ and db are integers, this contradicts with the assumption that y is an irrational number.

Supplementary Explanation

Answer. The sum is irrational.

Derivation/proof. Let y be irrational and $x = a/b$ rational, where $a, b \in \mathbb{Z}$ and $b \neq 0$. If $x + y = c/d$ were rational with $d \neq 0$, then $y = c/d - a/b = (bc - ad)/(bd)$ would be rational, since its numerator and nonzero denominator are integers. This contradicts the choice of y.

L169

Inference and Proofs

2017 Midterm | Question 3(b) | Source page 5 | Difficulty 2
Q-2017-MID-P03-b | Answer provenance: official

> Assuming that any number that is not rational is an irrational number, prove the following:
>
> (b) If x^2 is irrational, then x is irrational.

Official Answer

Source: physical PDF page 5 (2017-midterm.pdf).

We use a proof by contraposition to show that if x is rational, then x^2 is rational. Since x is rational, let $x = \frac{a}{b}$ for some integers a and b. $x = (\sqrt{x})^2 = (\frac{a}{b})^2 = \frac{a^2}{b^2}$; both a^2 and b^2 are integers. Thus x^2 is rational.

Supplementary Explanation

Answer. The implication is true.

Derivation/proof. Prove its contrapositive. If x is rational, write $x = a/b$ for integers a, b with $b \neq 0$. Then $x^2 = a^2/b^2$ is rational, because a^2, b^2 are integers and $b^2 \neq 0$. Hence an irrational square cannot have a rational base.

L170

Inference and Proofs

2015 Midterm | Question 2(a) | Source page 5 | Difficulty 2 | 4 marks
Q-2015-MID-P02-a | Answer provenance: official

> In this question, p, q, r and s are logic propositions. In each part, a set of premises and a conclusion are given. You are asked to derive the conclusion from the premises using the rules of inference that you have learned in class. Show all steps of your derivations, and state clearly the assumption or the inference rule used in each step.
>
> (a) Premises:
>
> 1. $\neg p$
> 2. $\neg r$
> 3. $\neg(p \vee q) \rightarrow r$
>
> Conclusion: q

Official Answer

Source: PDF physical page 6 (2015-midterm.pdf).

(a)

Step	Statement	Reason
1	$\neg(p \vee q) \rightarrow r$	Premise
2	$\neg r$	Premise
3	$(p \vee q)$	Modus tollens (1,2)
4	$\neg p$	Premise
5	q	Disjunctive syllogism (3,4)

Supplementary Explanation

Answer. q follows.

Derivation/proof. 1. $\neg(p \vee q) \rightarrow r$ (premise). 2. $\neg r$ (premise). 3. $\neg\neg(p \vee q)$ (modus tollens, 1–2). 4. $p \vee q$ (double negation, 3). 5. $\neg p$ (premise). 6. q (disjunctive syllogism, 4–5).

L171

Inference and Proofs

2015 Midterm | Question 2(b) | Source page 5 | Difficulty 2 | 4 marks
Q-2015-MID-P02-b | Answer provenance: official

> In this question, p, q, r and s are logic propositions. In each part, a set of premises and a conclusion are given. You are asked to derive the conclusion from the premises using the rules of inference that you have learned in class. Show all steps of your derivations, and state clearly the assumption or the inference rule used in each step.
>
> (b) Premises:
>
> 1. $\neg q$
> 2. $p \rightarrow q$
> 3. $\neg p \rightarrow (r \wedge s)$
>
> Conclusion: r

Official Answer

Source: PDF physical page 6 (2015-midterm.pdf).

(b)

Step	Statement	Reason
1	$p \rightarrow q$	Premise
2	$\neg q$	Premise
3	$\neg p$	Modus tollens (1,2)
4	$\neg p \rightarrow (r \wedge s)$	Premise
5	$(r \wedge s)$	Modus Ponens (3,4)
6	r	Simplification (5)

Supplementary Explanation

Answer. r follows.

Derivation/proof. 1. $\neg q$ and $p \rightarrow q$ (premises). 2. $\neg p$ (modus tollens, 1). 3. $\neg p \rightarrow (r \wedge s)$ (premise). 4. $r \wedge s$ (modus ponens, 2–3). 5. r (simplification, 4).

L172

Inference and Proofs

2015 Midterm | Question 4(a) | Source page 9 | Difficulty 2 | 4 marks
Q-2015-MID-P04-a | Answer provenance: official

> A real number is rational if and only if it can be written as a/b, where a and b are integers. Let x and y be two real numbers such that $xy \neq 0$.
>
> (a) Show, by direct proof, that if x and y are rational, then the product xy is also rational.

Official Answer

Source: PDF physical page 9 (2015-midterm.pdf).

(a) Since x and y are rational, they can be written as $x = \frac{a_1}{b_1}$ and $y = \frac{a_2}{b_2}$, where a_1, b_1, a_1 and b_2 are integers. Then we have $xy = \frac{a_1 a_2}{b_1 b_2}$. Since integers are closed under multiplication operations, $a_1 a_2$ and $b_1 b_2$ are both integers. Therefore xy is rational.

Supplementary Explanation

Answer. xy is rational.

Derivation/proof. Write $x = a/b$ and $y = c/d$ with integers a, b, c, d and $b, d \neq 0$. Then $xy = ac/(bd)$. Closure of integers under multiplication gives integer ac, bd, and $bd \neq 0$. This is a rational representation of xy.

L173

Inference and Proofs

2015 Midterm | Question 4(b) | Source page 9 | Difficulty 2 | 4 marks
Q-2015-MID-P04-b | Answer provenance: official

> A real number is rational if and only if it can be written as a/b, where a and b are integers. Let x and y be two real numbers such that $xy \neq 0$.
>
> (b) Show, by contraposition, that if x is irrational, then $x^{1/5}$ is irrational.

Official Answer

Source: PDF physical page 9 (2015-midterm.pdf).

(b) Assume $x^{1/5}$ were rational. By part (a), $x = x^{1/5}x^{1/5}x^{1/5}x^{1/5}x^{1/5}$ must also be rational, which contradicts the premise of the statement. Therefore if x is irrational, then $x^{1/5}$ is irrational.

Supplementary Explanation

Answer. An irrational real x has irrational real fifth root.

Derivation/proof. Prove the contrapositive. Suppose $x^{1/5} = a/b$ with integers a, b and $b \neq 0$. Then $x = (a/b)^5 = a^5/b^5$ is rational. Thus x irrational implies $x^{1/5}$ irrational. The odd root exists for every real x.

Taking fifth powers gives $x = a^5/b^5$, a ratio of integers with nonzero denominator.

L174

Inference and Proofs

2025 Fall Midterm | Question 1(c) | Source page 4 | Difficulty 3 | 5 marks
Q-2025F-MID-P01-c | Answer provenance: official

> (c) [5 points] Let p, q, r, t denote four propositions. You are given the following premises:
>
> 1. $p \to (r \to q)$
> 2. $\neg q$
> 3. $(q \vee p) \wedge t$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion "$\neg r \wedge t$" is true.

Official Answer

Source: physical PDF page 5 (COMP2711_F2025_Midterm.pdf).

(c)

	Statement	Reason
1	$p \to (r \to q)$	premise
2	$\neg q$	premise
3	$(q \vee p) \wedge t$	premise
4	t	from 3
5	$q \vee p$	from 3
6	p	from 5, 2
7	$r \to q$	from 6, 1
8	$\neg r \vee q$	from 7
9	$\neg r$	from 8, 2
10	$t \wedge \neg r$	from 9,4

Supplementary Explanation

Answer. $\neg r \wedge t$.

Derivation/proof. 1. Obtain $q \vee p$ and t from premise 3 by simplification. 2. Obtain p from the disjunction in step 1 and premise 2, $\neg q$, by disjunctive syllogism. 3. Obtain $r \to q$ from step 2 and premise 1 by modus ponens. 4. Obtain $\neg r$ from step 3 and $\neg q$ by modus tollens. 5. Obtain $\neg r \wedge t$ from step 4 and the t in step 1 by conjunction. Every result follows from a premise or earlier formula. The printed final conjunction is in reverse order; commutativity gives exactly the requested conclusion.

L175

Inference and Proofs

2025 Fall Midterm | Question 2(c) | Source page 7 | Difficulty 3
Q-2025F-MID-P02-c-proof | Answer provenance: official

> (c) [5 points] Consider the following predicates.
>
> $S(x)$ be "x is a satisfactory"
>
> $C(x)$ be "x is a clear explanation"
>
> $E(x)$ be "x is an excuse"
>
> Assume the following premises:
>
> Premise 1: All clear explanations are satisfactory
>
> Premise 2: Some excuses are unsatisfactory
>
> then write a formal proof for the following statement:
>
> Some excuses are not clear explanations.

Official Answer

Source: physical PDF page 8 (COMP2711_F2025_Midterm.pdf).

> While conclusion is $\exists x\, E(x) \land \neg C(x)$. The proof goes as follows:
>
> 1. $\exists x\, E(x) \land \neg S(x)$ (premise)
> 2. $E(x_0) \land \neg S(x_0)$ (for some x_0, by existential specification of 1))
> 3. $\forall x(C(x) \to S(x))$ (premise)
> 4. $C(x_0) \to S(x_0)$ (by universal specification of 3))
> 5. $\neg S(x_0)$ (by simplification of 2))
> 6. $\neg C(x_0)$ (Modus Tollens of 4) and 5))
> 7. $E(x_0)$ (by simplification of 2))
> 8. $E(x_0) \land \neg C(x_0)$ (conjunction of 6) and 7))
> 9. $\exists x\, E(x) \land \neg C(x)$ (existential generalization of 8))

Supplementary Explanation

Answer. $\exists x(E(x) \land \neg C(x))$.

Derivation/proof. Use premises $\forall x(C(x) \to S(x))$ and $\exists x(E(x) \land \neg S(x))$. Existential instantiation introduces a fresh witness a with $E(a) \land \neg S(a)$. Simplification gives E(a) and $\neg S(a)$. Universal instantiation gives $C(a) \to S(a)$; modus tollens then gives $\neg C(a)$. Conjunction yields $E(a) \land \neg C(a)$, and existential generalization yields the required formula. The fresh witness is eliminated by the final generalization, and no claim about all excuses is made.

L176

Inference and Proofs

2025 Spring Midterm | Question 1(c) | Source page 2 | Difficulty 3
Q-2025S-MID-P01-c | Answer provenance: official

> (c) Let p, q, r, s denote four propositions. You are given the following set of premises:
>
> 1. $\neg s \leftrightarrow (q \rightarrow r)$
> 2. $\neg q$
> 3. $(q \rightarrow s) \rightarrow (p \vee s)$
>
> Using rules of inference for propositional logic, show that the truth of the above premises implies that the conclusion $\neg s \wedge p$ is true.

Official Answer

Source: physical PDF page 2 (COMP2711_S2025_Midterm.pdf).

(c)

1. $\neg q$ — Premise
2. $\neg q \vee s$ — From (1)
3. $(q \rightarrow s)$ — Equivalent of (2)
4. $(q \rightarrow s) \rightarrow (p \vee s)$ — Premise
5. $(p \vee s)$ — From (3) and (4)
6. $\neg q \vee r$ — From (1)
7. $q \rightarrow r$ — Equivalent of (6)
8. $\neg s \leftrightarrow (q \rightarrow r)$ — Premise
9. $\neg s$ — From (7) and (8)
10. p — From (5) and (9)
11. $p \wedge \neg s$ — From (9) and (10)

Supplementary Explanation

Answer. $\neg s \wedge p$.

Derivation/proof.

1. $\neg q$, premise 2.
2. $\neg q \vee r$, by addition from 1, hence $q \rightarrow r$.
3. $(q \rightarrow r) \rightarrow \neg s$, by eliminating the biconditional in premise 1.
4. $\neg s$, from 2 and 3 by modus ponens.
5. $\neg q \vee s$, by addition from 1, hence $q \rightarrow s$.
6. $p \vee s$, from 5 and premise 3 by modus ponens.
7. p, from 6 and 4 by disjunctive syllogism.
8. $\neg s \wedge p$, from 4 and 7 by conjunction. The printed answer ends with $p \wedge \neg s$; commutativity gives the requested order.

L177

Inference and Proofs

2024 Spring Midterm | Question 6 | Source page 8 | Difficulty 3 | 10 marks
Q-2024S-MID-P06 | Answer provenance: official

> Prove that $\sqrt{2}+\sqrt{3}$ is irrational.

Official Answer

Source: physical PDF page 8 (COMP2711_S2024_Midterm.pdf).

> Answer. For the sake of contradiction, suppose that $\sqrt{2}+\sqrt{3}$ is rational, that is, $\sqrt{2}+\sqrt{3}=a/b$, where a and b are both integers. Squaring both sides, we obtain $5+2\sqrt{6}=a^2/b^2$, which implies that $\sqrt{6}=\frac{1}{2}\left(\frac{a^2}{b^2}-5\right)=\frac{a^2-5b^2}{2b^2}$. As a^2-5b^2 and $2b^2$ are both integers, it follows that $\sqrt{6}$ is rational. Thus, there exist integers p and q such that $\sqrt{6}=p/q$ and $\gcd(p,q)=1$. Hence $6=p^2/q^2$. It follows that $p^2=6q^2$, which implies that p^2 is even. It follows that p is even, that is, there exists an integer k such that $p=2k$. Substituting into the equation $p^2=6q^2$, we obtain $4k^2=6q^2$. This yields $2k^2=3q^2$. It follows that q^2 is even, which implies that q is even. We have shown that both p and q are even, which contradicts our assumption that $\gcd(p,q)=1$. This proves that $\sqrt{2}+\sqrt{3}$ is irrational.

Supplementary Explanation

Answer. $\sqrt{2}+\sqrt{3}$ is irrational.

Derivation/proof. Suppose $\sqrt{2}+\sqrt{3}=a/b$ with integers a, b and $b\neq 0$. Squaring gives $5+2\sqrt{6}=a^2/b^2$, hence $\sqrt{6}=(a^2-5b^2)/(2b^2)$ is rational. Write it in lowest terms as p/q, with $q\neq 0$ and $\gcd(p,q)=1$. Then $p^2=6q^2$, so p is even: write $p=2k$. Substitution gives $2k^2=3q^2$. The left side is even and three is odd, so q^2 is even and q is also even. This contradicts coprimality in the lowest-terms fraction, disproving the original assumption.

L178

Inference and Proofs

2024 Spring Midterm | Question 7 | Source page 9 | Difficulty 3 | 10 marks
Q-2024S-MID-P07 | Answer provenance: official

> Let k be an even integer. Prove that it is not possible to write 1 as the sum of the reciprocals of k odd integers. Note: As an example, if $k = 4$, then the k odd integers selected could be 3, 3, 5, and 7. Note that, in this case, the sum of the reciprocals of these k odd integers is given by
>
> $$\frac{1}{3} + \frac{1}{3} + \frac{1}{5} + \frac{1}{7},$$
>
> which is not equal to 1. In this problem, you need to show that no matter how we pick the k odd integers, where k is an arbitrary even integer, the sum of the reciprocals of these k odd integers cannot equal 1. Hint: Try to prove this by contradiction.

Official Answer

Source: physical PDF page 9 (COMP2711_S2024_Midterm.pdf).

> Answer. Fix an even integer k. Suppose for the sake of contradiction that there exist k odd integers $a_1, a_2, \ldots, a_k$ where $\frac{1}{a_1} + \frac{1}{a_2} + \cdots + \frac{1}{a_k} = 1$. Fractions seem to complicate the problem, so let's eliminate them by multiplying both sides by the product $a_1 \cdot a_2 \cdots a_k$:
>
> $$\frac{a_1 \cdots a_k}{a_1} + \frac{a_1 \cdots a_k}{a_2} + \cdots + \frac{a_1 \cdots a_k}{a_k} = a_1 \cdot a_2 \cdots a_k. \quad (*)$$
>
> Note that every term on the LHS is an integer because the denominator is always present in the numerator, and is therefore cancelled out. We can use the fact that k is even and that each a_i is odd to observe the following: (a) The RHS of (??) is the product of odd numbers, which must be odd. (b.1) Each term in the LHS is of the form $\frac{a_1 \cdots a_k}{a_i} = a_1 \cdots a_{i-1} \cdot a_{i+1} \cdots a_k$, which is the product of odd integers, and hence must be odd. (b.2) The LHS is the sum of k terms, each of which is odd by the previous observation. Since k is even, that sum must be even. Observation (a) tells us that the RHS of (??) is odd, but observation (b.2) tells us that the LHS is even. This contradicts the fact that LHS and RHS are equal. Therefore, we conclude that our hypothesis was wrong and that such k integers $a_1, \ldots, a_k$ do not exist.

Supplementary Explanation

Answer. The reciprocals of any even number of odd integers cannot sum to 1.

Derivation/proof. Suppose odd integers $a_1, \ldots, a_k$ satisfy $\sum_i 1/a_i = 1$. Odd integers are nonzero, so multiply by $D = \prod_i a_i$ to obtain $\sum_i (D/a_i) = D$. Every $D/a_i = \prod_{j \neq i} a_j$ is odd. There are an even number k of terms, so the left side is even, while the right side D is odd: a contradiction. Negative odd integers and repeated denominators do not change the parity argument. For $k = 0$ the empty sum is zero, not one; a negative k cannot describe a number of terms and yields no such representation either.

L179

Inference and Proofs

2022 Midterm | Question 6(b) | Source page 3 | Difficulty 3 | 10 marks
Q-2022-MID-P06-b | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that $p \equiv 1 \pmod 4$. (Hint: Analyze $(2n)^{p-1}$ using Fermat's Little Theorem and part (a))

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (b) $4n^2 + 1$ is an odd number so we know that p has to be odd. So either $p \equiv 1 \pmod 4$ or $p \equiv 3 \pmod 4$. Assume $p \equiv 3 \pmod 4$ which means $p = 4k + 3$. Since $p \nmid n$ and p is an odd prime, so $p \nmid 2n$. By Fermat's Little Theorem, we have $(2n)^{p-1} \equiv (2n)^{4k+2} \equiv (4n^2)^{2k+1} \equiv 1 \pmod p$. Recall that $p \mid 4n^2 + 1 \implies 4n^2 \equiv -1 \pmod p$. So $(4n^2)^{2k+1} \equiv (-1)^{2k+1} \equiv -1 \pmod p$. Now we have $1 \equiv -1 \pmod p$, which is impossible since $p \neq 2$. This proves that $p \equiv 1 \pmod 4$.

Supplementary Explanation

Answer. $p \equiv 1 \pmod 4$.

Derivation/proof. Because 4n²+1 is odd, p is odd, and p∤n implies gcd(2n,p)=1. If p≡3 mod 4, write p=4k+3. Fermat gives (2n)^(p−1)≡1 modp, but (2n)^(4k+2)=(4n²)^(2k+1)≡(−1)^(2k+1)=−1 modp by the divisibility hypothesis. Hence p divides2, contradicting oddness. The only other odd residue modulo 4 is 1, proving the result.

The required hypothesis is $\gcd(2n, p) = 1$. The power $(4n^2)^{2k+1}$ is -1 modulo p, so comparison with Fermat gives $p \mid 2$, contradicting oddness.

L180

Inference and Proofs

2021 Midterm | Question 3 | Source page 2 | Difficulty 3 | 10 marks
Q-2021-MID-P03 | Answer provenance: official

> Show that there is no largest prime number.

Official Answer

Source: physical PDF page 2 (2021-midterm.pdf).

> We prove this by contradiction.
>
> Suppose there is a largest prime number n. Note that $n! + 1$ has no factors between 1 and n. Thus, if all primes were n or less, $n! + 1$ would have no prime factors, and it would be prime itself. Therefore, $n! + 1$ is a prime number greater than n, which is a contradiction.

Supplementary Explanation

Answer. There is no largest prime.

Derivation/proof. Assume a largest prime p exists and form N=p!+1>1. Every integer greater than one has a prime divisor q (take its smallest divisor greater than one, which must be prime). If q≤p, then q divides p! as well as N and thus divides their difference 1, impossible. Therefore q>p, contradicting maximality. We use a prime divisor of N; we do not assume N itself is prime.

The least divisor $q > 1$ of N is prime: if it were composite, a proper divisor strictly between 1 and q would also divide N, contradicting minimality. The factorial construction need not itself be prime; for example $5! + 1 = 121 = 11^2$.

If $q \leq p$, then $q \mid (N - p!) = 1$, impossible; hence $q > p$.

L181

Inference and Proofs

2021 Midterm | Question 7 | Source page 3 | Difficulty 3 | 10 marks
Q-2021-MID-P07 | Answer provenance: official

> Show that $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$ for any two distinct prime numbers p and q.

Official Answer

Source: physical PDF page 3 (2021-midterm.pdf).

> By Fermat's little theorem, $p^{q-1} \equiv 1 \pmod{q}$. Clearly, $q^{p-1} \equiv 0 \pmod{q}$. Therefore, $p^{q-1} + q^{p-1} \equiv 1 + 0 \equiv 1 \pmod{q}$. Similarly, $p^{q-1} + q^{p-1} \equiv 0 + 1 \equiv 1 \pmod{p}$. By the Chinese remainder theorem, we have $p^{q-1} + q^{p-1} \equiv 1 \pmod{pq}$.

Supplementary Explanation

Answer. The congruence holds.

Derivation/proof. Modulo q, p is nonzero because the primes are distinct, so Fermat gives $p^{q-1} \equiv 1$. Also $q^{p-1} \equiv 0$ since p−1≥1. Their sum is 1 modulo q. Similarly it is 1 modulo p. Thus p and q both divide the sum minus one; because they are coprime, pq divides it. This proves the desired congruence, including the case where one prime is 2.

With $A = p^{q-1} + q^{p-1}$, the separate congruences give $q \mid A - 1$ and $p \mid A - 1$; coprimality gives $pq \mid A - 1$.

L182

Inference and Proofs

2020 Midterm | Question Bonus Problem | Source page 5 | Difficulty 3 | 10 marks
Q-2020-MID-BONUS | Answer provenance: official

> A quasi-square number $n \in \mathbb{N}$ is one that is divisible by a square number. For example, 24 is a quasi-square number because it is divisible by $4 = 2^2$, while 15 is not a quasi-square number.
>
> Prove that for any $k \in \mathbb{N}$, there exist k consecutive numbers all of which are quasi-square numbers. [Hint: Use the Chinese Remainder Theorem.]

Official Answer

Source: physical PDF page 5 (2020-midterm.pdf).

We give a constructive proof. Let p_i be the ith prime number. We know that $\gcd(p_i^2, p_j^2) = 1$ for $i \neq j$. Consider the following system of linear congruences:

$$\begin{aligned} x &\equiv -1 \pmod{p_1^2} \\ x &\equiv -2 \pmod{p_2^2} \\ &\cdots \\ x &\equiv -k \pmod{p_k^2}. \end{aligned}$$

Based on the Chinese Remainder Theorem, there is a solution $x_0 \in \mathbb{Z}_n$ to this system for $n = p_1^2 \cdots p_k^2$. We know that

$$p_1^2 \mid x_0 + 1, \quad p_2^2 \mid x_0 + 2, \quad \ldots, \quad p_k^2 \mid x_0 + k.$$

So $x_0 + 1, \ldots, x_0 + k$ are all quasi-square numbers.

Supplementary Explanation

Answer. For every positive k, there is a block $x_0 + 1, \ldots, x_0 + k$ such that $p_i^2 \mid x_0 + i$ for distinct primes $p_1, \ldots, p_k$.

Derivation/proof. Fix k≥1 and choose k distinct primes $p_1, \ldots, p_k$. Their squares are pairwise coprime: a common prime divisor of p_i^2 and p_j^2 would force $p_i = p_j$. By the Chinese Remainder Theorem, the simultaneous congruences $x \equiv -i \pmod{p_i^2}$ for i=1,...,k have a solution modulo $M = \prod_{i=1}^{k} p_i^2$. Choose a nonnegative representative x_0, or add a multiple of M if a larger positive starting value is desired. For each i, the congruence is exactly $p_i^2 \mid x_0 + i$. Thus all k numbers are positive, consecutive, and divisible by square numbers strictly greater than 1. This proves the stronger nontrivial-square claim, so it also proves the literal existence statement despite the defective example. If a convention includes k=0 in the naturals, the empty block case is vacuous. Independently, for k=3 choose primes 2,3,5: $x_0 = 547$ satisfies residues 3 mod 4, 7 mod 9, and 22 mod 25; the block 548,549,550 equals $4 \cdot 137, 9 \cdot 61, 25 \cdot 22$.

Choose a nonnegative representative x_0 so all $x_0 + i$ are positive. If $k = 0$ is permitted, the empty block is automatic. For $k = 3$, primes $2, 3, 5$ and $x_0 = 547$ give $548 = 4 \cdot 137$, $549 = 9 \cdot 61$, and $550 = 25 \cdot 22$.

L183

Inference and Proofs

2019 Final | Question 7 | Source page 9 | Difficulty 3 | 8 marks
Q-2019-FINAL-P07 | Answer provenance: official

> Let P be a set of 5 distinct points in the plane with integer coordinates. Draw a line segment between every two points of P (note that a total of 10 line segments will be drawn). Show that at least one of these line segments has its midpoint with integer coordinates.

Official Answer

Source: physical PDF page 9 (2019-final.pdf).

The midpoint of the segment whose endpoints are (a, b) and (c, d) is $((a + c)/2, (b + d)/2)$. We are concerned only with integer values of the original coordinates. Clearly the coordinates of these fractions will be integers as well if and only if a and c have the same parity (both odd or both even) and b and d have the same parity. Thus what matters in this problem is the parities of the coordinates. There are four possible pairs of parities: (odd, odd), (odd, even), (even, odd), and (even, even). Since we are given five points, the pigeonhole principle guarantees that at least two of them will have the same pair of parities. The midpoint of the segment joining these two points will therefore have integer coordinates.

Supplementary Explanation

Answer. At least one of the ten segments has an integer-coordinate midpoint.

Derivation/proof. Classify a point (a, b) by $(a \bmod 2, b \bmod 2)$. There are four classes: $(0, 0), (0, 1), (1, 0), (1, 1)$. Five points force two distinct points $(a, b), (c, d)$ in one class by pigeonhole principle. Then a+c and b+d are even, so the midpoint $((a + c)/2, (b + d)/2)$ has both coordinates integral. The required segment is present because every pair is joined.

Every pair of points is joined in the question, so the guaranteed same-parity pair supplies one of the ten drawn segments.

If $a \equiv b \pmod 2$ and $b \equiv c \pmod 2$, then adding the two even differences gives $a \equiv c \pmod 2$. This verifies transitivity of the parity relation.

L184

Inference and Proofs

2019 Midterm | Question 7(b) | Source page 9 | Difficulty 3 | 3 marks
Q-2019-MID-P07-b | Answer provenance: official

> Give an example of two uncountable sets A and B such that $A - B$ is
>
> (b) countably infinite.

Official Answer

Source: physical PDF page 9 (2019-midterm.pdf).

In each case, let us take A to be the set of real numbers. We can let B be the set of real numbers that are not positive integers; in symbols, $B = A - \mathbb{Z}^+$. Then, $A - B = \mathbb{Z}^+$, which is countably infinite.

Supplementary Explanation

Answer. $A = \mathbb{R}$, $B = \mathbb{R} \setminus \mathbb{Z}_{>0}$; then $A - B = \mathbb{Z}_{>0}$.

Derivation/proof. The difference is exactly the positive integers, so it is countably infinite. The input A is uncountable. If B were countable, $\mathbb{R} = B \cup \mathbb{Z}_{>0}$ would be a union of two countable sets and therefore countable, a contradiction. Thus B is also uncountable, verifying every requirement.

For these sets, $A - B = \mathbb{Z}^+$, which is countable. If B were countable too, $\mathbb{R} = B \cup \mathbb{Z}^+$ would be countable, contradicting the uncountability of the reals.

L185

Inference and Proofs

2018 Final | Question 6(b) | Source page 8 | Difficulty 3
Q-2018-FINAL-P06-b | Answer provenance: official

In this question, we consider simple graphs $G = (V, E)$ with $|V| \geq 3$ where all vertices satisfy $deg(v) \geq 2$. Let us call them two-dense graphs.

Consider the following statement:

All two-dense graphs have a triangle subgraph (a cycle of length 3).

Here is an inductive proof of the statement.

Proof. Let the proposition $P(n)$ be the above statement for $|V| = n$. We will prove $P(n)$ is true for $n \geq 3$.

Base case ($P(3)$): For the two-dense graph G with three vertices, each vertex has a degree of 2, so G is simply a triangle and has itself as a subgraph.

Inductive step: Suppose $P(k)$ is true, i.e. any two-dense graph with k vertices has a triangle subgraph. Extend any such graph G to obtain $G' = (V', E')$ by adding one vertex and at least two new edges from the new vertex to any different vertices in $|V|$. G' is a simple graph with $k + 1$ vertices and all vertices satisfy $deg(v) \geq 2$, so G' is two-dense. G' contains G as a subgraph, so using the inductive assumption, G' also has a triangle subgraph. Therefore, $P(k + 1)$ is true.

(b) Since the statement is wrong, the proof must be wrong as well. Choose one of the following statements to explain why the proof is wrong, and explain it.

1. There is an error in the base case.
2. The proof requires another base case, which has not been proven.
3. In the inductive step, G' is wrongly constructed; sometimes it is not a two-dense graph.
4. In the inductive step, G' may not always have a triangle subgraph.
5. In the inductive step, G' is correct and always has a triangle subgraph, but that does not show $P(k + 1)$.

Official Answer

Source: physical PDF page 9 (2018-final.pdf).

The correct choice is (5). Not all graphs with $k + 1$ vertices satisfying $\deg(v) \geq 2$ can be constructed using the procedure in the inductive step. However, $P(k + 1)$ is a statement on all such graphs. Therefore, $P(k + 1)$ being true does not follow from G' containing a triangle.

Supplementary Explanation

Answer. Choice 5.

Derivation/proof. The extension really is two-dense and retains the old triangle. However $P(k + 1)$ quantifies over every two-dense graph with $k + 1$ vertices, not merely the outputs of that extension. For example, C_4 cannot arise by adding a vertex and edges to a triangle, because such an operation never deletes the triangle. The induction step covers only a proper subclass and so cannot establish the universal statement.

The base case and the constructed graph are valid, so choices 1, 3, and 4 misidentify the issue. Adding another base case would not make the construction cover every graph either.

Choose 5, the option identifying the missing universal coverage of the claimed induction extension. The step only constructs a special family; it does not establish the property for all $k + 1$-vertex graphs, so it cannot justify the universal claim.

L186

Inference and Proofs

2018 Final | Question 12(a) | Source page 17 | Difficulty 3
Q-2018-FINAL-P12-a | Answer provenance: official

> a) Let $f_1(x), f_2(x), g(x)$ be functions from the set of real numbers to the set of positive real numbers. If $f_1(x)$ and $f_2(x)$ are both $\Theta(g(x))$, then $f_1(x) + f_2(x)$ is also $\Theta(g(x))$.

Official Answer

Source: physical PDF page 17 (2018-final.pdf).

This is true. There exist positive constraints $C_1, C_1', C_2, C_2', k_1, k_1', k_2, k_2'$, such that $f_1(x) \geq C_1 g(x)$ for all $x > k_1$, $f_1(x) \leq C_1' g(x)$ for all $x > k_1'$, $f_2(x) \geq C_2 g(x)$ for all $x > k_2$, and $f_2(x) \leq C_2' g(x)$ for all $x > k_2'$. Adding the first and third inequality, we get that $f_1(x) + f_2(x) \geq (C_1 + C_2)g(x)$ for all $x > \max\{k_1, k_2\}$. Adding the second and fourth inequality, we get that $f_1(x) + f_2(x) \leq (C_1' + C_2')g(x)$ for all $x > \max\{k_1', k_2'\}$. Hence $f_1(x) + f_2(x)$ is $\Theta(g(x))$ with constraints $(C_1 + C_2), (C_1' + C_2'), \max\{k_1, k_2\}, \max\{k_1', k_2'\}$.

Supplementary Explanation

Answer. True.

Derivation/proof. There are positive constants a_i, b_i and thresholds N_i with $a_i g(x) \leq f_i(x) \leq b_i g(x)$ for $x \geq N_i$, for $i = 1, 2$. For $x \geq \max(N_1, N_2)$, addition yields $(a_1 + a_2)g(x) \leq f_1(x) + f_2(x) \leq (b_1 + b_2)g(x)$. Both new constants are positive and independent of x, proving the required Θ bound.

Choose $N = \max(N_1, N_2)$ so that both asymptotic inequalities hold simultaneously. Beyond that threshold their constants can be combined to prove the asserted bound.

L187

Inference and Proofs

2018 Final | Question 12(b) | Source page 17 | Difficulty 3
Q-2018-FINAL-P12-b | Answer provenance: derived

> b) Let $f(x), g(x)$ be functions from the set of real numbers to the set of positive real numbers. If $f(x)$ is $O(g(x))$ then $2^{f(x)}$ is $O(2^{g(x)})$.

Official Answer

Source: physical PDF page 17 (2018-final.pdf).

This is false. Consider $f(x) = 2x$ and $g(x) = x$. Clearly $f(x)$ is $O(g(x))$. Then $2^{f(x)} = 2^{2x} = 4^x$, whereas $2^{g(x)} = 2^x$. So $2^{f(x)}/2^{g(x)} = 2^x$, i.e., the ratio of the two functions is not bound by any constant C, therefore $2^{f(x)}$ is not $O(2^{g(x)})$.

Supplementary Explanation

Answer. False. For example $g(x) = 2^x$ and $f(x) = 2 \cdot 2^x$.

Derivation/proof. Both functions are positive for all real x, and $f = 2g$, so $f = O(g)$. But $2^{f(x)}/2^{g(x)} = 2^{f(x)-g(x)} = 2^{2^x}$, which tends to infinity. No constant can upper-bound this ratio eventually, so the proposed implication fails. The source's $f = 2x, g = x$ shows the same growth on positive x but does not satisfy the stated codomain on all reals; the displayed example does.

The counterexample must be positive everywhere on the stated real domain. Exponentials $f(x) = 2 \cdot 2^x$ and $g(x) = 2^x$ meet this condition while keeping the ratio and the desired contradiction fixed.

L188

Inference and Proofs

2018 Midterm | Question 3(c) | Source page 5 | Difficulty 3 | 2 marks
Q-2018-MID-P03-c | Answer provenance: official

> Consider the set $S = \{1, 2, \ldots, 2n\}$, and subsets $P = \{p_1, p_2, \ldots, p_m\} \subseteq S$ such that they satisfy the following criterion:
>
> $\forall p_i \in P \forall p_j \in P (i \neq j \rightarrow p_i \nmid p_j)$
>
> In the above, $p_i \nmid p_j$ means p_j is not divisible by p_i.
>
> (c) Prove that there is no $P' \subseteq S$ satisfying the above criterion such that $|P'| > n$. (Hint: By contradiction, suppose $|P'| = n + 1$ satisfies the above criterion. Following (b), put each p_i in the pigeonhole c_i.)

Official Answer

Source: physical PDF page 5 (2018-midterm.pdf).

Let each hole be an odd number in S. The largest odd number in S is $2n - 1$, so the holes are $\{1, 3, 5, \ldots, 2n - 1\}$, so there are n holes, and the odd factor of each number in P' fits into one of these holes. Therefore, there are $n + 1$ numbers for n holes, and by the pigeonhole principle, there exists two different $p_i, p_j \in P'$ in the same hole, i.e. $c_i = c_j$. But we know that this is impossible from (b), giving a contradiction. (2 marks)

Supplementary Explanation

Answer. Every such subset has size at most n.

Derivation/proof. There are exactly n possible odd parts in S, namely $1, 3, \ldots, 2n - 1$. If P' had more than n elements, pigeonhole principle would place two distinct elements in the same odd-part class. Writing them as $c2^r, c2^s$ makes the smaller divide the larger, contradicting the criterion. Thus $|P'| \leq n$.

If there are more than n elements, select any $n + 1$ of them before applying pigeonhole principle.

L189

Inference and Proofs

2018 Midterm | Question 4(c) | Source page 6 | Difficulty 3
Q-2018-MID-P04-c | Answer provenance: official

> (c) Consider a lossless data compression algorithm, like zip or rar: compressed files can always be decompressed to obtain the original files. Prove that it is impossible for a lossless data compression algorithm to reduce the size of every file. (Hint: Use (a), and prove by contradiction on the set of all files of size n bits or less.)

Official Answer

Source: physical PDF page 6 (2018-midterm.pdf).

By contradiction, consider such a function f defined on $F_n \to T$, where F_n is the set of all files of size n bits or less. As f reduces the sizes of all files, T satisfies $T \subseteq F_{n-1}$, the size of all files of size $n-1$ bits or less. Clearly $F_{n-1} \subset F_n$, so $|T| < |F_n|$. However, since f is invertible, f must be injective. Therefore, using (a), we have $|T| \geq |F_{n-1}|$. This is a contradiction.

Supplementary Explanation

Answer. No lossless compressor can shorten every file.

Derivation/proof. Let F_n contain all bit strings of length at most n, including the empty string. Then $|F_n| = \sum_{i=0}^{n} 2^i = 2^{n+1} - 1$, whereas $|F_{n-1}| = 2^n - 1$. If every input became shorter, compression would map F_n into F_{n-1}. Lossless decoding implies injectivity: equal compressed outputs must decode to equal inputs. This contradicts pigeonhole principle because $|F_n| > |F_{n-1}|$.

More explicitly, $|T| \leq 2^n - 1 < 2^{n+1} - 1 = |F_n|$, while lossless decoding requires $|T| \geq |F_n|$.

A decoder must reconstruct every original string from its compressed representation. Thus two distinct inputs cannot share one output; injectivity and the pigeonhole principle supply the cardinality obstruction.

L190

Inference and Proofs

2018 Midterm | Question 5(a) | Source page 7 | Difficulty 3
Q-2018-MID-P05-a | Answer provenance: official

> (a) Let x, y, z be positive integers. Show that, if $z \mid xy$ and y, z are relatively prime, then $z \mid x$.

Official Answer

Source: physical PDF page 7 (2018-midterm.pdf).

Since y, z are relatively prime, $\gcd(y, z) = 1$. Using the extended Euclidean algorithm, we can calculate integers s, t such that

$$sy + tz = 1 \tag{1}$$

Multiplying both sides of (1) with x, we get

$$syx + tzx = x \tag{2}$$

From the problem statement, $z \mid xy$, thus $z \mid syx$. Moreover, it is obvious that $z \mid tzx$. We know that from this it follows that z divides their sum, that is, $z \mid (syx + tzx)$. But, from (2), $(syx + tzx) = x$, therefore $z \mid x$. Students may choose to explicitly prove the fact that if $a \mid b$ and $a \mid c$ then $a \mid (b + c)$. This is fine but not necessary.

Supplementary Explanation

Answer. $z \mid x$.

Derivation/proof. Since $\gcd(y, z) = 1$, Bezout's identity supplies integers s, t with $sy + tz = 1$. Multiply by x to obtain $sxy + tzx = x$. The hypothesis $z \mid xy$ gives $z \mid sxy$, and $z \mid tzx$ directly. Hence z divides their sum x.

Multiplying the Bezout equation by x gives $sxy + tzx = x$. Both summands on the left are divisible by z, so $z \mid (sxy + tzx) = x$, proving the required divisibility.

L191

Inference and Proofs

2017 Final | Question 4(c) | Source page 6 | Difficulty 3 | 3 marks
Q-2017-FINAL-P04-c | Answer provenance: derived

> Recall that in RSA, we need to choose two primes p and q, and the public key is (n, e) where $n = pq$. Let $T = (p-1)(q-1)$. The private key is d, where $ed \equiv 1 \pmod{T}$. The encryption of x is $x^e \bmod n$. This means that d must be an integer in $\mathbb{Z}_T$ that is relatively prime with T. We can find d using the following algorithm:
>
> 1. Randomly pick $d \in \mathbb{Z}_T$.
> 2. If d and T are relatively prime, then return d. If they are not, then pick a new d and loop from step 1.
>
> Assume that we will never pick the same d twice.
>
> (c) Bob says: "Instead of picking a random d, I will iterate over the primes from smallest to largest: I will test $d = 2, d = 3, d = 5, d = 7, \ldots$ until I get a valid d." Prove that in the worst case, this algorithm will terminate in $O(\log n)$ number of loops.

Official Answer

Source: physical PDF page 7 (2017-final.pdf).

Suppose $P_{\log_2 n}$ consists of the first $\log_2 n$ primes. Then $\prod_{x \in P_{\log_2 n}} x > \prod_{x \in P_{\log_2 n}} 2 > n$. So using the fundamental theorem of arithmetic, there cannot be more than $\log_2 n$ primes which are factors of n. This means that Bob will have to try at most $\log_2 n + 1 = O(\log n)$ primes.

Supplementary Explanation

Answer. At most $\lfloor \log_2 T \rfloor + 1 = O(\log n)$ prime candidates are tested.

Derivation/proof. A prime candidate fails precisely when it divides T. If r distinct primes fail, their product divides T and is at least 2^r. Therefore $2^r \leq T$, giving $r \leq \log_2 T$. The next tested prime succeeds, so there are at most $r + 1$ tests. Since $T = (p-1)(q-1) < pq = n$, this is $O(\log n)$. This counts candidate tests, not prime generation time.

L192

Inference and Proofs

2017 Final | Question 14(c) | Source page 18 | Difficulty 3 | 4 marks
Q-2017-FINAL-P14-c | Answer provenance: official

> In this question, we want to find the minimum number of edges in a connected graph with n vertices.
>
> (c) Using the above, prove that there is no connected graph with n vertices and fewer than $n-1$ edges.

Official Answer

Source: physical PDF page 18 (2017-final.pdf).

Suppose in contradiction that G is a connected graph that has n vertices and k edges where $k < n-1$. Then using part (b), we have a connected graph that has $n-1$ vertices and $k-1$ edges. We can repeat this process until we reach a connected graph with at least 2 vertices and 0 edges. This is a contradiction, as such a graph is clearly not connected.

Supplementary Explanation

Answer. Every connected n-vertex graph has at least $n-1$ edges.

Derivation/proof. The $n = 1$ case holds since edge counts are nonnegative. Suppose $n \geq 2$ and a connected graph has $k < n-1$ edges. Connectivity gives every degree at least one. The degree sum $2k < 2n$ ensures some vertex has degree one. Delete it and its incident edge; a simple path between remaining vertices cannot use a degree-one vertex internally, so connectivity is preserved. Repeat while edges remain. After k deletions, at least $n-k > 1$ vertices remain and no edges remain, impossible for a connected graph. The reduction is used only for at least two vertices, its valid domain.

At each deletion, the difference between the vertex count and edge count stays $n-k > 1$, so the reduction is invoked only within its valid range.

If a graph with n vertices and $k < n$ edges had every vertex of degree at least two, the handshake identity would instead give $\sum \deg v \geq 2n$. But $\sum \deg v = 2k < 2n$, a contradiction; some vertex has degree at most one.

L193

Inference and Proofs

2016 Midterm | Question 5(a) | Source page 9 | Difficulty 3 | 5 marks
Q-2016-MID-P05-a | Answer provenance: official

> There are n dots painted on a paper. A kid draws some line segments on the paper and each line segment connects exactly two dots as the two end-points of this line segment. For example, the following figure shows 3 line segments and 6 dots.
>
> Note that if one line segment (connecting two dots) is an extension of another line segment (connecting two dots where one of them is the dot connected by the previous line segment mentioned), we will still refer them as two separate line segments (not one line segment). For example, the following figure shows 3 line segments and 6 dots.
>
> We say that the n dots are all connected if for every pair of two dots x and y, there exists a sequence of line segments, namely $(d_1, d_2), (d_2, d_3), (d_3, d_4), \ldots, (d_{l-1}, d_l)$ where d_i is a dot for $i \in [1, l]$, such that $d_1 = x$ and $d_l = y$. For example, the following figure shows that the 6 dots are all connected but the above 2 figures shows that 6 dots are not all connected.
>
> The following shows an example that the line segments form a loop.
>
> (a) Prove by contraposition that if the kid only draws $n - 2$ line segments, then the n dots are not all connected.

Official Answer

Source: PDF physical page 9 (2016-midterm.pdf).

(a) Let p be the kid only draws $n - 2$ line segments.

Let q be the n dots are not all connected.

To prove $p \Rightarrow q$ using contraposition, we state $\neg q$ which is "n dots are all conntected". Since all n dots are connected, each dot must have a least one line segment attached to it. We use a highlighter pen to highlight one arbitrary line segment and the two dots connected by the line segment. Then we highlight another line segment such that it connects one of the "highlighted" dots and one of the "unhighlighted" dots. Thus we will have two line segments and three dots highlighted and $n - 3$ dots unhighligthed. The process repeats until all n dots are highlighted. The process is repeatable since every dot is connected. That means there must be some unhighlighted dots connecting with a highlighted dot at any state of the process. After n dots are highlighted, there are $n - 1$ line segments highlighted. That implies the kid has drawn at least $n - 1$ line segments ($\neg p$).

Supplementary Explanation

Answer. With only $n - 2$ segments, the dots cannot all be connected.

Derivation/proof. Assume the graph is connected. Start with one highlighted vertex. While fewer than n vertices are highlighted, a path to an unhighlighted vertex has a first edge leaving the highlighted set. Select that edge and its new endpoint. Each step adds one previously unhighlighted vertex using a distinct edge, so reaching all n vertices selects $n - 1$ edges. Thus a connected graph cannot have only $n - 2$ edges. This proves the contrapositive for the meaningful range $n \geq 2$.

For $n = 1$, the antecedent of drawing -1 segments is impossible, so the statement is vacuous. For $n \geq 2$, each chosen crossing edge is new because it reaches a previously unhighlighted vertex.

The construction selects $n - 1$ distinct edges, one for each newly reached vertex after the starting vertex.

L194

Inference and Proofs

2016 Midterm | Question 5(b) | Source page 9 | Difficulty 3 | 5 marks
Q-2016-MID-P05-b | Answer provenance: official

> There are n dots painted on a paper. A kid draws some line segments on the paper and each line segment connects exactly two dots as the two end-points of this line segment. For example, the following figure shows 3 line segments and 6 dots.
>
> Note that if one line segment (connecting two dots) is an extension of another line segment (connecting two dots where one of them is the dot connected by the previous line segment mentioned), we will still refer them as two separate line segments (not one line segment). For example, the following figure shows 3 line segments and 6 dots.
>
> We say that the n dots are all connected if for every pair of two dots x and y, there exists a sequence of line segments, namely $(d_1, d_2), (d_2, d_3), (d_3, d_4), \ldots, (d_{l-1}, d_l)$ where d_i is a dot for $i \in [1, l]$, such that $d_1 = x$ and $d_l = y$. For example, the following figure shows that the 6 dots are all connected but the above 2 figures shows that 6 dots are not all connected.
>
> The following shows an example that the line segments form a loop.
>
> (b) Prove by contradiction that if the kid draws a loop that connects the n dots, then s/he needs at least n line segments to draw the loop.

Official Answer

Source: PDF physical page 9 (2016-midterm.pdf).

(b) Assume that it is possible to draw a loop with smaller than n line segments for n dots. Since each dot needs to be connected with exactly two line segments in a loop, we count the total number of connections between a line segment and a dot, there are $2n$ connections. If the loop has smaller than n line segment, say, there are only k line segments where $k < n$, and thus each line segment on average needs to connect $2n/k$ dots (more than 2 dots). However, each line segment can exactly connect two dots. By contradiction, it requires at least n line segments to draw a loop that connects n dots.

Supplementary Explanation

Answer. At least n line segments are needed.

Derivation/proof. Suppose such a loop has $m < n$ segments. Every one of the n cycle vertices has two incident loop edges, so there are $2n$ endpoint incidences. Each segment has exactly two endpoints and contributes two incidences, giving only $2m < 2n$. This contradiction establishes the bound (a simple cycle has exactly n edges).

Equivalently, $2n = 2m < 2n$ is impossible. Count the loop's own edges; extra chords do not reduce how many edges its loop needs.

L195

Inference and Proofs

2015 Midterm | Question 2(c) | Source page 5 | Difficulty 3 | 4 marks
Q-2015-MID-P02-c | Answer provenance: official

> In this question, p, q, r and s are logic propositions. In each part, a set of premises and a conclusion are given. You are asked to derive the conclusion from the premises using the rules of inference that you have learned in class. Show all steps of your derivations, and state clearly the assumption or the inference rule used in each step.
>
> (c) Premises:
>
> 1. $s \to r$
> 2. $p \wedge q$
> 3. $p \to \neg(q \wedge r)$
>
> Conclusion: $\neg s$

Official Answer

Source: PDF physical page 6 (2015-midterm.pdf).

(c)

Step	Statement	Reason
1	$p \wedge q$	Premise
2	p	Simplification
3	q	Simplification
4	$p \to \neg(q \wedge r)$	Premise
5	$\neg(q \wedge r)$	Modus Ponens (3,4)
6	$\neg q \vee \neg r$	DeMorgan's law, 5
7	$\neg r$	Disjunctive syllogism (3,6)
8	$s \to r$	Premise
9	$\neg s$	Modus tollens (7,8)

Supplementary Explanation

Answer. $\neg s$ follows.

Derivation/proof. 1. p, q (simplification of the premise $p \wedge q$). 2. $\neg(q \wedge r)$ (modus ponens with $p \to \neg(q \wedge r)$ and p). 3. $\neg q \vee \neg r$ (De Morgan, 2). 4. $\neg r$ (disjunctive syllogism using q, 3). 5. $\neg s$ (modus tollens using $s \to r$, 4).

The printed solution's step 5 cites its line 3 (q) where line 2 (p) is needed; the derivation above uses the correct premise.

L196

Inference and Proofs

2015 Midterm | Question 4(c) | Source page 9 | Difficulty 3 | 4 marks
Q-2015-MID-P04-c | Answer provenance: official

> A real number is rational if and only if it can be written as a/b, where a and b are integers. Let x and y be two real numbers such that $xy \neq 0$.
>
> (c) Show, by contradiction, that if x is irrational and xy is rational, then y is irrational.

Official Answer

Source: PDF physical page 9 (2015-midterm.pdf).

(c) Assume x is irrational and xy is rational, and y were rational.

Let $y = \frac{a_2}{b_2}$. Then $\frac{1}{y} = \frac{b_2}{a_2}$, and hence would be rational. By part (1), $xy\frac{1}{y} = x$ would also be rational, which contradicts the assumption that x is irrational. Therefore, if x is irrational and xy is rational, then y is irrational.

Supplementary Explanation

Answer. y must be irrational.

Derivation/proof. Assume x irrational, xy rational, and y rational. Write $xy = a/b$ and $y = c/d$ with nonzero b, d, c; $c \neq 0$ follows from $y \neq 0$. Then $x = (a/b)/(c/d) = ad/(bc)$ is rational, contradicting the premise. Therefore y is irrational.

The shared source condition $xy \neq 0$ supplies $y \neq 0$, hence $c \neq 0$ in the displayed supplementary ratio. The nonzero condition is part of the problem, not an added assumption.

The nonzero factors b and c ensure $bc \neq 0$, making the rational representation valid.

L197

Inference and Proofs

2014 Final | Question 6(a) | Source page 10 | Difficulty 3
Q-2014-FINAL-P06-a | Answer provenance: official

> Prove the following statements. You may use a direct proof, a proof by contraposition or a proof by contradiction as appropriate.
>
> (a) $\sqrt{3}$ is an irrational number.

Official Answer

Source: PDF physical page 10 (2014-final.pdf).

(a) Suppose $\sqrt{3}$ is rational. Then there are integers m and n with no common factors so that $\sqrt{3} = m/n$. Squaring both sides gives $3 = \frac{m^2}{n^2}$, or $m^2 = 3n^2$. This implies that 3 divides m^2. Together with the fact that 3 is prime, we have 3 divides m. Let $m = 3k$ for some integer k. Substituting into $3n^2 = m^2$, we have $3n^2 = (3k)^2$, thus $n^2 = 3k^2$, which implies 3 divides n^2. Therefore, 3 divides n. Hence both m and n have a common factor of 3. But this contradicts the supposition that m and n have no common factors.

Another way to argue for contradiction: m^2 has even number of factors in its prime factorization, while $3n^2$ has odd number of factors.Therefore, they cannot be equal.

Supplementary Explanation

Answer. $\sqrt{3}$ is irrational.

Derivation/proof. Assume $\sqrt{3} = m/n$ in lowest terms. Then $m^2 = 3n^2$, so the prime 3 divides m^2 and hence m; write $m = 3k$. Substitution gives $n^2 = 3k^2$, so 3 also divides n, contradicting coprimality.

The denominator satisfies $n \neq 0$, and lowest terms means $\gcd(m, n) = 1$. In the alternative prime-factor argument, factors are counted with multiplicity; more specifically, the exponent of the prime 3 is even in m^2 and odd in $3n^2$.

L198

Inference and Proofs

2014 Final | Question 6(b) | Source page 10 | Difficulty 3
Q-2014-FINAL-P06-b | Answer provenance: official

> Prove the following statements. You may use a direct proof, a proof by contraposition or a proof by contradiction as appropriate.
>
> (b) If m and n are integers such that mn is even, then m is even or n is even.

Official Answer

Source: PDF physical page 10 (2014-final.pdf).

(b) Suppose m and n are both odd. $m = 2k + 1$ for some integer k and $n = 2h + 1$ for some integer h. Then $mn = (2k + 1)(2h + 1) = 2(2kh + k + h) + 1$, which is odd.

Supplementary Explanation

Answer. The statement follows by contraposition.

Derivation/proof. Assume m, n are both odd: $m = 2k + 1$, $n = 2h + 1$. Then $mn = 4kh + 2k + 2h + 1 = 2(2kh + k + h) + 1$, odd. Thus if mn is even, they cannot both be odd, so at least one is even.

The odd representations are $m = 2k + 1, n = 2h + 1$ with integer k, h.

L199

Inference and Proofs

2023 Fall Final | Question 4(d) | Source page 7 | Difficulty 4 | 4 marks
Q-2023F-FINAL-P04-d | Answer provenance: official

> The CSE Department of HKUST has a total of n light bulbs and m switches. We know that each light bulb is connected to at least one switch. By flipping a switch, the status of every light bulb connected to it will change (so if it was OFF it will be turned ON and vice versa). Assume X_i is the random variable that shows the status of the ith light bulb. $X_i = 0$ means light bulb i is OFF and $X_i = 1$ means it is ON. Also, let Y be the random variable that shows the number of light bulbs that are ON in the department. Note that the initial status of every light bulb is arbitrary.
>
> Recall the CSE Department with its light bulbs and switches again. Prove that it is possible to have at least half of the light bulbs being turned ON.
>
> (Hint: You might want to use some facts from the previous parts. You are also allowed to use the fact in part (c) even if you could not prove it)

Official Answer

Source: physical PDF page 7 (COMP2711_F2023_Final.pdf).

> (d) If we flip every switch with 50% probability, then on expectation half of the light bulbs will be on. So based on (c) there exists a way for switches such that at least half of the light bulbs are on.

Supplementary Explanation

Answer. There exists a switch configuration with at least $n/2$ bulbs ON (equivalently at least $\lceil n/2 \rceil$).

Derivation/proof. Choose independently for each of the m switches whether to flip it, with probability $1/2$. This is a deliberately constructed uniform distribution on all 2^m configurations. For bulb i, select one incident switch; toggling just that choice pairs each configuration with another in which bulb i has the opposite final state. Thus $E(X_i) = 1/2$ regardless of its initial state. Since $Y = \sum_i X_i$, linearity gives $E(Y) = n/2$, even though the bulb indicators may depend on one another. If every configuration had $Y < n/2$, their finite uniform average would be strictly below $n/2$, contradicting $E(Y) = n/2$. Therefore some configuration has $Y \geq n/2$. Equivalently, apply the finite-average lower-witness argument to $K = -Y$; that argument is proved locally, without importing the quarantined claim (c).

The 2^m equally likely switch configurations admit a one-to-one pairing by toggling any chosen switch incident to a fixed bulb. Exactly one member of each pair leaves that bulb on. Linearity of expectation holds even when the bulb indicators are dependent. Apply part (c) to $K = -Y$ to obtain $Y \geq n/2$, hence at least $\lceil n/2 \rceil$ lit bulbs.

L200

Inference and Proofs

2023 Fall Midterm | Question 4(d) | Source page 9 | Difficulty 4 | 4 marks
Q-2023F-MID-P04-d | Answer provenance: official

> Imagine having an equilateral triangle with each side being 3 cm. Prove that for all possible ways that you can place 10 points inside this equilateral triangle, there exist at least two points with their distance being less than 1 cm.

Official Answer

Source: physical PDF page 10 (COMP2711_F2023_Midterm.pdf).

> (d) We split the equilateral triangle of sides 3 into 9 equilateral triangles of side 1cm as follows:
>
> page=10 (COMP2711_F2023_Midterm.pdf)
>
> Now we want to put 10 points inside this triangle. Which means at least two points will be either inside one of the $1 \times 1 \times 1$ triangle cells (based on pigeonhole principle and that there are 10 points and 9 triangles) which means their distance is less than 1. Also it is possible for both points to be on the internal lines (not on the border lines though) but even in that case the results remains the same because the distance between any two points on the internal lines (and not on the border lines) are still less than 1 and also if one point is one the internal line and another point is completely inside the $1 \times 1 \times 1$ related triangle, their distance is still less than 1cm.

Supplementary Explanation

Answer. Some pair of the ten interior points has distance strictly less than 1 cm.

Derivation/proof. Subdivide the side-3 triangle into nine equilateral triangles of side 1 by drawing parallels through the trisection points. Assign each point on a shared edge to one adjacent cell, so every point belongs to exactly one of nine cells. Ten points force two into the same cell. A unit equilateral triangle has diameter 1, and equality is possible only between two distinct vertices. In this subdivision only one lattice vertex lies strictly inside the original triangle; every other small-triangle vertex is on the original boundary. Consequently two allowed interior points cannot be such a distinct vertex pair. Their distance is therefore strictly less than 1.

There are nine side-one triangles. Only one grid vertex lies strictly inside the original triangle, so the distance must be strictly less than one for the selected pair. If the outer boundary were allowed, the ten grid vertices would be a counterexample.

L201

Inference and Proofs

2022 Midterm | Question 6(c) | Source page 3 | Difficulty 4 | 10 marks
Q-2022-MID-P06-c | Answer provenance: official

> Assume p is a prime and n is a positive integer such that $p \mid (4n^2 + 1)$.
>
> Show that there are infinitely many primes of the form $4k + 1$.

Official Answer

Source: physical PDF page 3 (2022-midterm.pdf).

> (c) Assume we have finitely many primes of form $4k + 1$, say $p_1, p_2, \cdots, p_t$. Consider $N = 4(p_1 p_2 \ldots p_t)^2 + 1$. Let p be any prime factor of N (p may be N itself if N is prime). Based on part (b), $p = 4k + 1$ for some k. Also note that p must be different from $p_1, \ldots, p_t$, since none of them divides N. Then we have found a new prime p in the form of $4k + 1$, which is a contradiction.

Supplementary Explanation

Answer. Infinitely many primes are $1 \pmod 4$.

Derivation/proof. Suppose the complete finite list is p1,...,pt and put N=4(p1...pt)²+1. (The class is nonempty because5 belongs to it.) Choose a prime divisor q of N. Applying the preceding divisor result with n=p1...pt gives q≡1 mod 4. For each listed pi, N≡1 modpi, so none divides N. Thus q is in the required class but absent from the allegedly complete list, a contradiction. N itself need not be prime.

With $M = p_1 \cdots p_t$, we have $N = 4M^2 + 1 > 1$. A prime divisor satisfies $q \equiv 1 \pmod 4$, but $N \equiv 1 \pmod{p_i}$ gives $p_i \nmid N$ for every listed prime.

L202

Inference and Proofs

2019 Final | Question 4(b) | Source page 6 | Difficulty 4 | 4 marks
Q-2019-FINAL-P04-b | Answer provenance: official

> (b) Give an example of two increasing functions $f(n)$ and $g(n)$ from the set of positive integers to the set of positive integers such that neither $f(n)$ is $O(g(n))$ nor $g(n)$ is $O(f(n))$. Justify your answer.

Official Answer

Source: physical PDF page 6 (2019-final.pdf).

We want the functions to play leap-frog, with first one much bigger than the other. Something like this will do the trick: Let $f(n) = n^{2\lfloor n/2 \rfloor + 1}$ and $g(n) = n^{2\lceil n/2 \rceil}$. Thus, $\{f(1) = 1^1, g(1) = 1^2\}, \{f(2) = 2^3, g(2) = 2^2\}, \{f(3) = 3^3, g(3) = 3^4\}, \{f(4) = 4^5, g(4) = 4^4\}$, and so on. Then for even n we have $f(n)/g(n) = n$, and for odd n we have $g(n)/f(n) = n$. Because both ratios are unbounded, neither function is big-O of the other.

Supplementary Explanation

Answer. $f(n) = n^{2\lfloor n/2 \rfloor + 1}$ and $g(n) = n^{2\lceil n/2 \rceil}$.

Derivation/proof. Both exponent sequences are positive and nondecreasing, and the base n strictly increases, so each function is strictly increasing on positive integers. For even n, the exponents are n+1 and n, so $f(n)/g(n) = n$. For odd n, the exponents are n and n+1, so $g(n)/f(n) = n$. Each ratio is therefore unbounded along an infinite subsequence. Any purported eventual constant big-O bound fails on its corresponding subsequence.

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L096	Q-2023S-MID-P02-c	5
L097	Q-2022-FINAL2-P05-a-i	3
L098	Q-2022-MID-P02-b	1
L099	Q-2022-MID-P03-3	2
L100	Q-2021-MID-P01-b	1
L101	Q-2021-MID-P01-c	1
L102	Q-2021-MID-P01-d	1
L103	Q-2021-MID-P02-proof	1
L104	Q-2021-MID-P02-translation-conclusion	1
L105	Q-2020-FINAL1-P01-a-ii	1
L106	Q-2020-MID-P01-b	1

Exercise	Knowledge-base question ID	Source page
L107	Q-2020-MID-P01-d	1
L108	Q-2020-MID-P03-b	2
L109	Q-2020-MID-P03-e	2
L110	Q-2020-MID-P03-f	2
L111	Q-2019-FINAL-P01-a	3
L112	Q-2019-FINAL-P01-d	3
L113	Q-2019-MID-P02-c	4
L114	Q-2019-MID-P02-d	4
L115	Q-2019-MID-P02-e	4
L116	Q-2018-FINAL-P01-b	3
L117	Q-2018-MID-P02-c	4
L118	Q-2018-MID-P02-d	4
L119	Q-2017-MID-P01-c	3
L120	Q-2017-MID-P01-d	3
L121	Q-2017-MID-P02-2	4
L122	Q-2017-MID-P02-4	4
L123	Q-2016-MID-P06-a	10
L124	Q-2016-MID-P06-b	10
L125	Q-2016-MID-P06-d	10
L126	Q-2015-MID-P03-statement-a	7
L127	Q-2015-MID-P03-statement-b	7
L128	Q-2015-MID-P03-statement-c	7
L129	Q-2013-FINAL-P06-b	9
L130	Q-2025F-FINAL-P01-c-ii	3
L131	Q-2025F-MID-P02-b	6
L132	Q-2025S-MID-P02-b-v	4
L133	Q-2024F-MID-P02-c	5

Exercise	Knowledge-base question ID	Source page
L134	Q-2023F-FINAL-P01-a-ii	3
L135	Q-2023F-FINAL-P01-b	3
L136	Q-2023F-MID-P02-a	5
L137	Q-2023F-MID-P02-b	5
L138	Q-2023S-MID-P02-d	5
L139	Q-2022-MID-P02-c	1
L140	Q-2022-MID-P03-proof	2
L141	Q-2020-FINAL1-P01-a-i	1
L142	Q-2020-MID-P04	2
L143	Q-2019-MID-P04	6
L144	Q-2019-MID-P05	7
L145	Q-2016-MID-P06-c	10
L146	Q-2015-MID-P01-c	3
L147	Q-2015-MID-P01-d	3
L148	Q-2015-MID-P03-equivalence	7
L149	Q-2014-FINAL-P05-b	9
L150	Q-2014-FINAL-P05-c	9
L151	Q-2019-MID-P06-b	8
L152	Q-2017-MID-P04-a	6
L153	Q-2017-MID-P04-b	6
L154	Q-2025F-FINAL-P01-b	3
L155	Q-2023F-MID-P03-a	7
L156	Q-2022-MID-P06-a	3
L157	Q-2020-MID-P08	4
L158	Q-2019-MID-P06-a	8
L159	Q-2019-MID-P08	10
L160	Q-2018-FINAL-P05-a	7

Exercise	Knowledge-base question ID	Source page
L161	Q-2018-FINAL-P06-a	8
L162	Q-2018-MID-P03-b	5
L163	Q-2018-MID-P04-a-proof	6
L164	Q-2018-MID-P04-b	6
L165	Q-2018-MID-P05-b	7
L166	Q-2017-FINAL-P13-a	17
L167	Q-2017-FINAL-P13-b	17
L168	Q-2017-MID-P03-a	5
L169	Q-2017-MID-P03-b	5
L170	Q-2015-MID-P02-a	5
L171	Q-2015-MID-P02-b	5
L172	Q-2015-MID-P04-a	9
L173	Q-2015-MID-P04-b	9
L174	Q-2025F-MID-P01-c	4
L175	Q-2025F-MID-P02-c-proof	7
L176	Q-2025S-MID-P01-c	2
L177	Q-2024S-MID-P06	8
L178	Q-2024S-MID-P07	9
L179	Q-2022-MID-P06-b	3
L180	Q-2021-MID-P03	2
L181	Q-2021-MID-P07	3
L182	Q-2020-MID-BONUS	5
L183	Q-2019-FINAL-P07	9
L184	Q-2019-MID-P07-b	9
L185	Q-2018-FINAL-P06-b	8
L186	Q-2018-FINAL-P12-a	17
L187	Q-2018-FINAL-P12-b	17

Exercise	Knowledge-base question ID	Source page
L188	Q-2018-MID-P03-c	5
L189	Q-2018-MID-P04-c	6
L190	Q-2018-MID-P05-a	7
L191	Q-2017-FINAL-P04-c	6
L192	Q-2017-FINAL-P14-c	18
L193	Q-2016-MID-P05-a	9
L194	Q-2016-MID-P05-b	9
L195	Q-2015-MID-P02-c	5
L196	Q-2015-MID-P04-c	9
L197	Q-2014-FINAL-P06-a	10
L198	Q-2014-FINAL-P06-b	10
L199	Q-2023F-FINAL-P04-d	7
L200	Q-2023F-MID-P04-d	9
L201	Q-2022-MID-P06-c	3
L202	Q-2019-FINAL-P04-b	6